

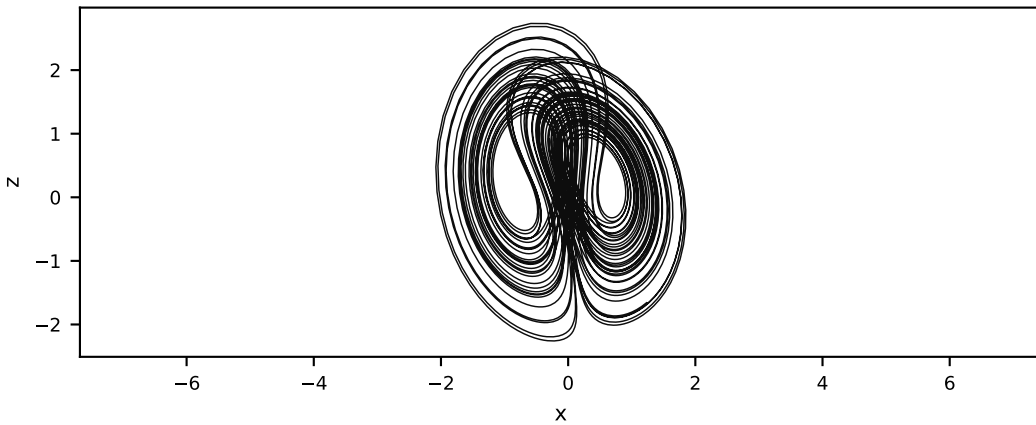
Multi-seed M-finescan: high-resolution trajectories

For each (M, P) cell we pick the seed whose ORTH0 PSE is the median of the three seeds, so the figure represents the typical run rather than the best or worst. Each page: 3-D views (top) and x-z 2-D projections (bottom) for ground truth / vanilla / orthogonal.

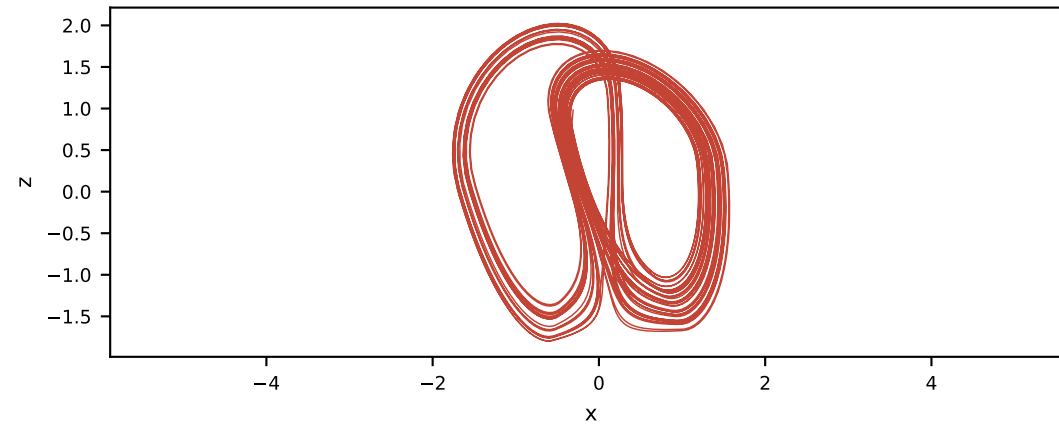
Header gives PSE and D_H_state mean +/- std over the 3 seeds and the relative ortho-vs-vanilla gap on the mean.

Top 4 ortho wins on D_H_state (state-space Hellinger) - median-PSE seed

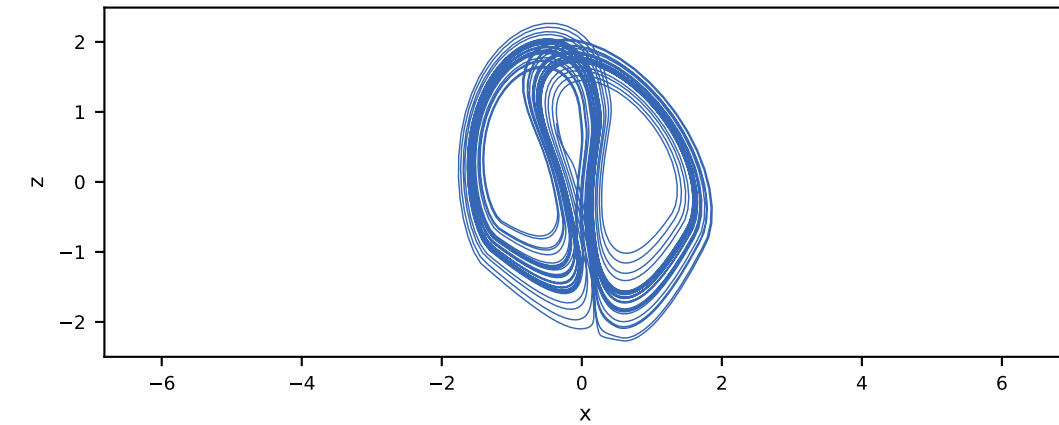
M=15 P=4 truth (D_H gap -21%)



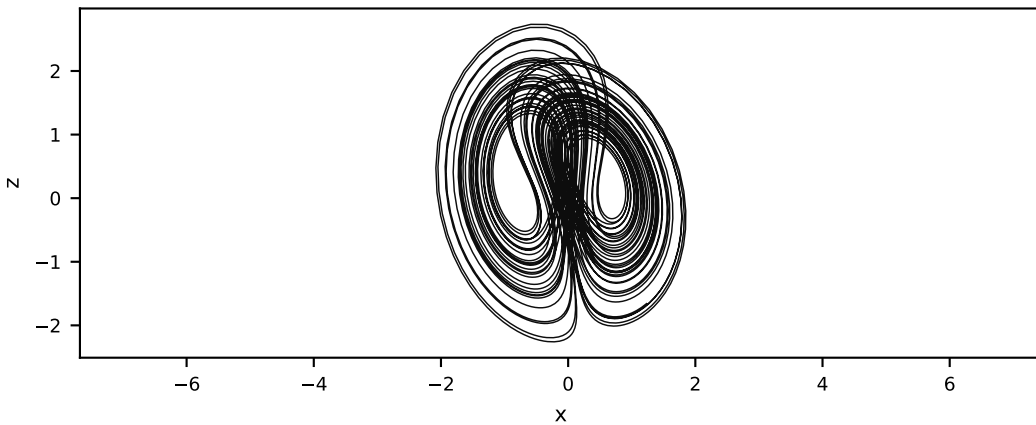
M=15 P=4 vanilla



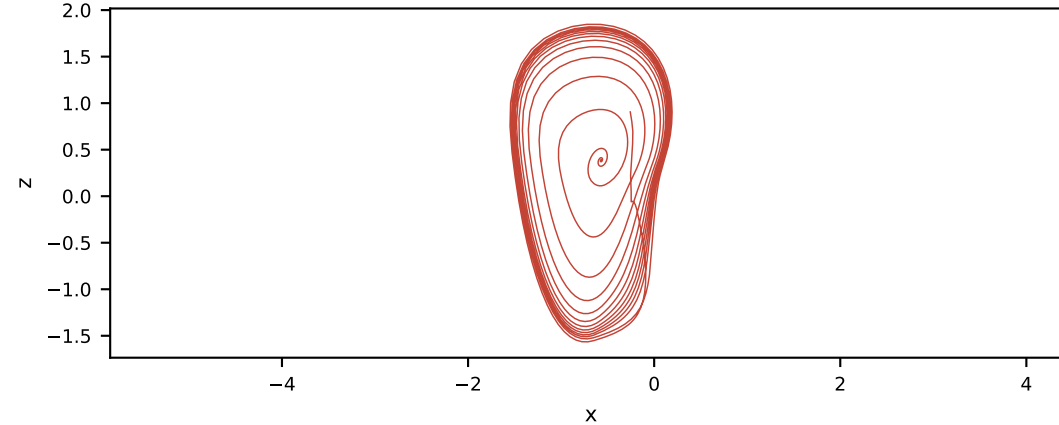
M=15 P=4 ortho



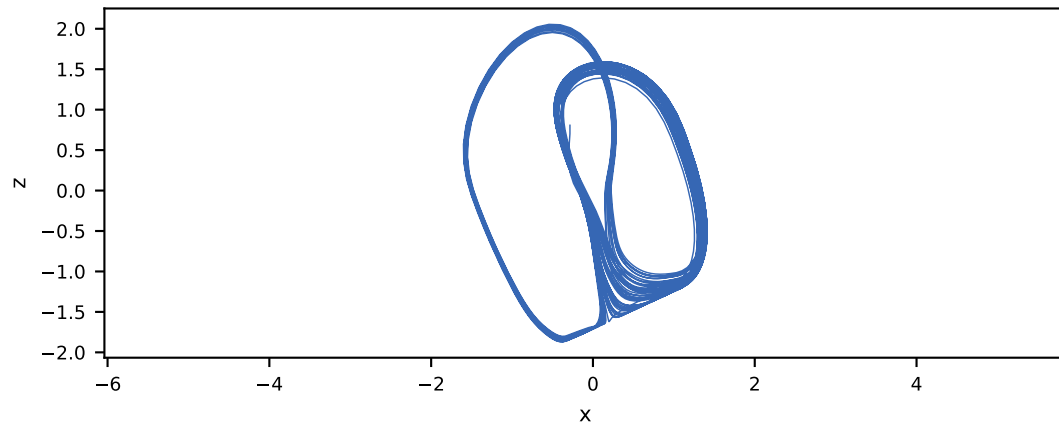
M=15 P=3 truth (D_H gap -17%)



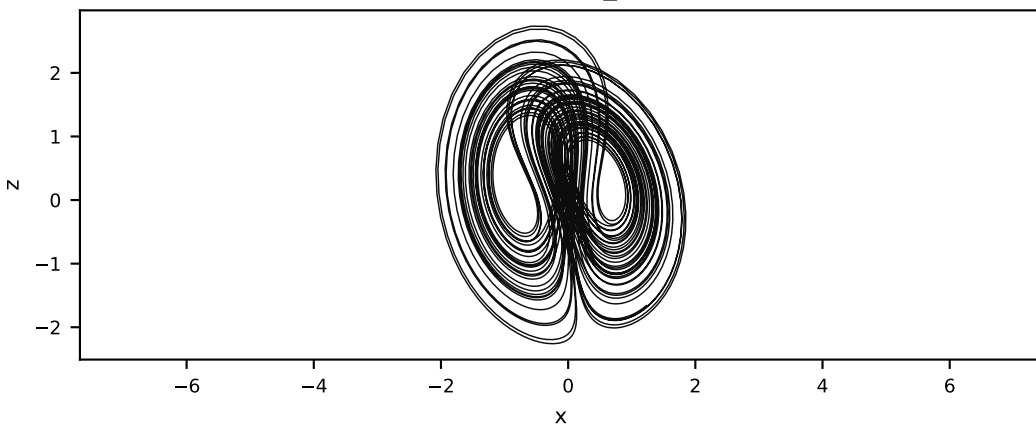
M=15 P=3 vanilla



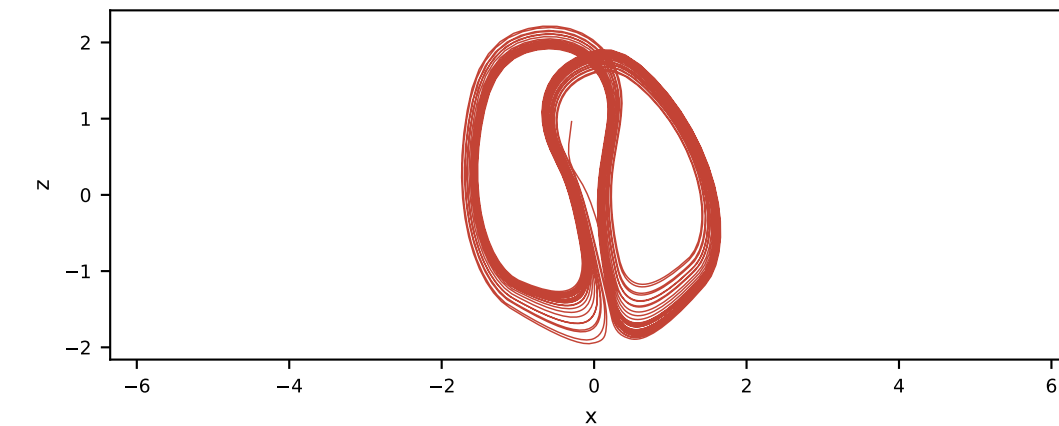
M=15 P=3 ortho



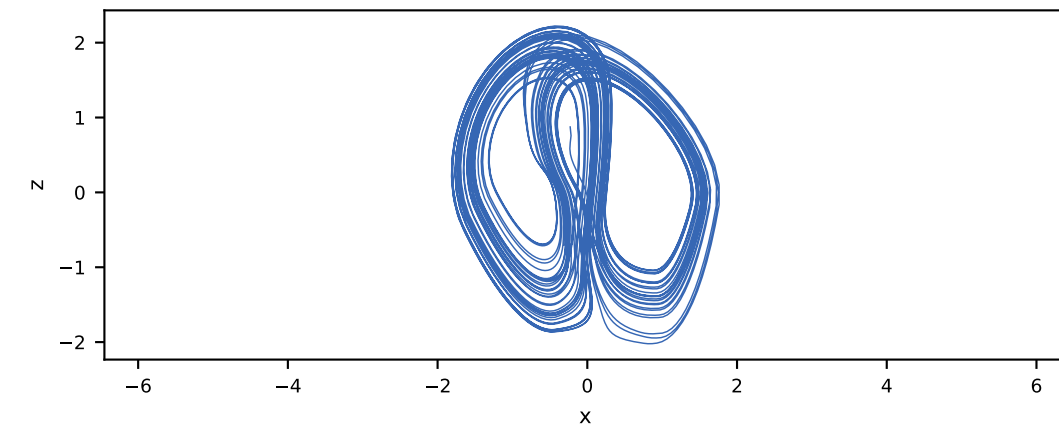
M=14 P=5 truth (D_H gap -16%)



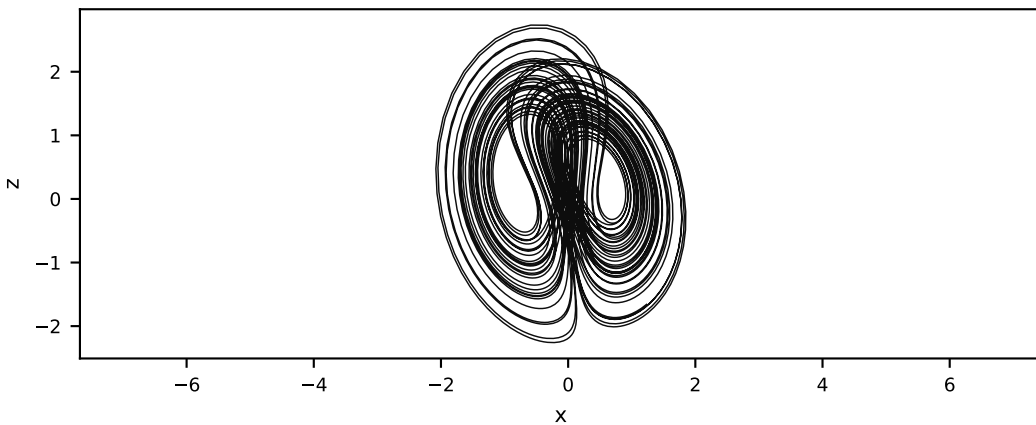
M=14 P=5 vanilla



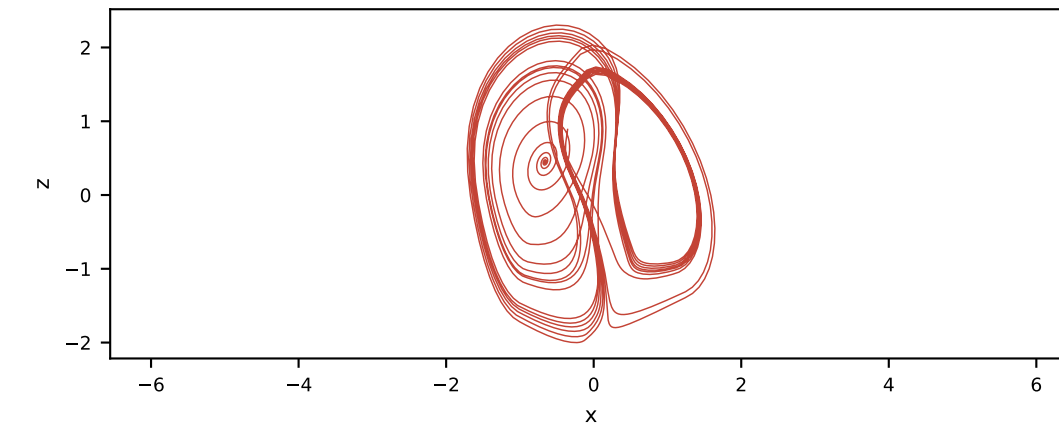
M=14 P=5 ortho



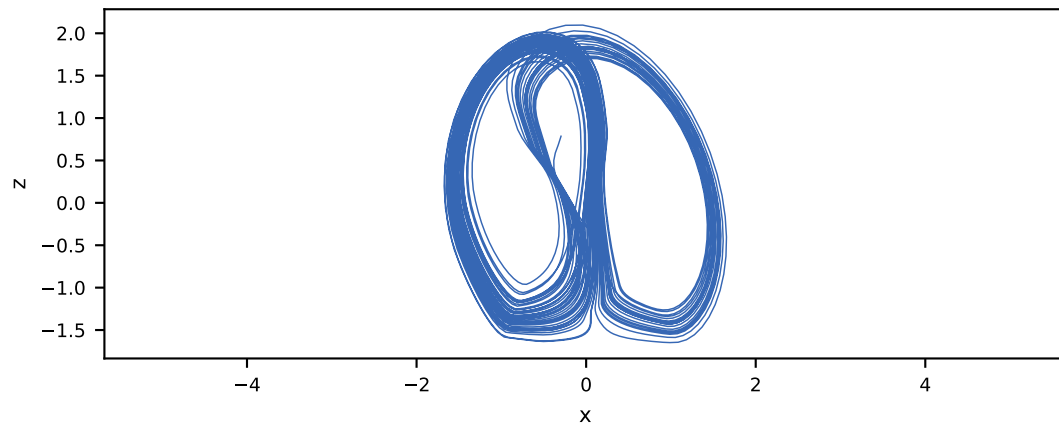
M=11 P=4 truth (D_H gap -15%)



M=11 P=4 vanilla

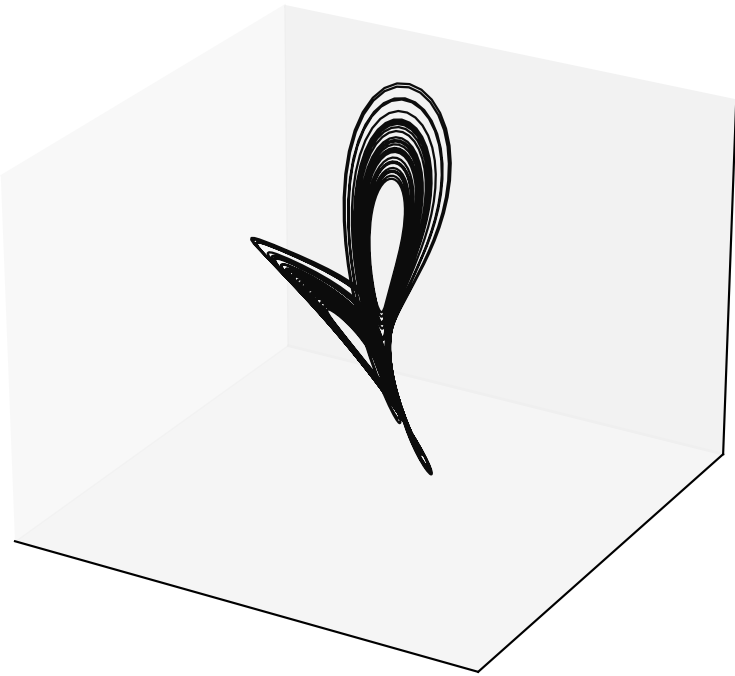


M=11 P=4 ortho

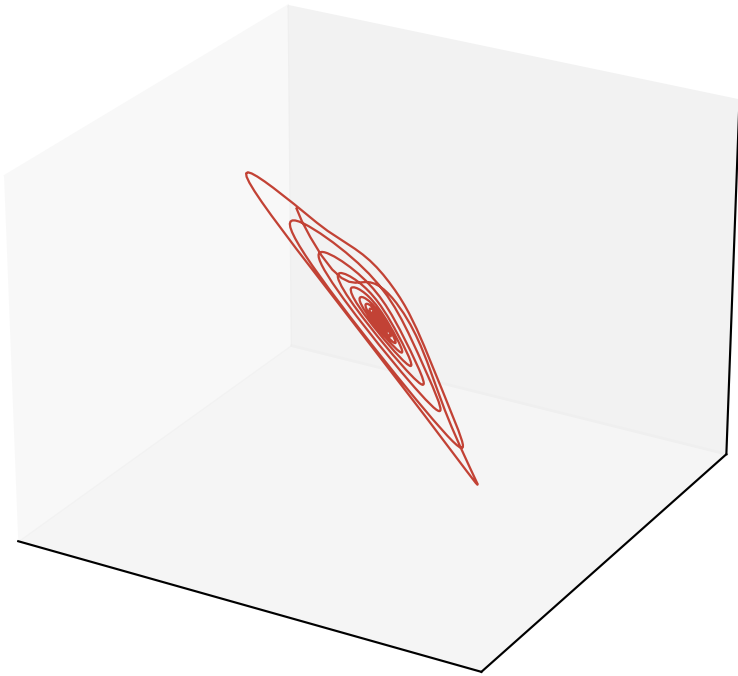


M = 10 P = 2 (median-PSE seed = 1 of [0, 1, 2])
PSE vanilla 0.3442±0.0596 ortho 0.3538±0.0117 mean rel gap +2.8% D_H_state vanilla 0.958±0.007 ortho 0.942±0.031 mean rel gap -1.6%

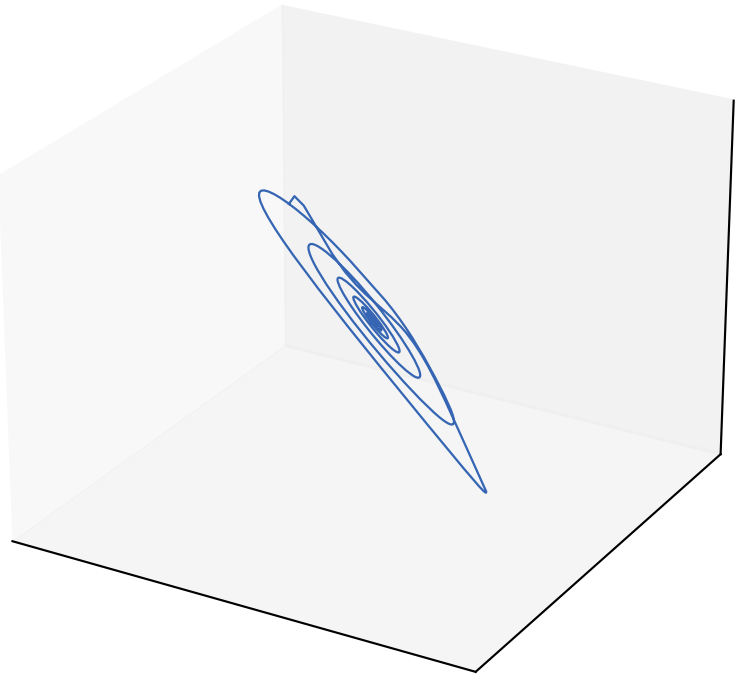
Ground truth (rotated)



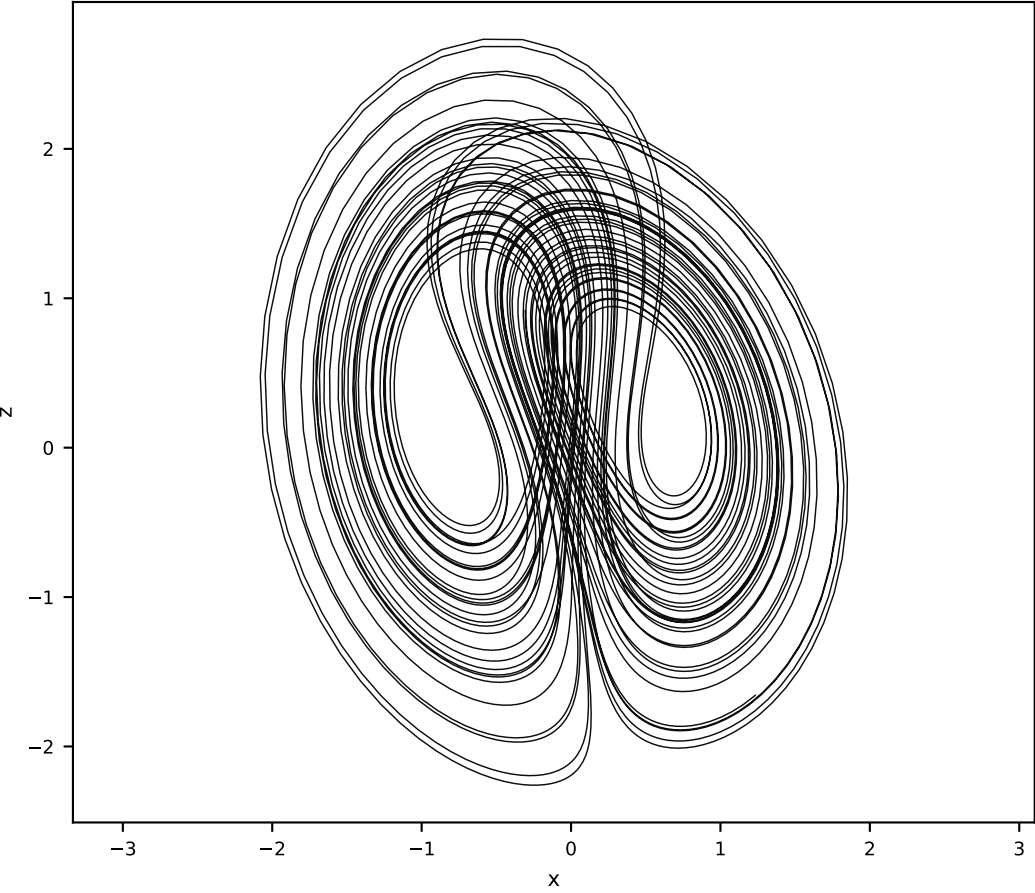
Vanilla AL-RNN (seed 1)



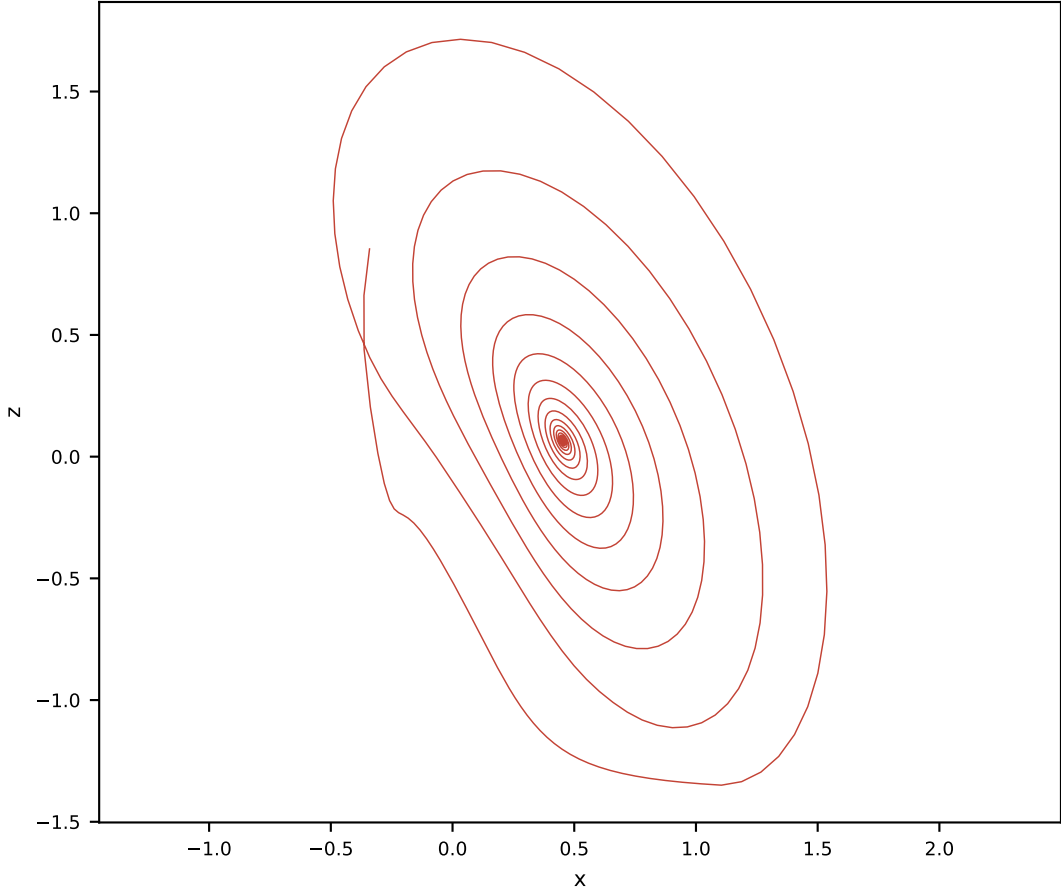
Orthogonal AL-RNN (seed 1)



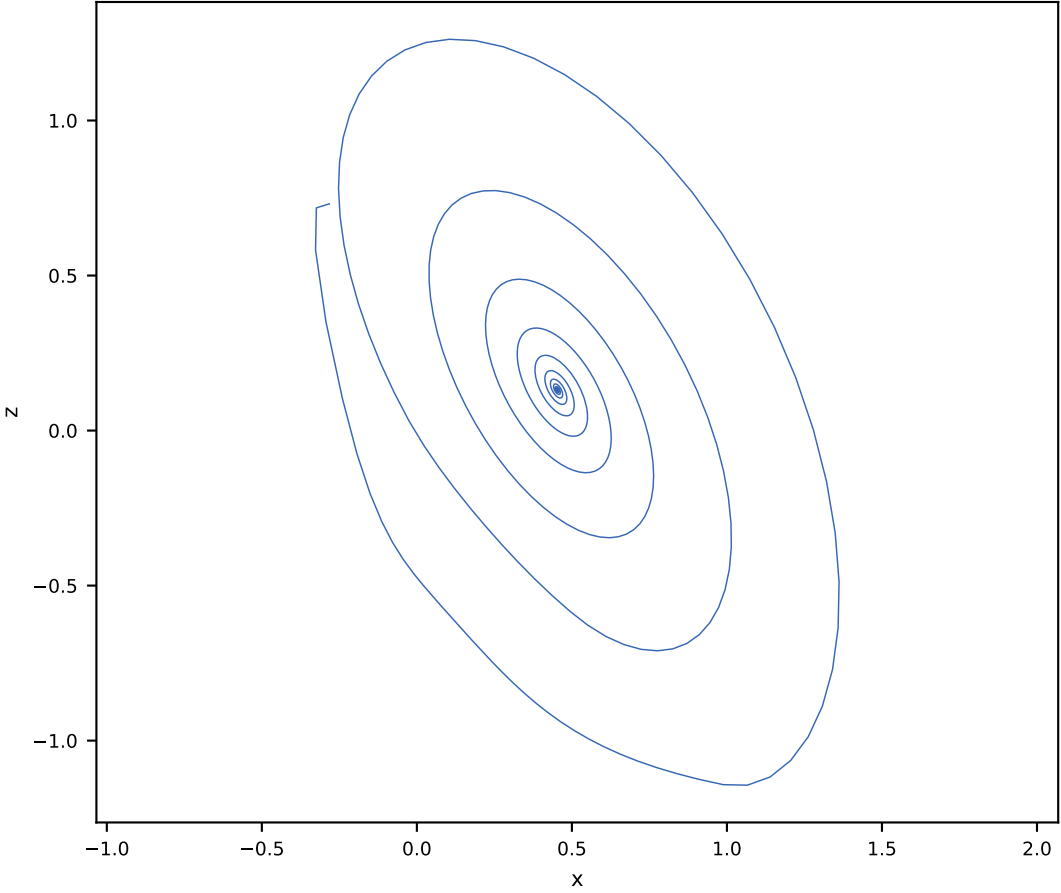
Ground truth x-z



Vanilla x-z (seed 1)

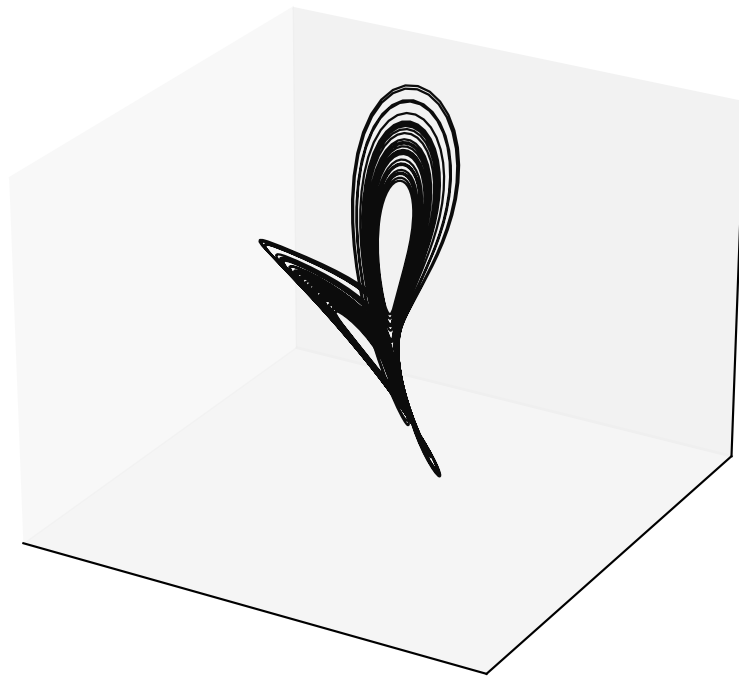


Orthogonal x-z (seed 1)

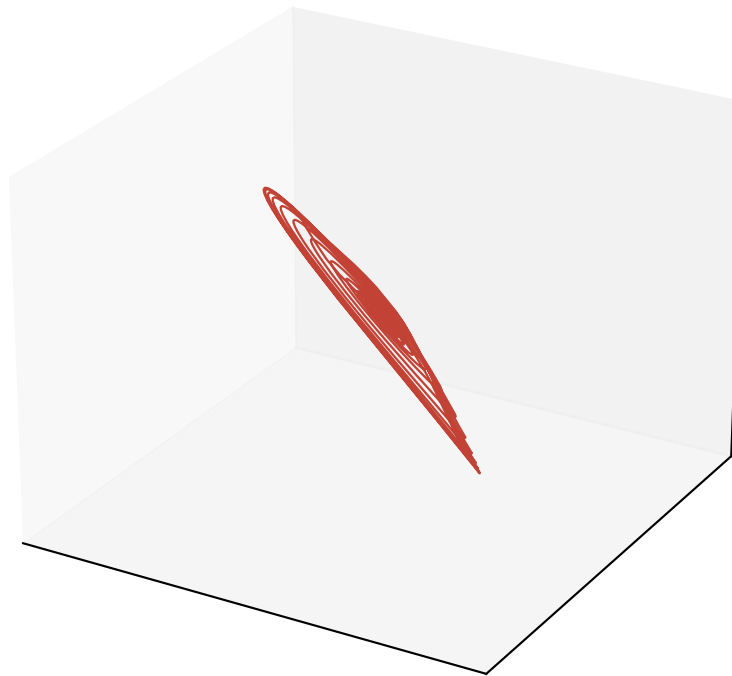


M = 10 P = 3 (median-PSE seed = 2 of [0, 1, 2])
PSE vanilla 0.2245 ± 0.1371 ortho 0.3376 ± 0.1719 mean rel gap +50.4% D_H_state vanilla 0.883 ± 0.089 ortho 0.939 ± 0.062 mean rel gap +6.4%

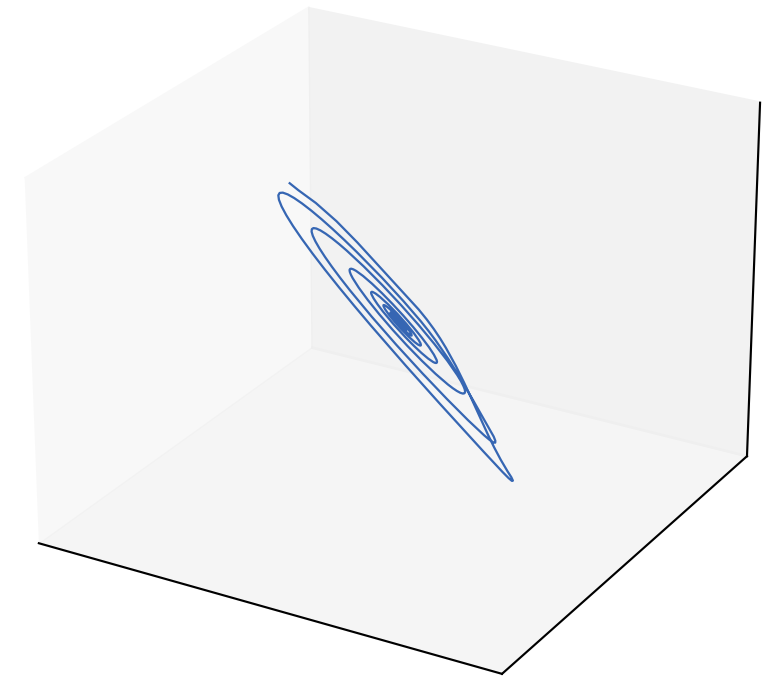
Ground truth (rotated)



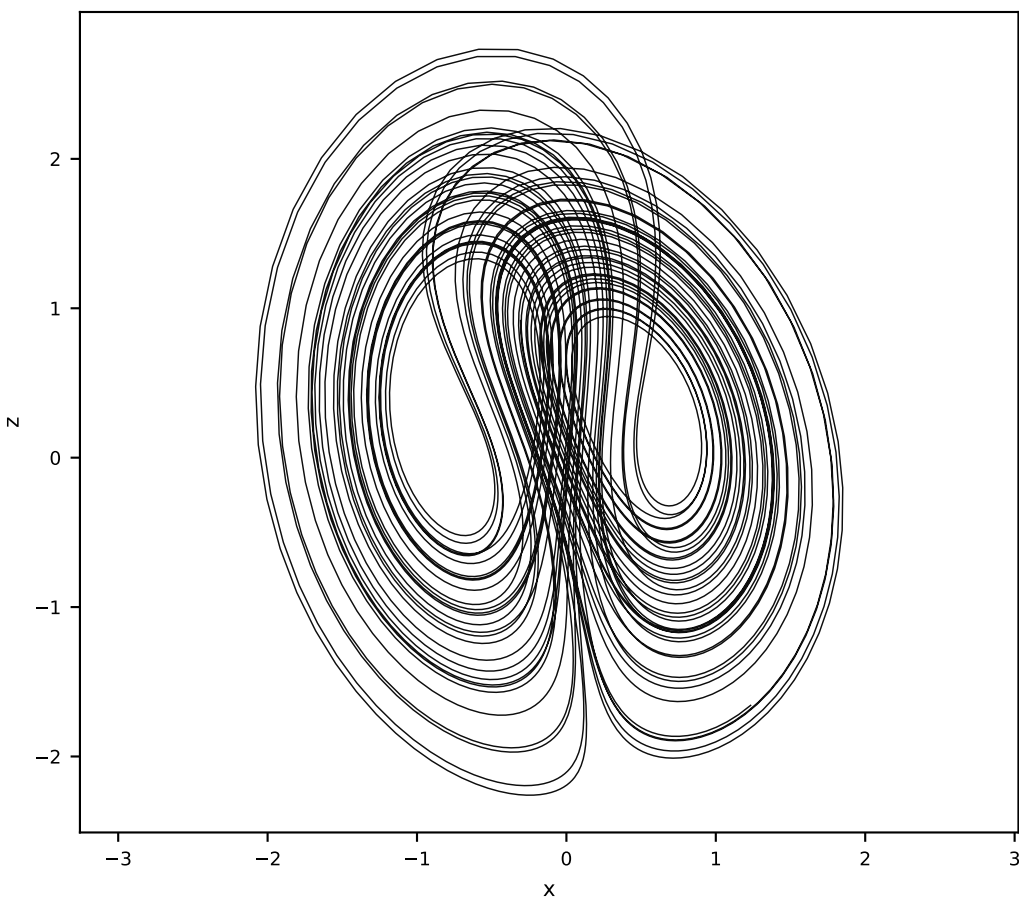
Vanilla AL-RNN (seed 2)



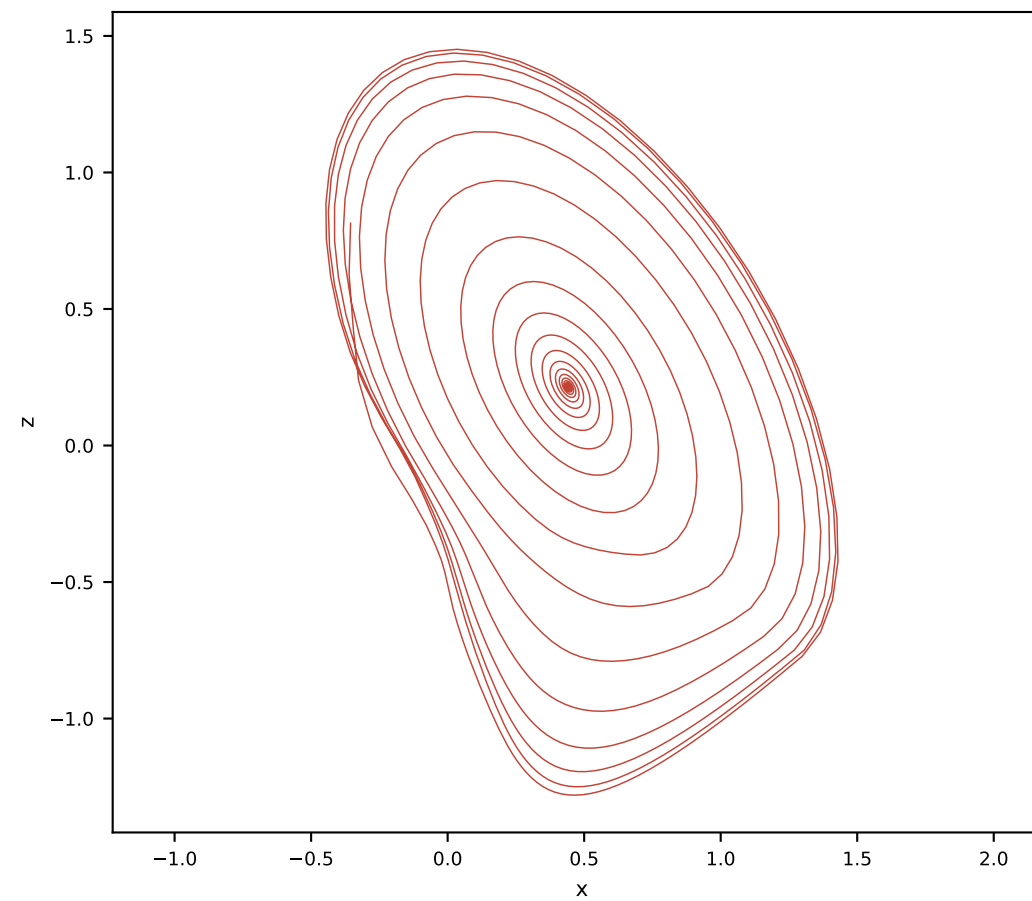
Orthogonal AL-RNN (seed 2)



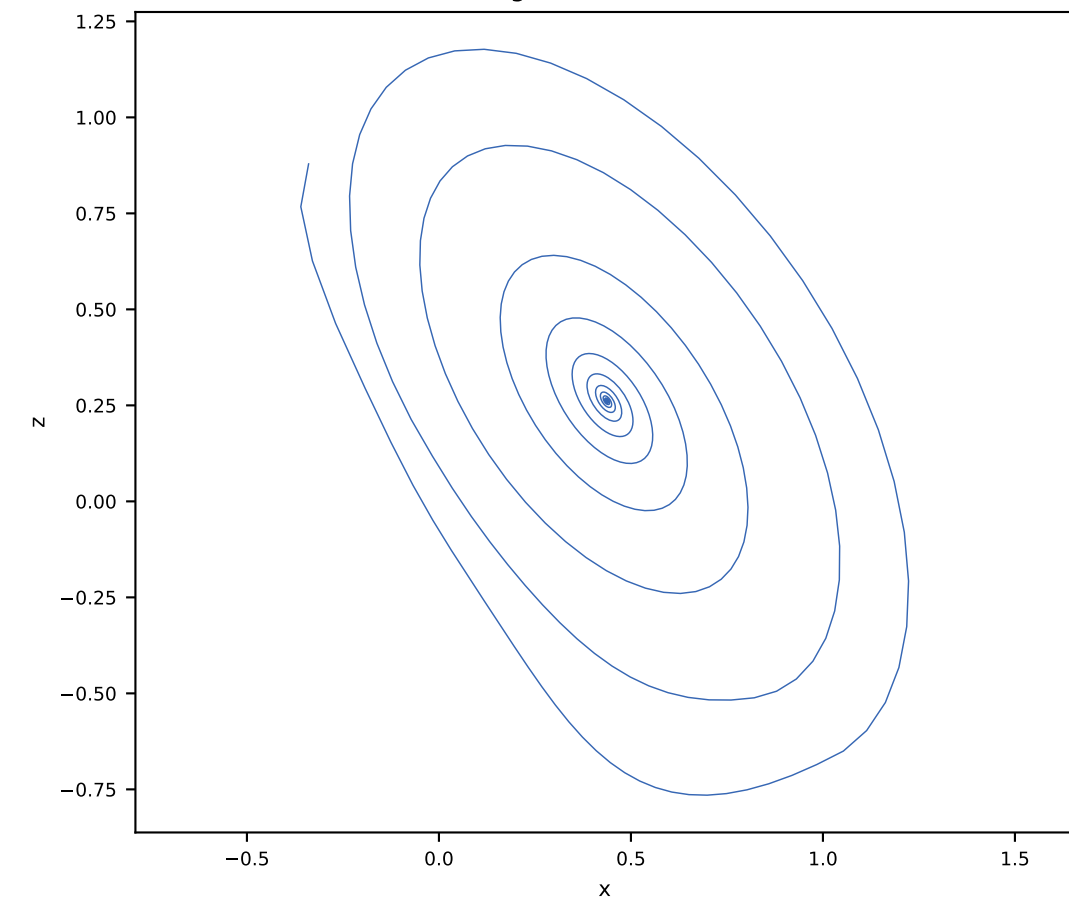
Ground truth x-z



Vanilla x-z (seed 2)

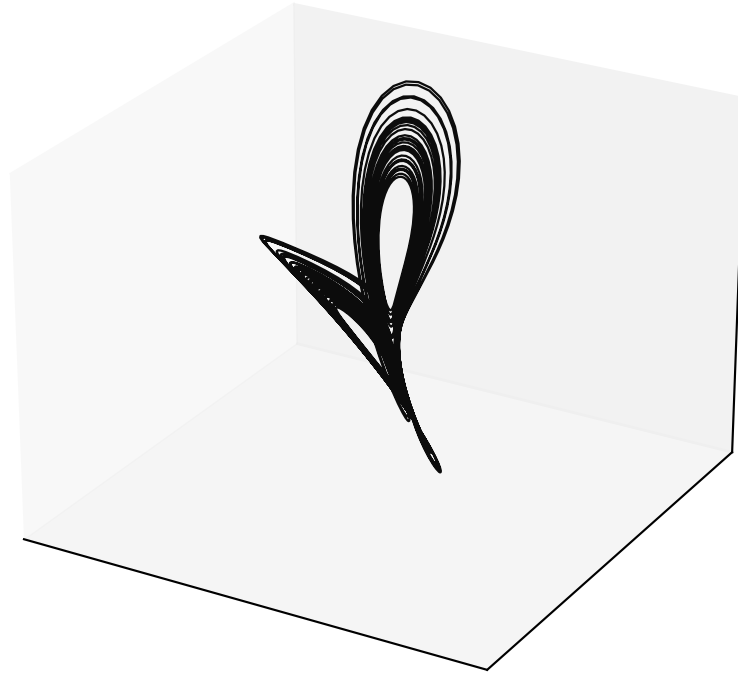


Orthogonal x-z (seed 2)

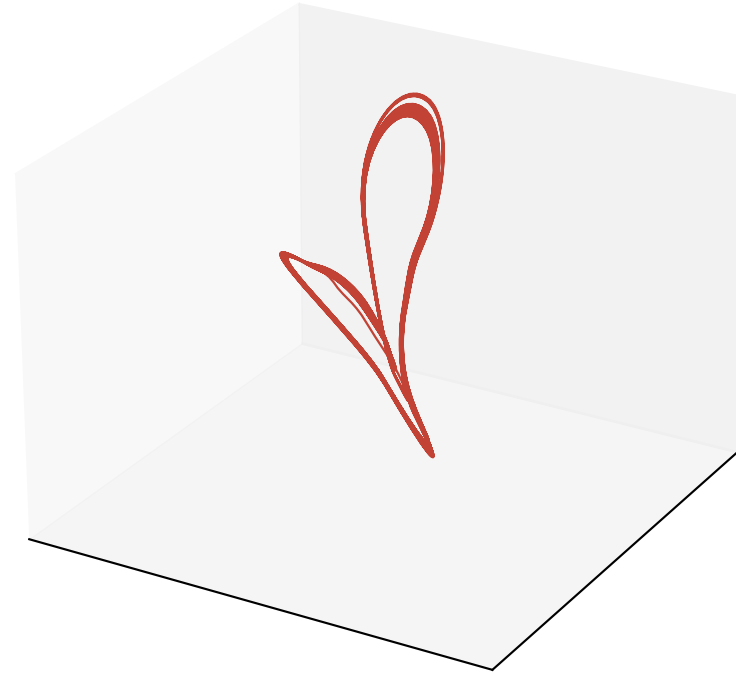


M = 10 P = 4 (median-PSE seed = 1 of [0, 1, 2])
PSE vanilla 0.0822 ± 0.0070 ortho 0.0912 ± 0.0132 mean rel gap +10.9% D_H_state vanilla 0.718 ± 0.054 ortho 0.663 ± 0.060 mean rel gap -7.7%

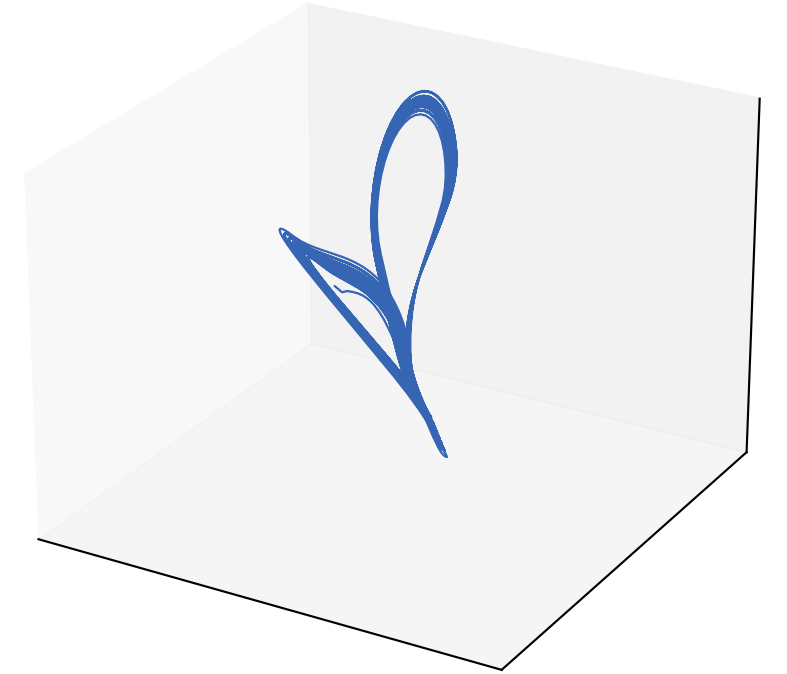
Ground truth (rotated)



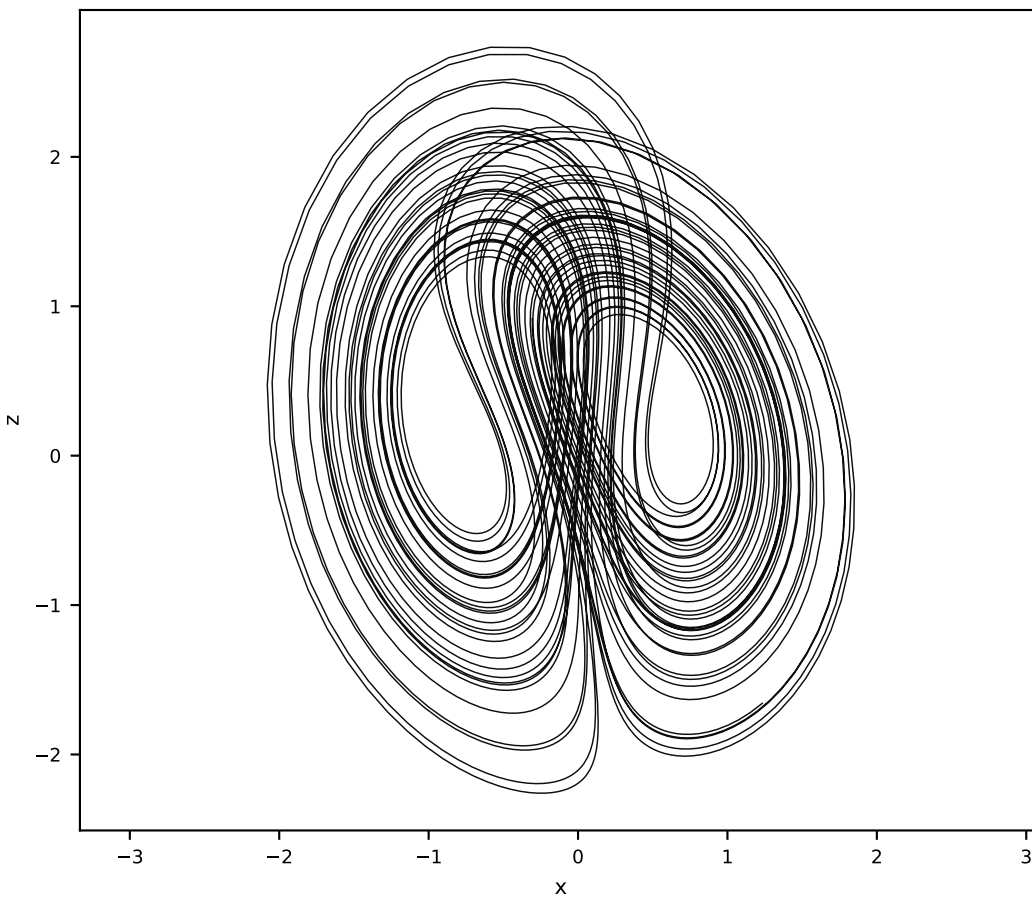
Vanilla AL-RNN (seed 1)



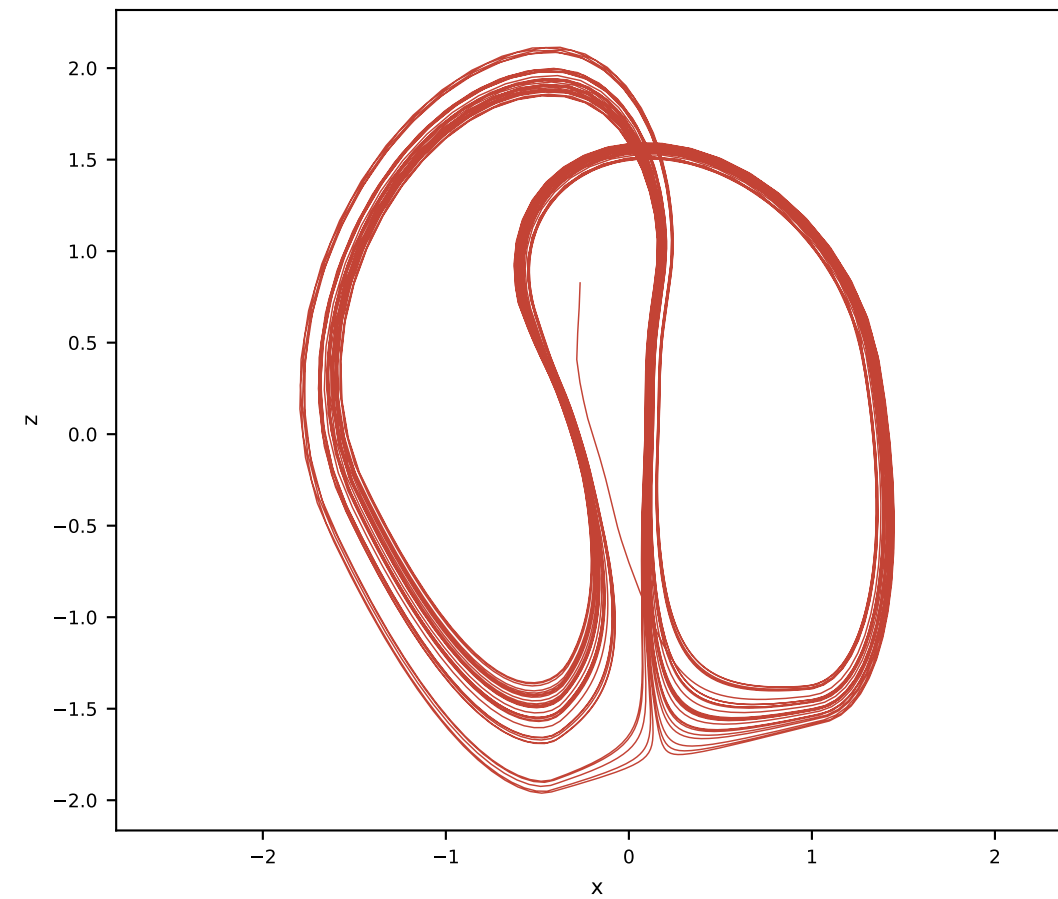
Orthogonal AL-RNN (seed 1)



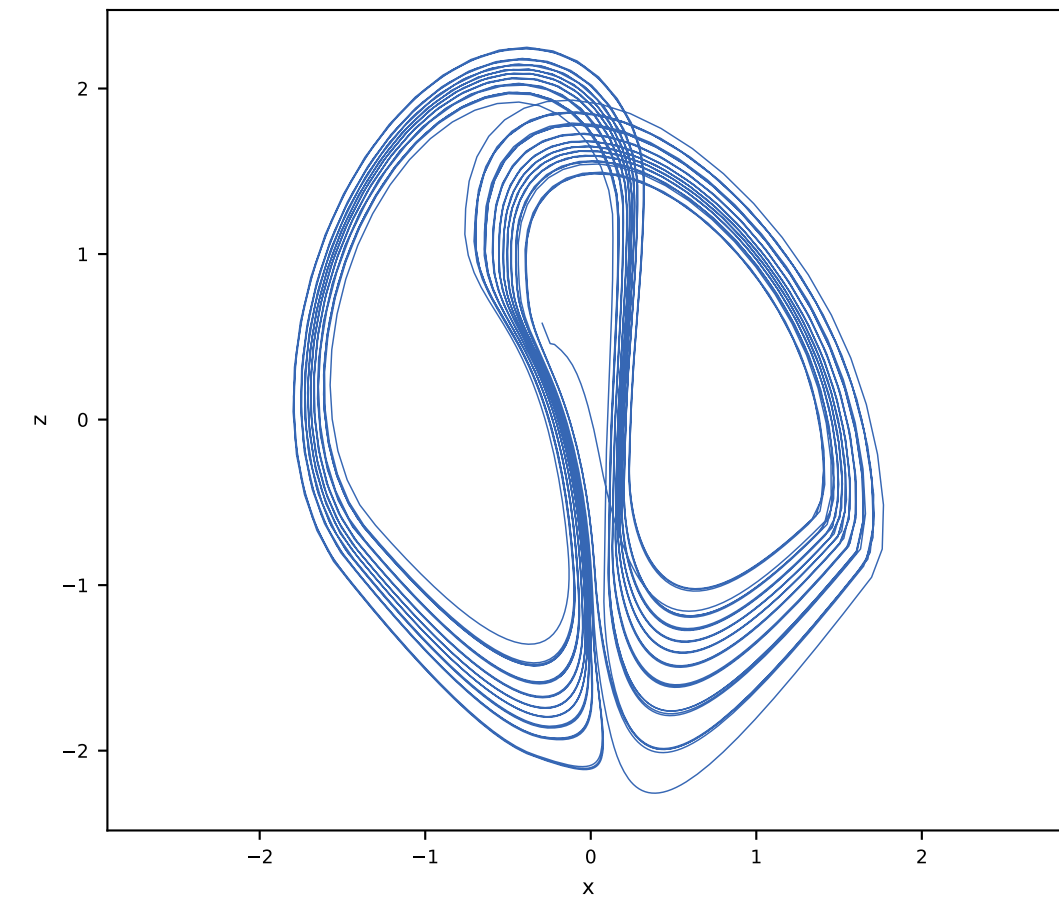
Ground truth x-z



Vanilla x-z (seed 1)

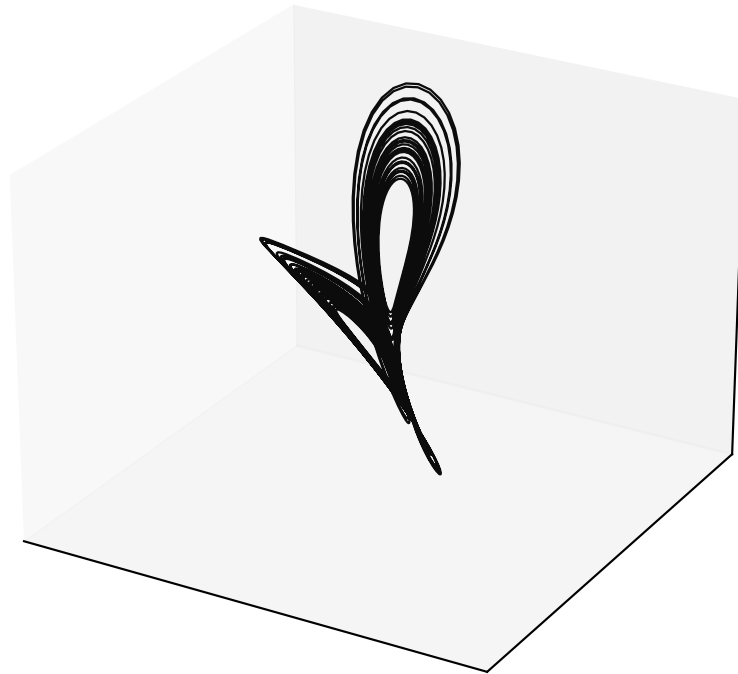


Orthogonal x-z (seed 1)

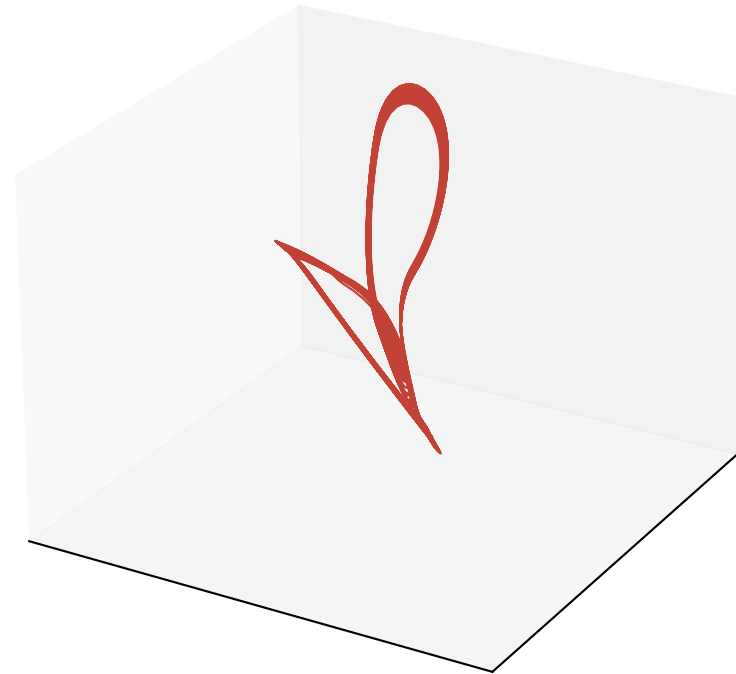


M = 10 P = 5 (median-PSE seed = 0 of [0, 1, 2])
PSE vanilla 0.0844 ± 0.0103 ortho 0.0694 ± 0.0139 mean rel gap -17.8% D_H_state vanilla 0.666 ± 0.033 ortho 0.581 ± 0.048 mean rel gap -12.7%

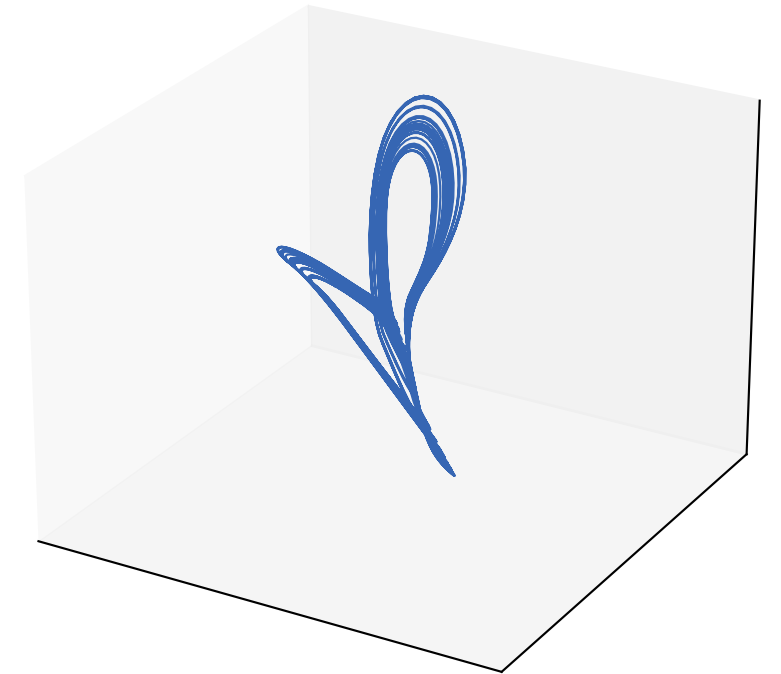
Ground truth (rotated)



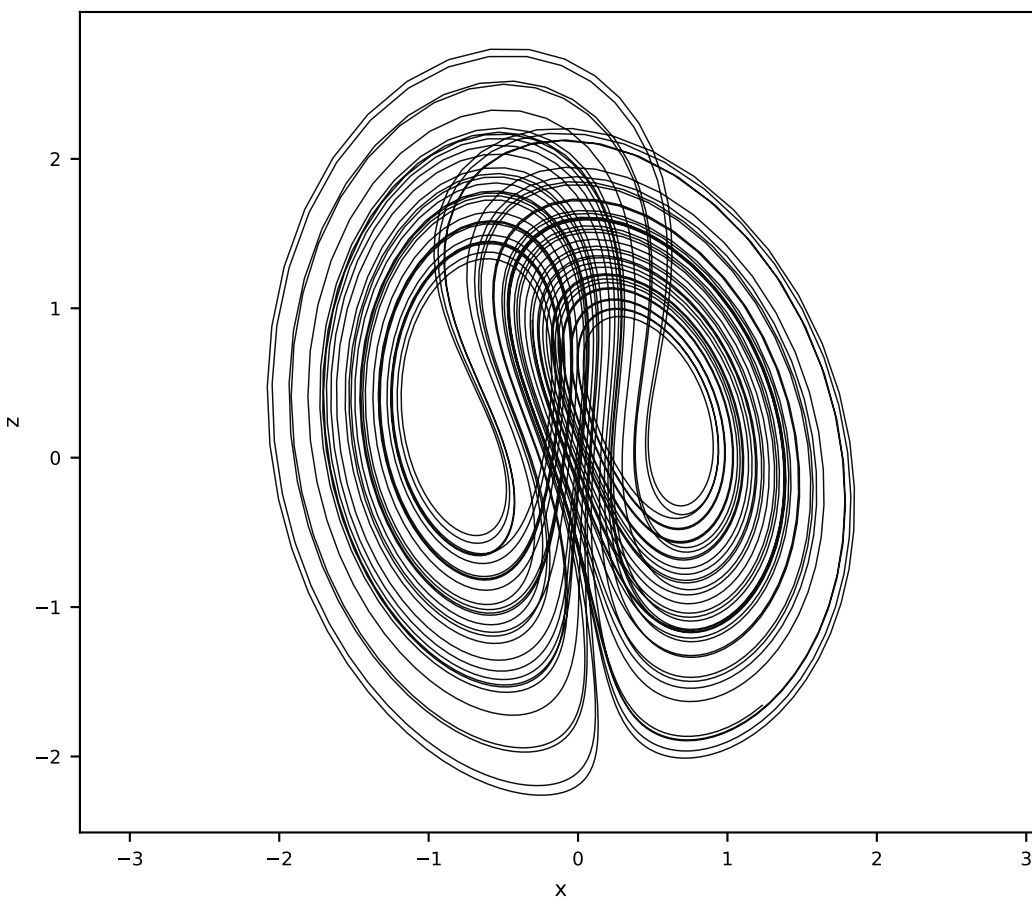
Vanilla AL-RNN (seed 0)



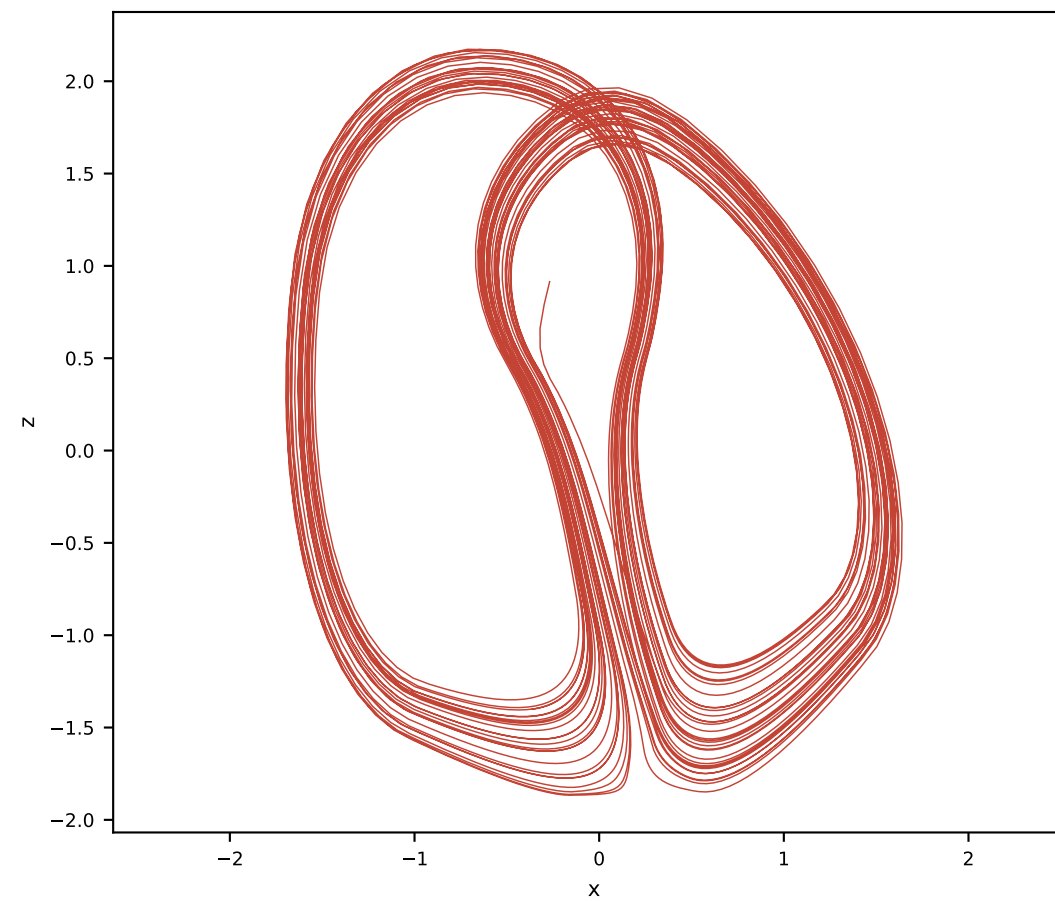
Orthogonal AL-RNN (seed 0)



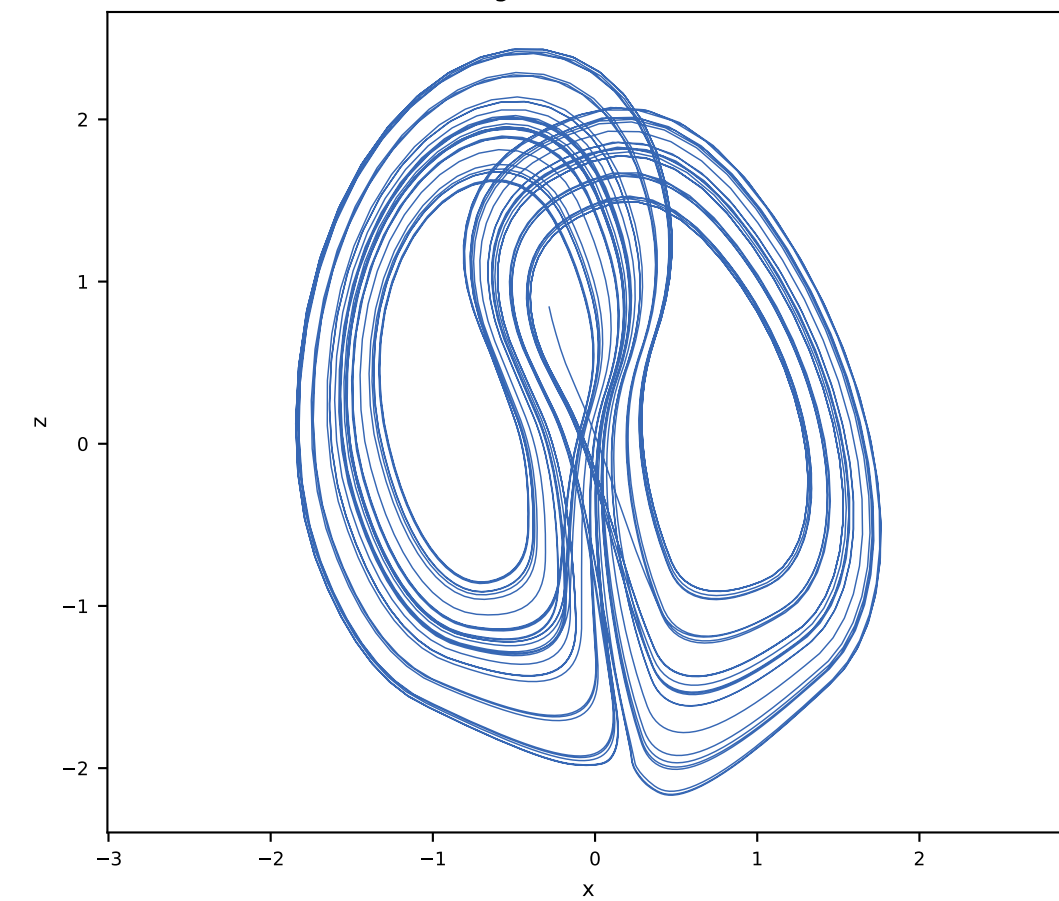
Ground truth x-z



Vanilla x-z (seed 0)

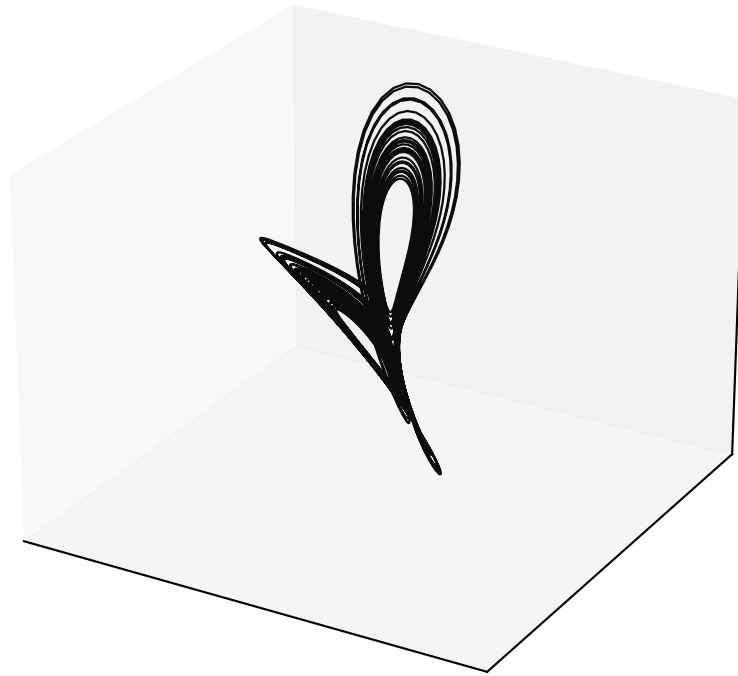


Orthogonal x-z (seed 0)

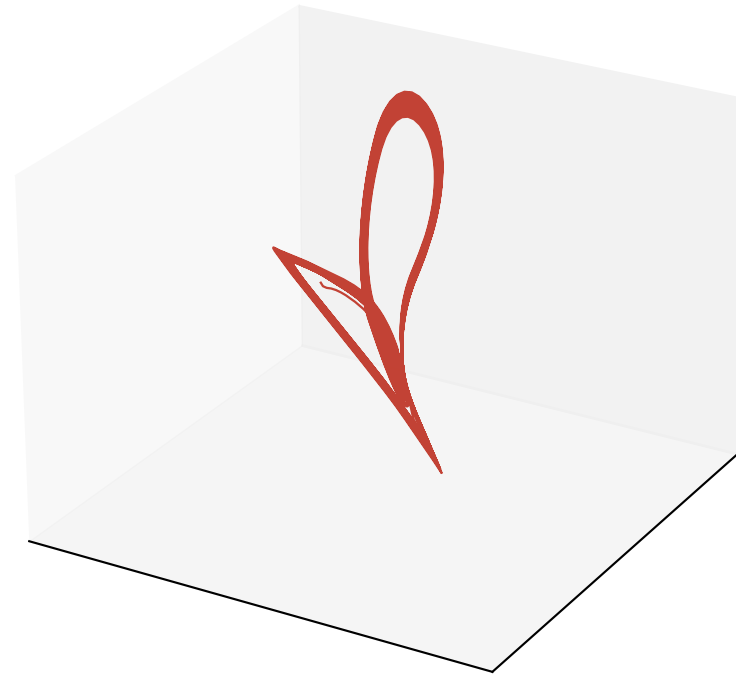


M = 10 P = 6 (median-PSE seed = 0 of [0, 1, 2])
PSE vanilla 0.0844 ± 0.0094 ortho 0.0778 ± 0.0041 mean rel gap -7.9% D_H_state vanilla 0.649 ± 0.045 ortho 0.635 ± 0.095 mean rel gap -2.2%

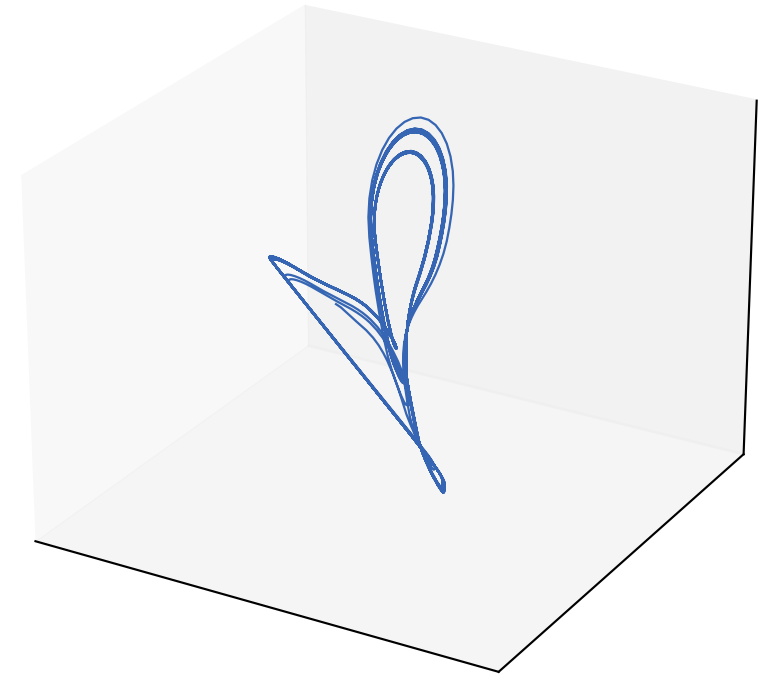
Ground truth (rotated)



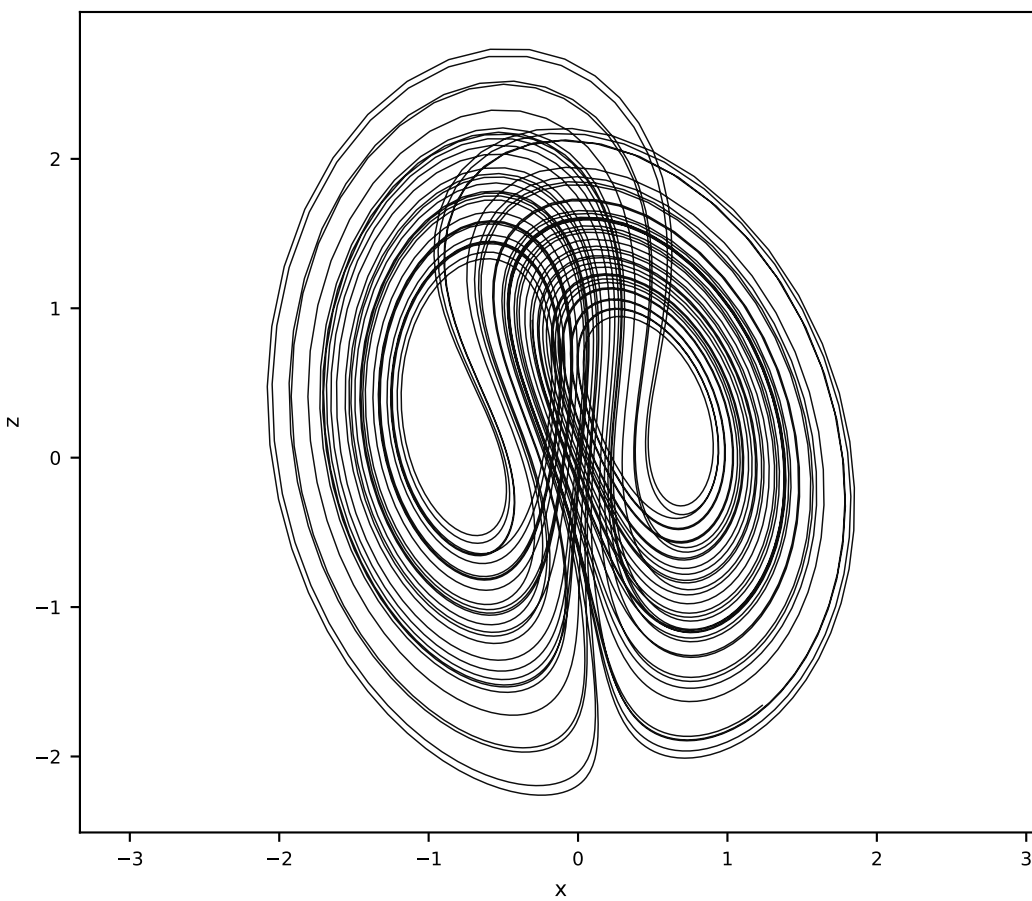
Vanilla AL-RNN (seed 0)



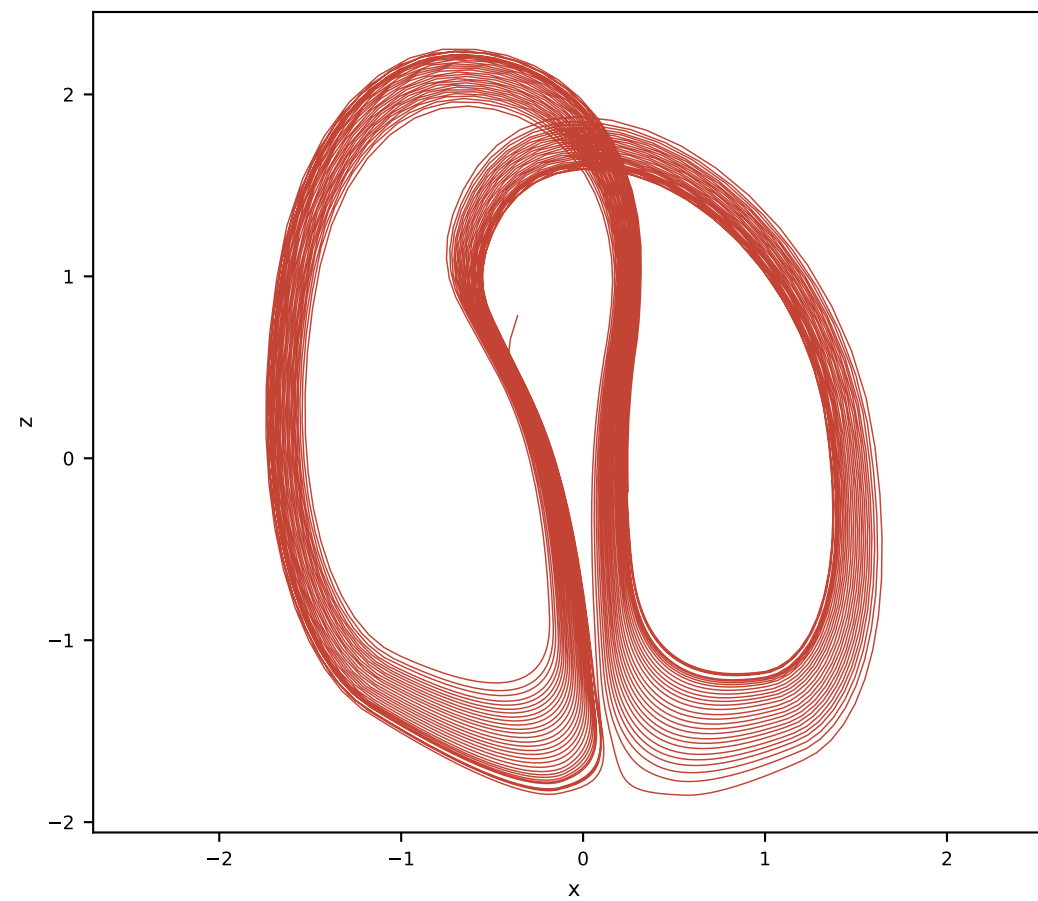
Orthogonal AL-RNN (seed 0)



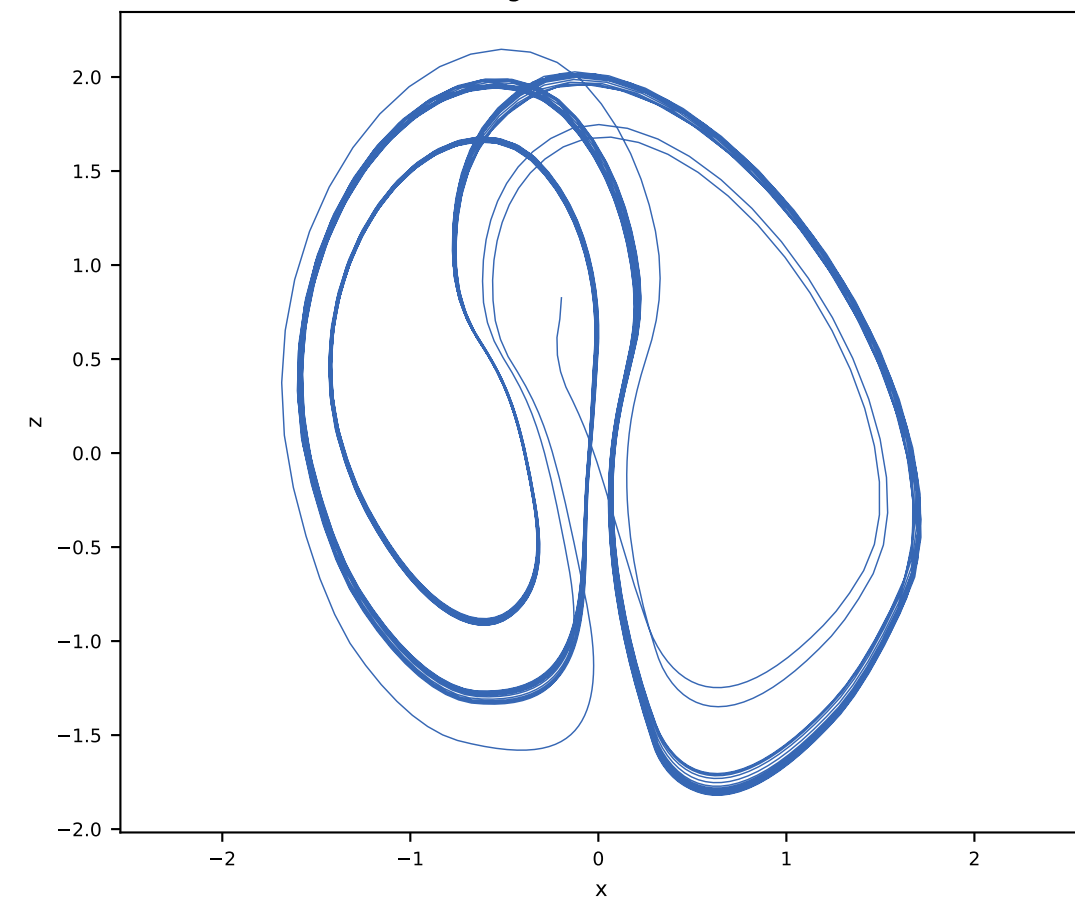
Ground truth x-z



Vanilla x-z (seed 0)

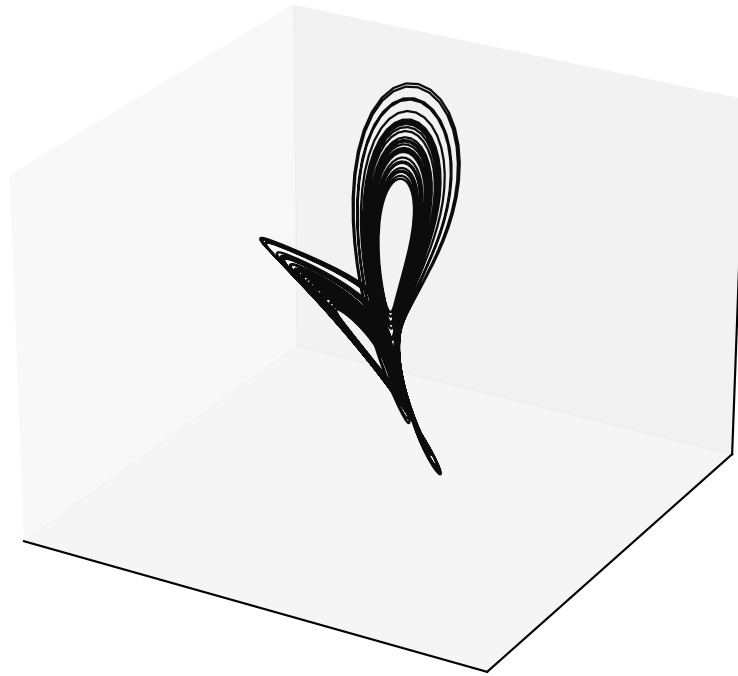


Orthogonal x-z (seed 0)

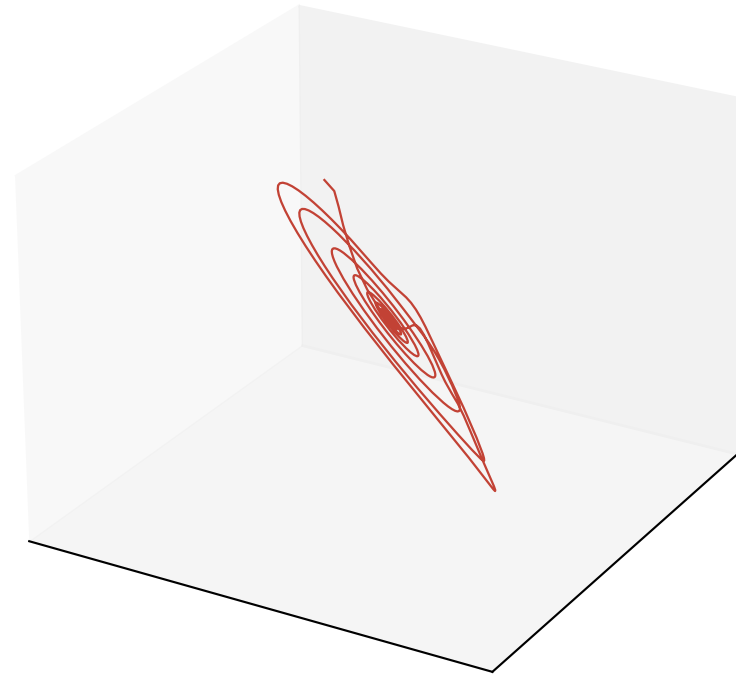


M = 11 P = 2 (median-PSE seed = 2 of [0, 1, 2])
PSE vanilla 0.3403 ± 0.0343 ortho 0.2310 ± 0.1494 mean rel gap -32.1% D_H_state vanilla 0.971 ± 0.010 ortho 0.872 ± 0.110 mean rel gap -10.1%

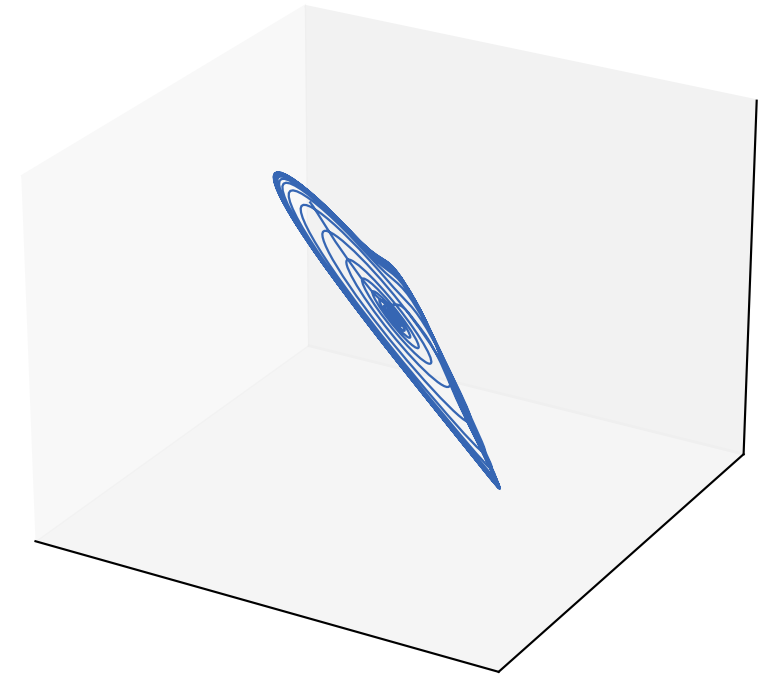
Ground truth (rotated)



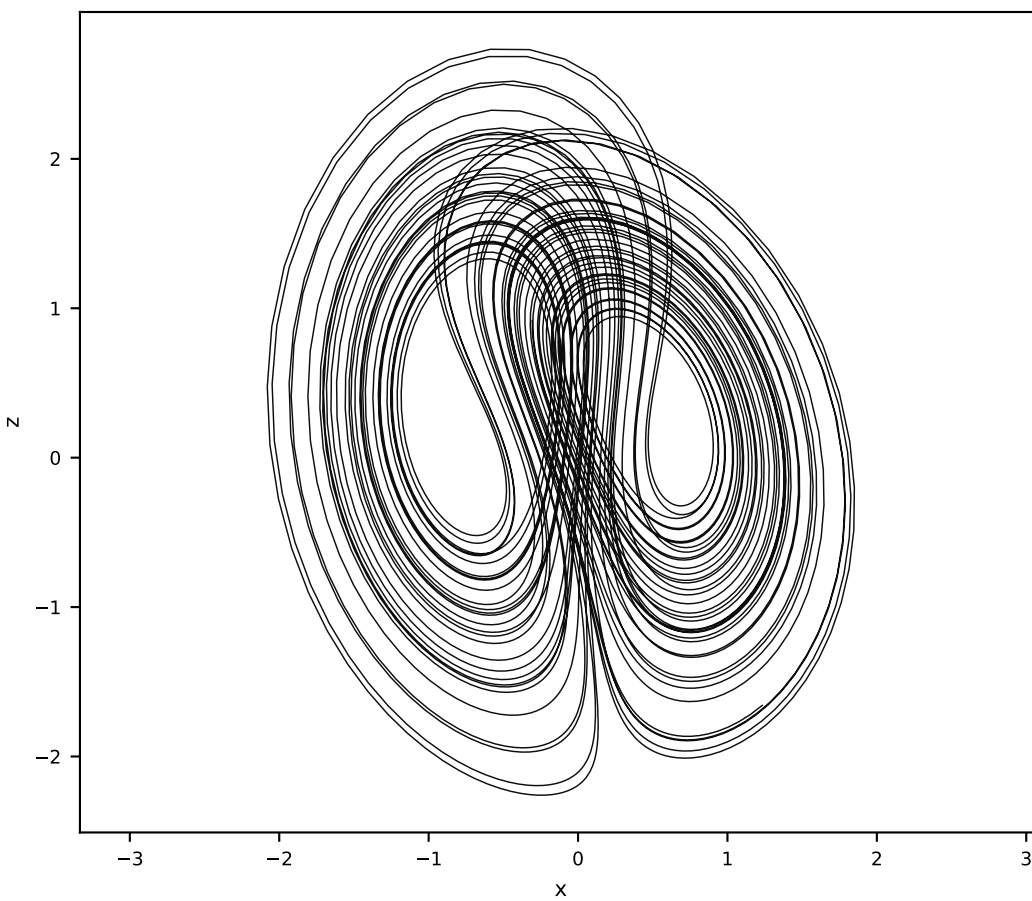
Vanilla AL-RNN (seed 2)



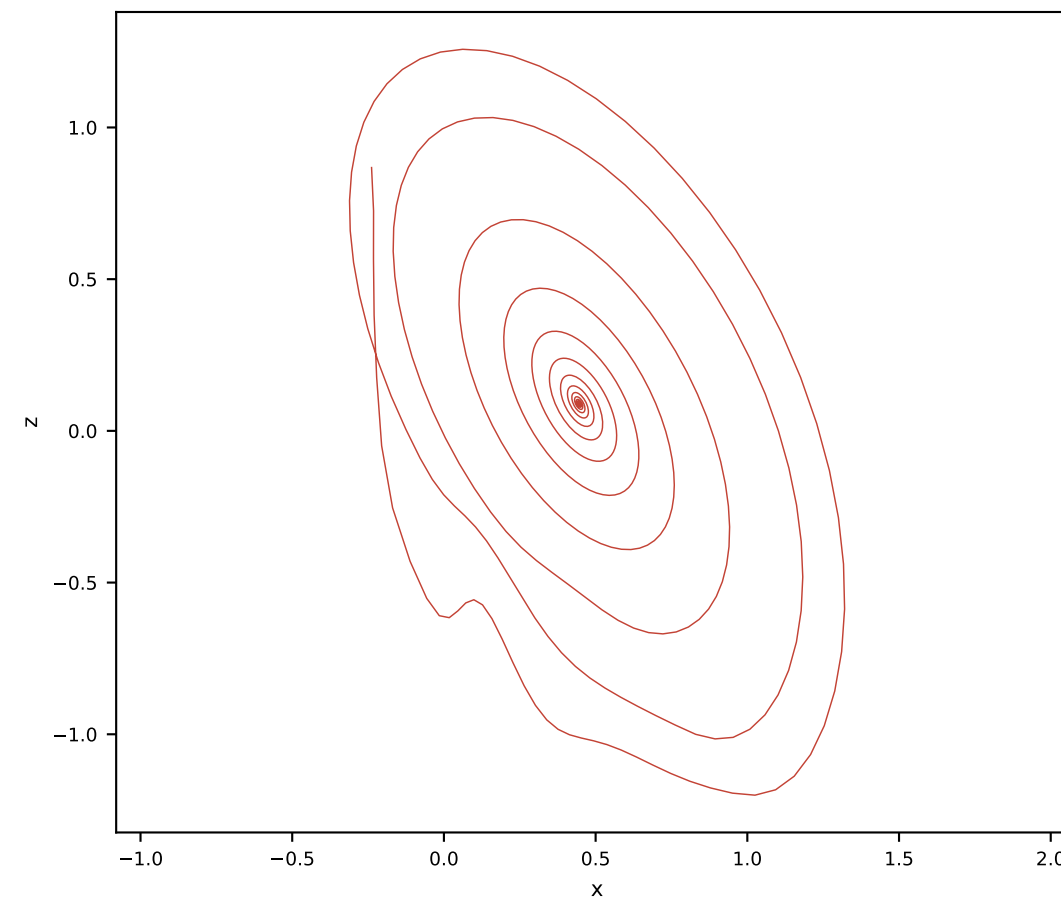
Orthogonal AL-RNN (seed 2)



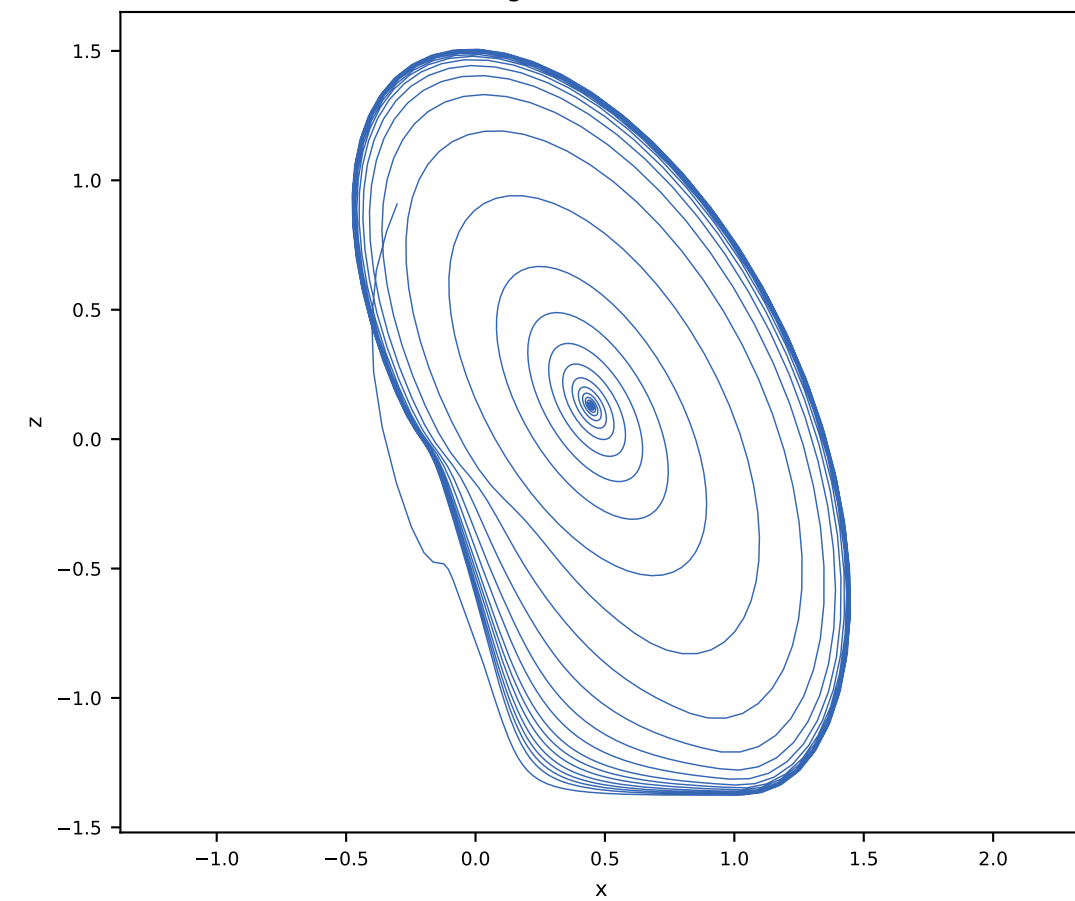
Ground truth x-z



Vanilla x-z (seed 2)

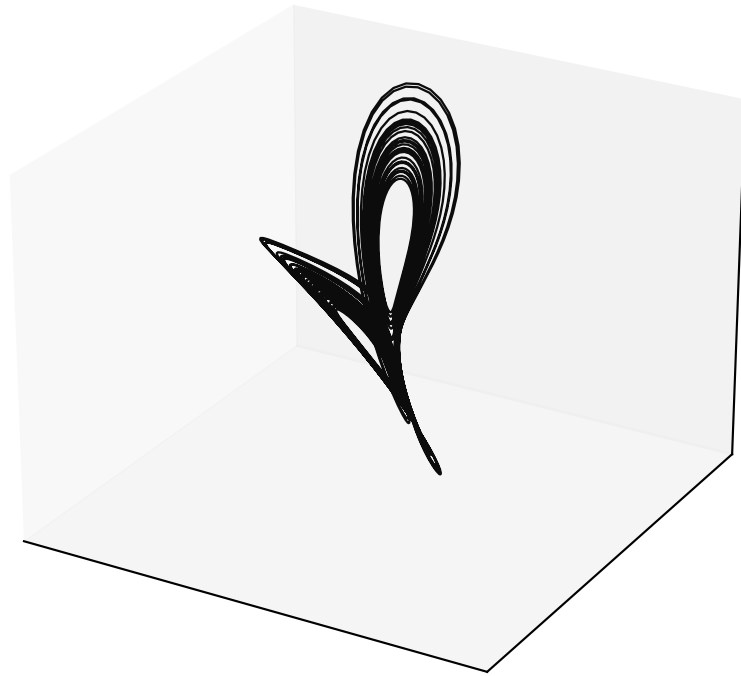


Orthogonal x-z (seed 2)

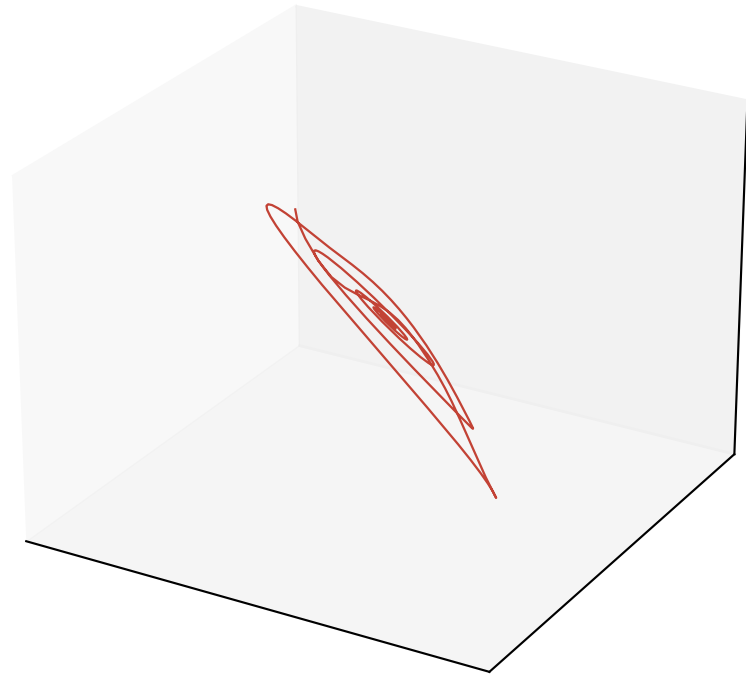


M = 11 P = 3 (median-PSE seed = 0 of [0, 1, 2])
PSE vanilla 0.2705 ± 0.1522 ortho 0.1868 ± 0.1683 mean rel gap -31.0% D_H_state vanilla 0.919 ± 0.101 ortho 0.843 ± 0.180 mean rel gap -8.3%

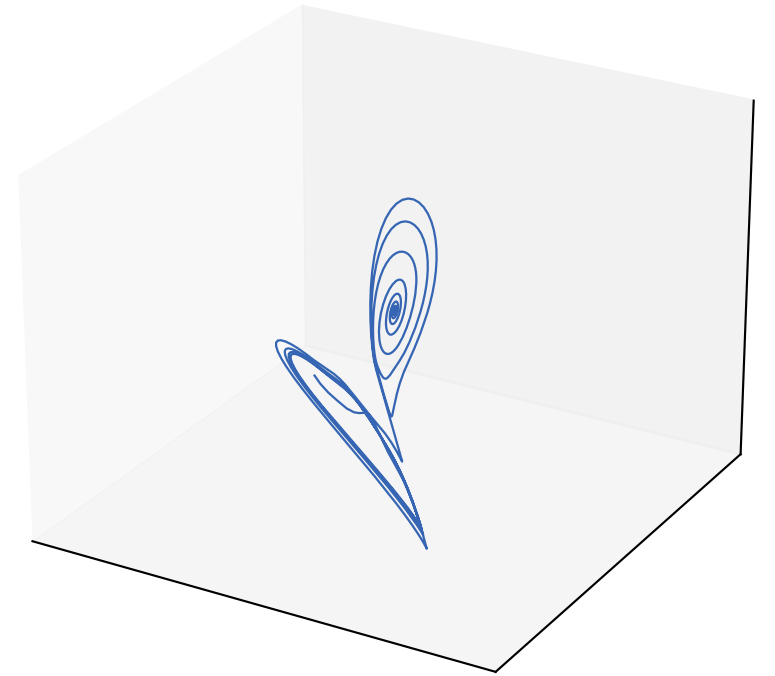
Ground truth (rotated)



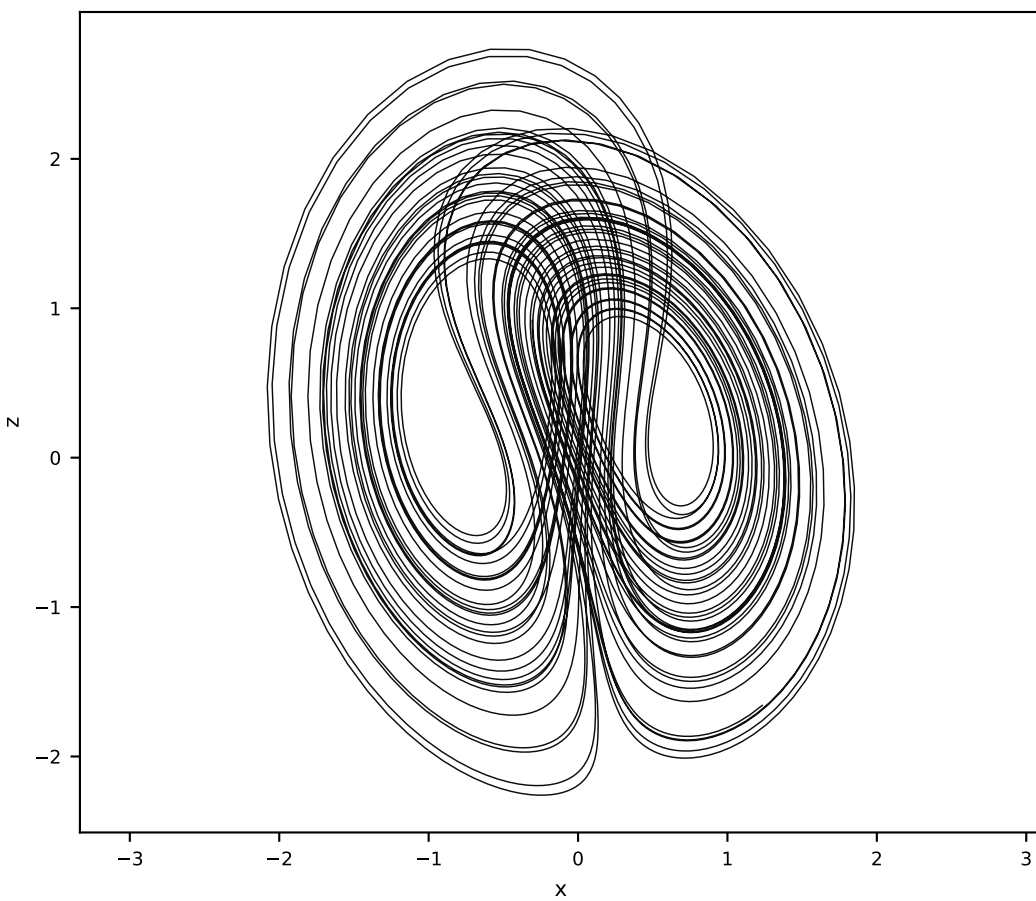
Vanilla AL-RNN (seed 0)



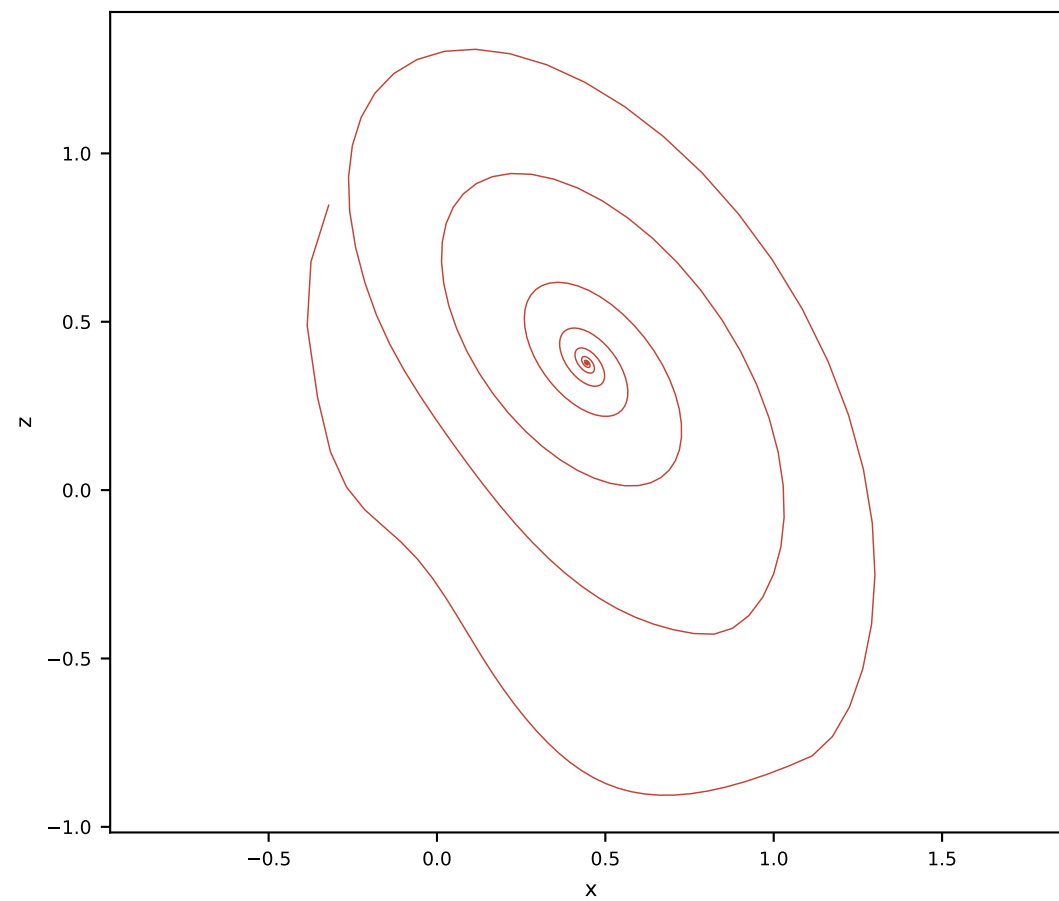
Orthogonal AL-RNN (seed 0)



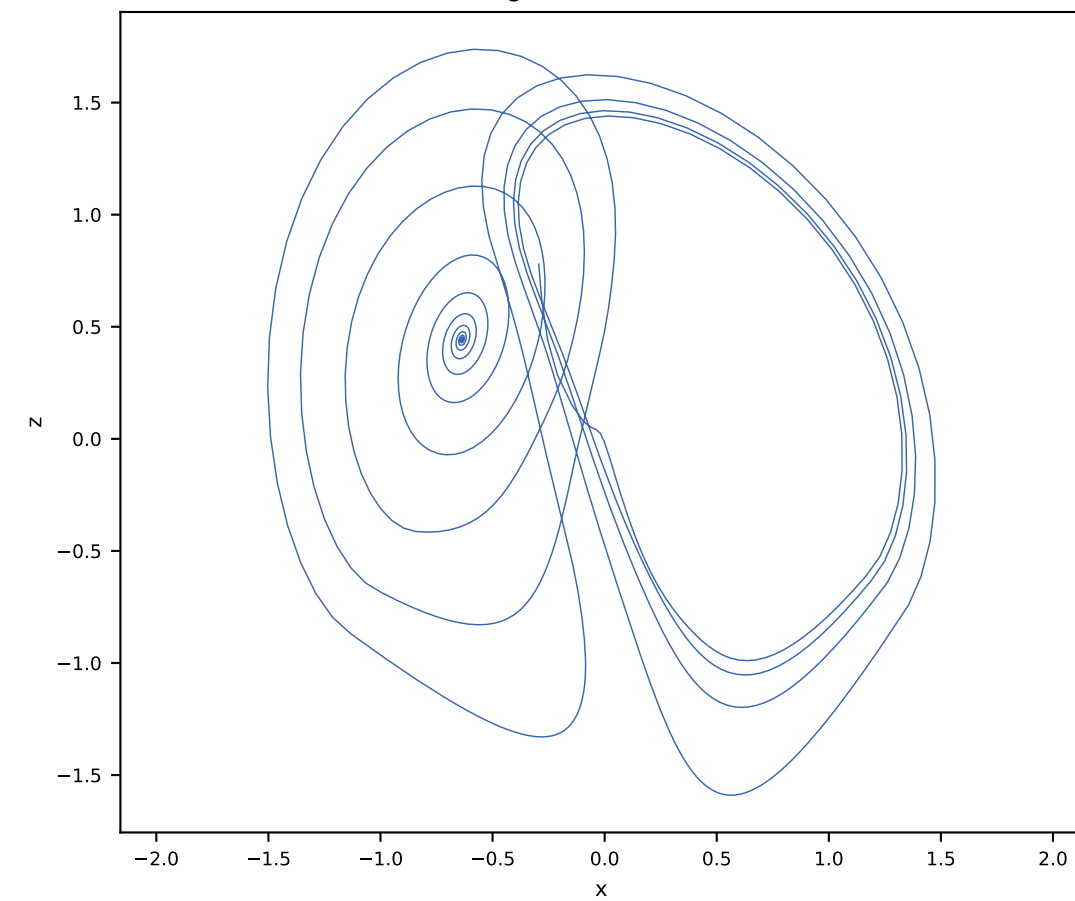
Ground truth x-z



Vanilla x-z (seed 0)

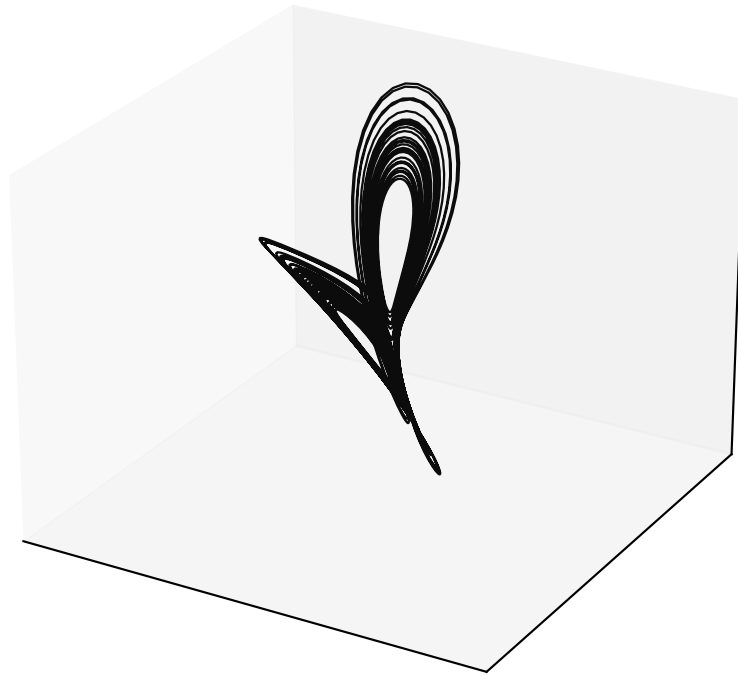


Orthogonal x-z (seed 0)

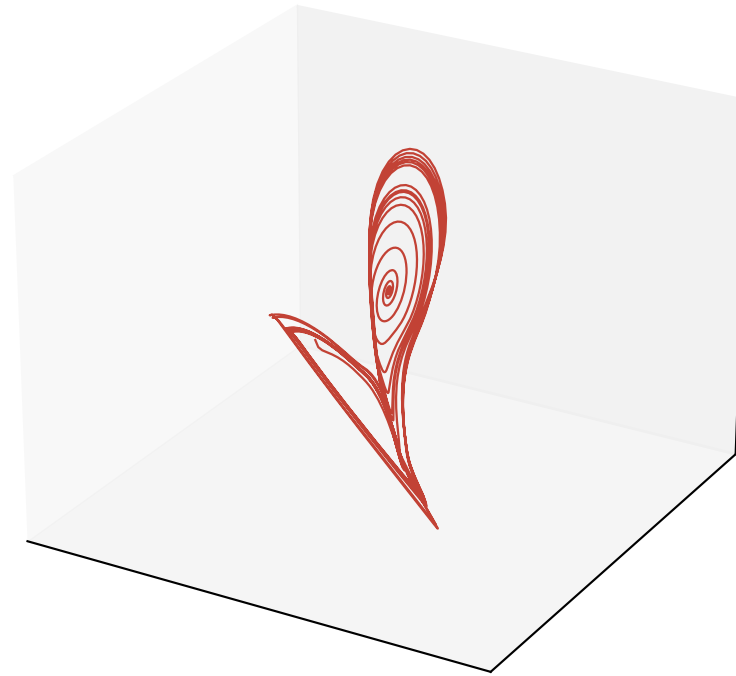


M = 11 P = 4 (median-PSE seed = 1 of [0, 1, 2])
PSE vanilla 0.0938 ± 0.0345 ortho 0.0714 ± 0.0151 mean rel gap -23.9% D_H_state vanilla 0.733 ± 0.155 ortho 0.626 ± 0.042 mean rel gap -14.6%

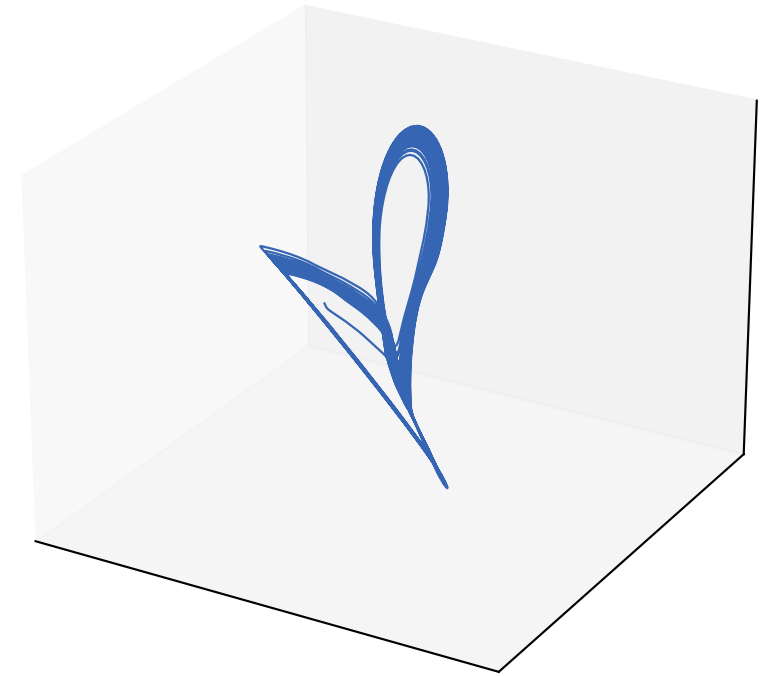
Ground truth (rotated)



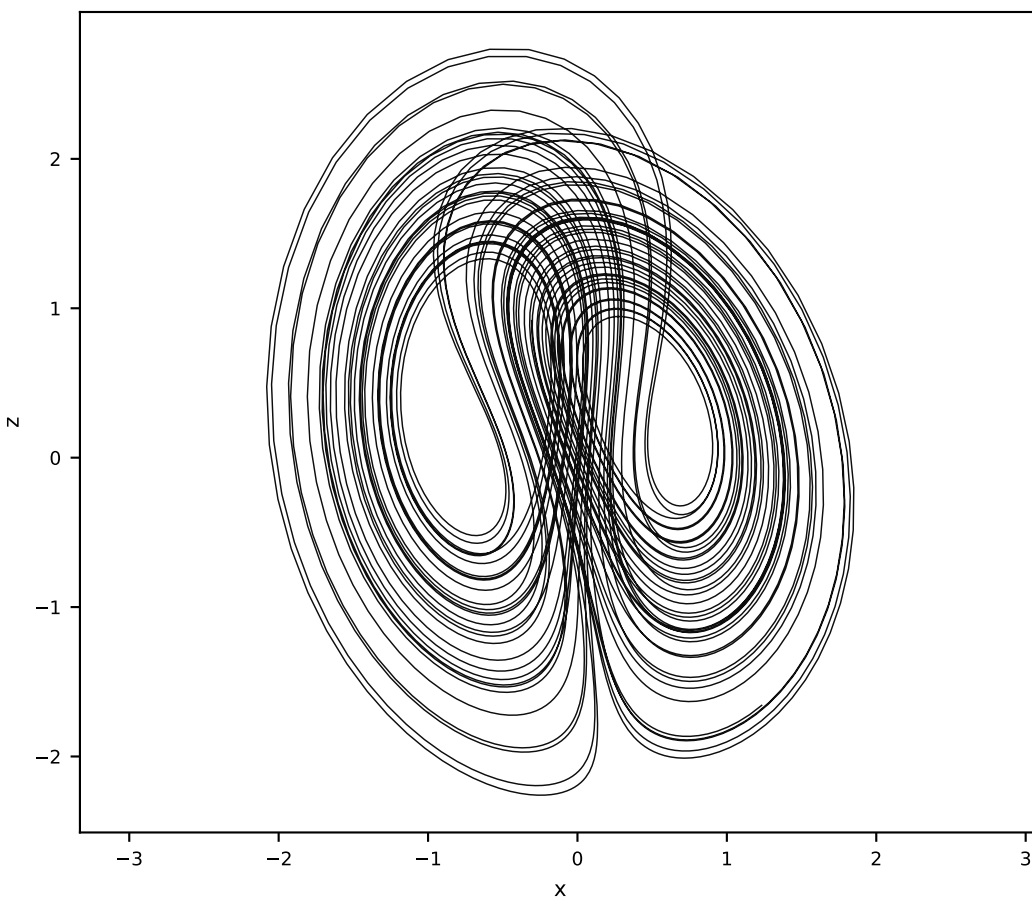
Vanilla AL-RNN (seed 1)



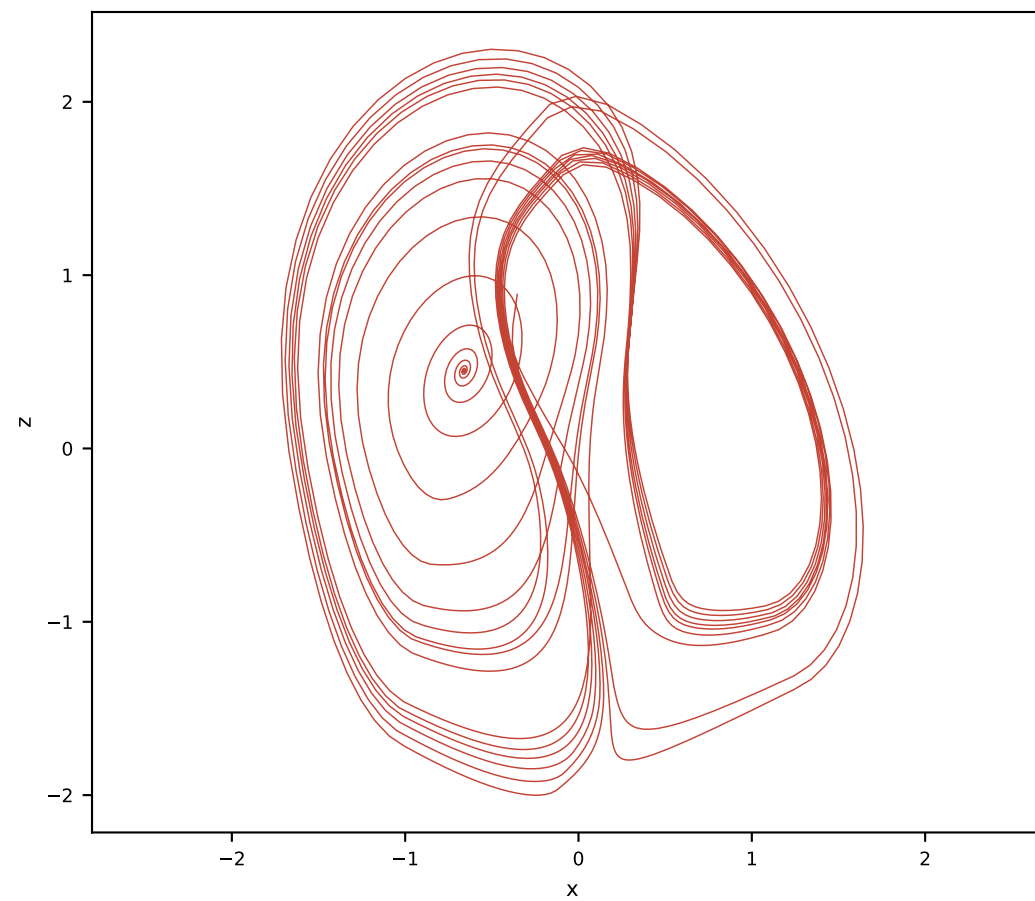
Orthogonal AL-RNN (seed 1)



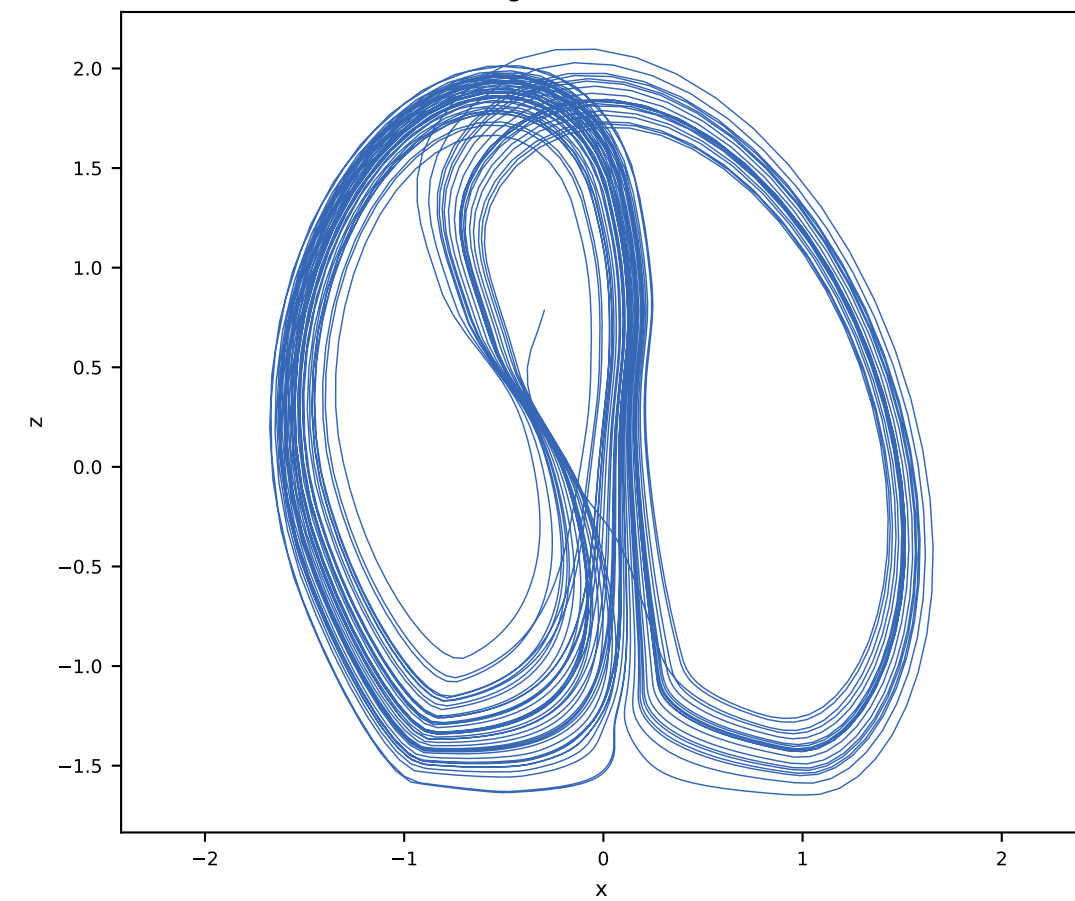
Ground truth x-z



Vanilla x-z (seed 1)

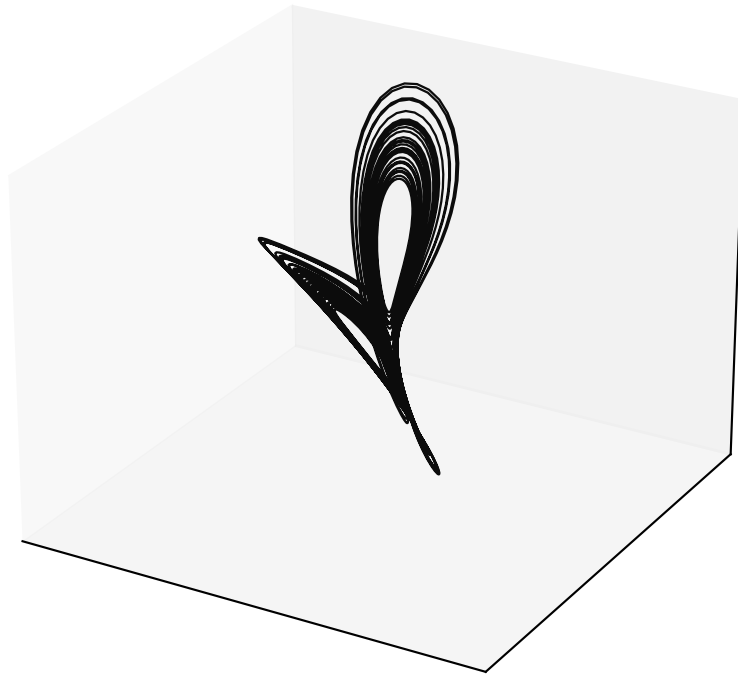


Orthogonal x-z (seed 1)

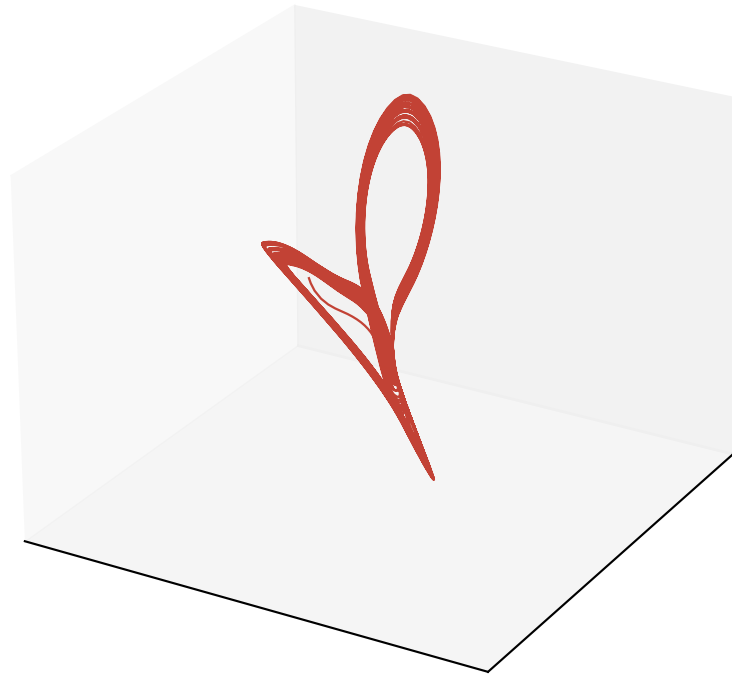


M = 11 P = 5 (median-PSE seed = 0 of [0, 1, 2])
PSE vanilla 0.0816 ± 0.0110 ortho 0.0828 ± 0.0084 mean rel gap +1.4% D_H_state vanilla 0.630 ± 0.015 ortho 0.654 ± 0.072 mean rel gap +3.8%

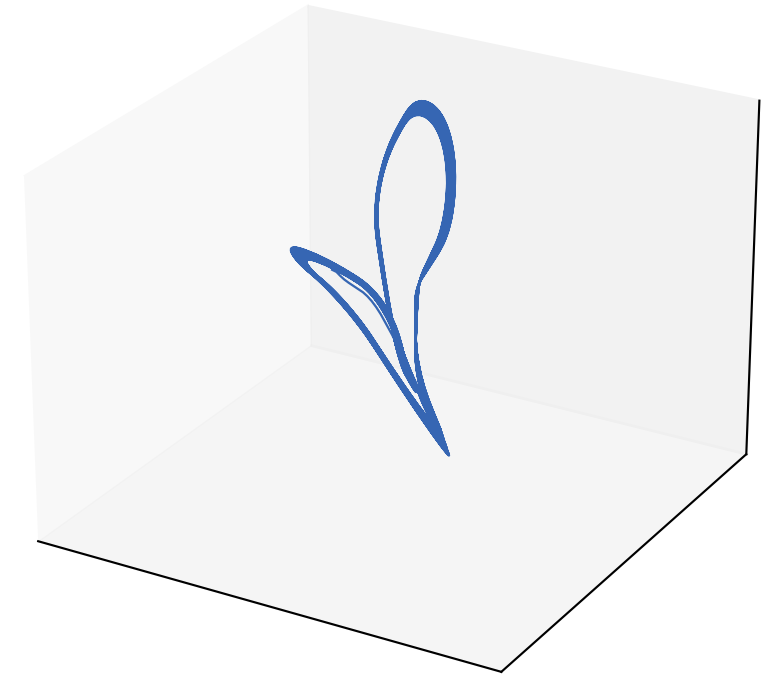
Ground truth (rotated)



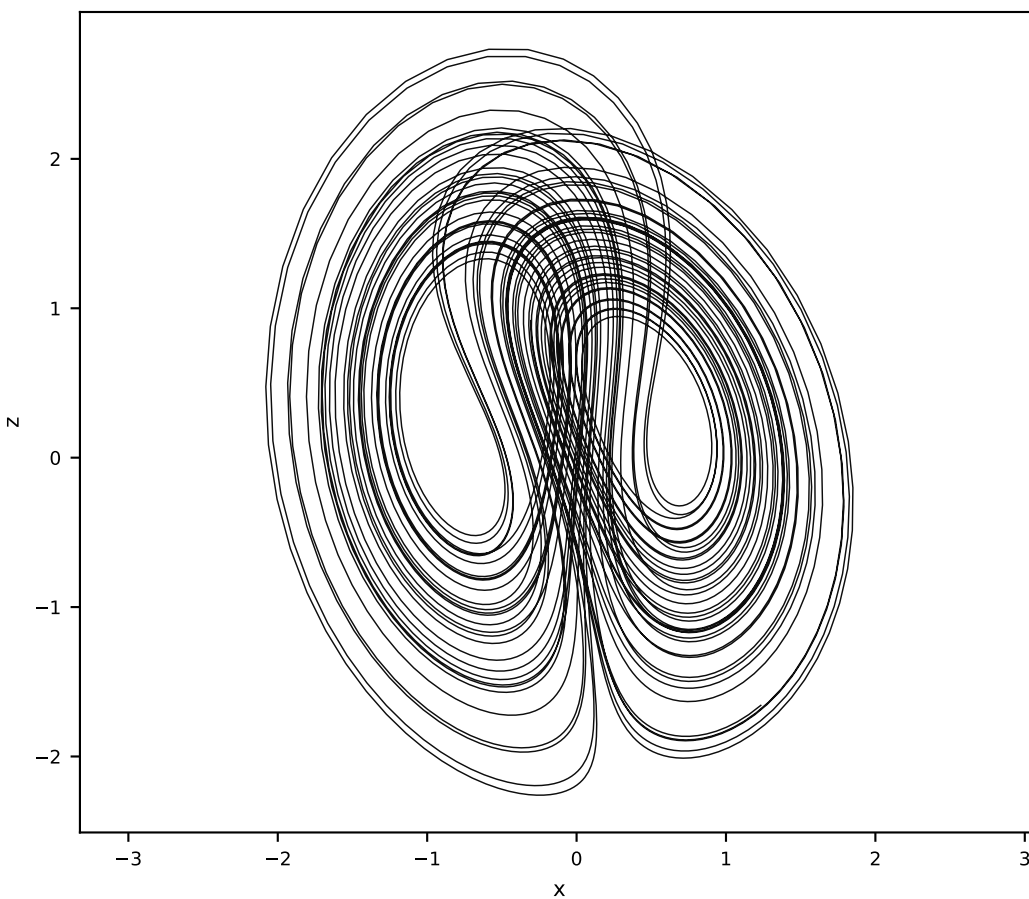
Vanilla AL-RNN (seed 0)



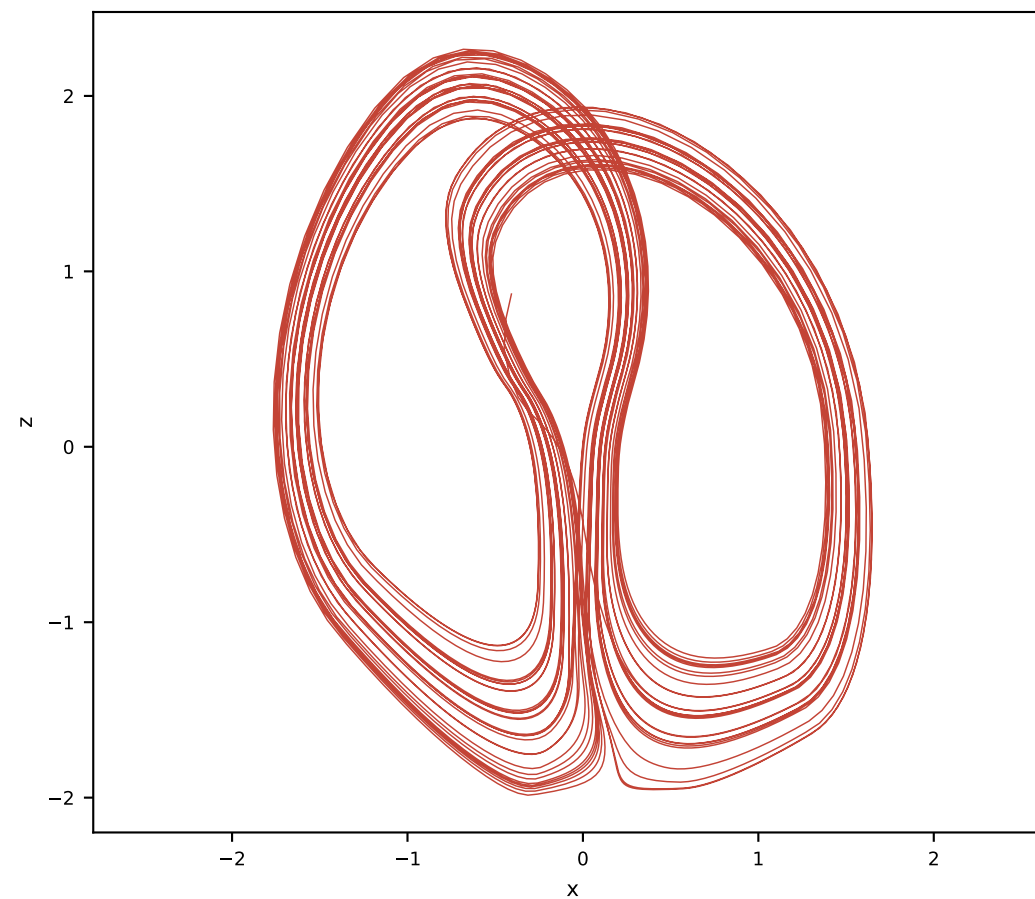
Orthogonal AL-RNN (seed 0)



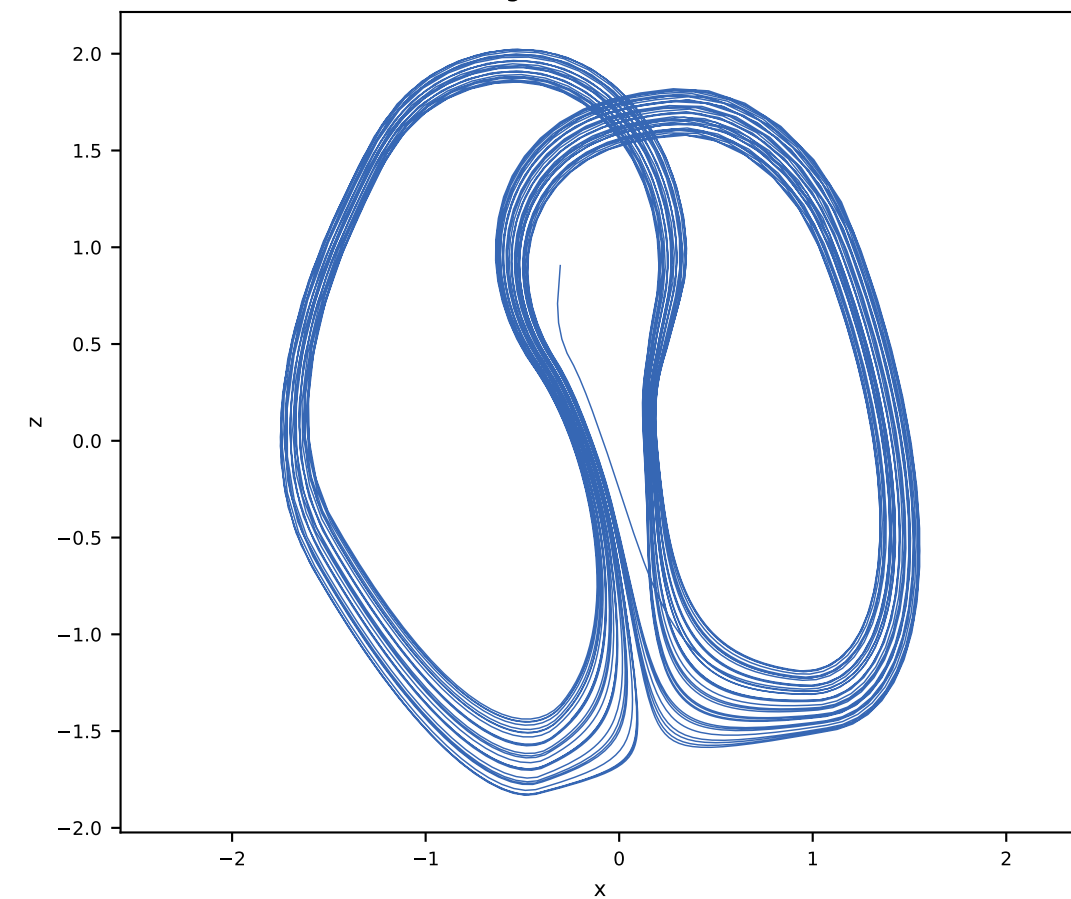
Ground truth x-z



Vanilla x-z (seed 0)

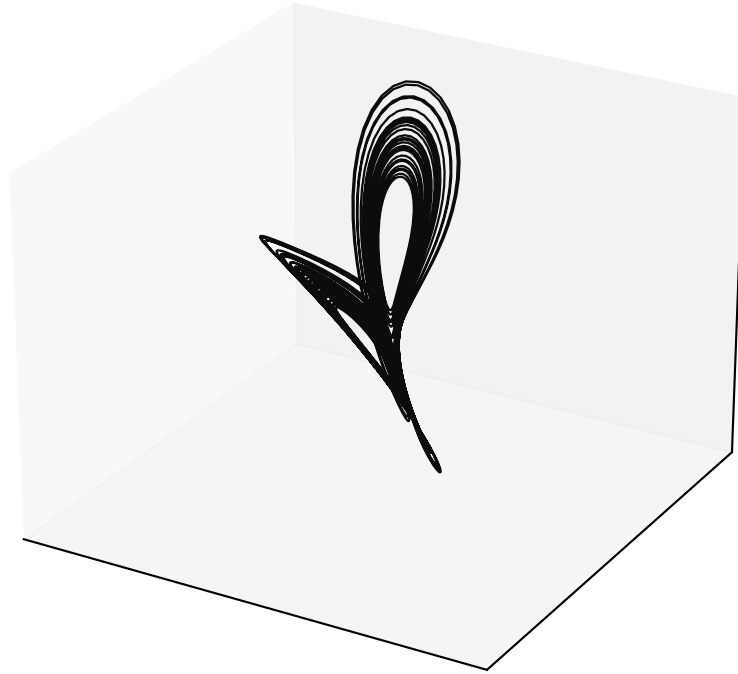


Orthogonal x-z (seed 0)

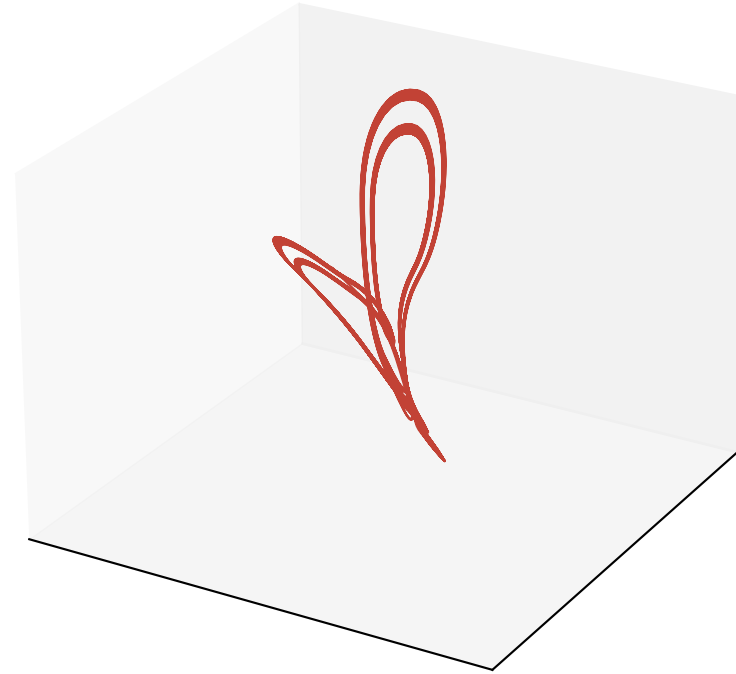


M = 11 P = 6 (median-PSE seed = 0 of [0, 1, 2])
PSE vanilla 0.0623 ± 0.0247 ortho 0.0699 ± 0.0151 mean rel gap +12.2% D_H_state vanilla 0.582 ± 0.078 ortho 0.539 ± 0.060 mean rel gap -7.4%

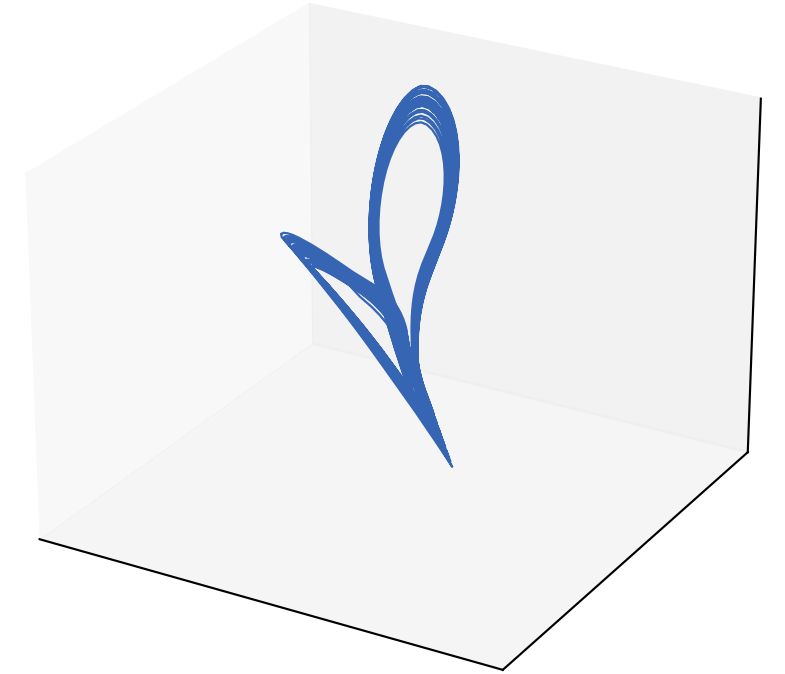
Ground truth (rotated)



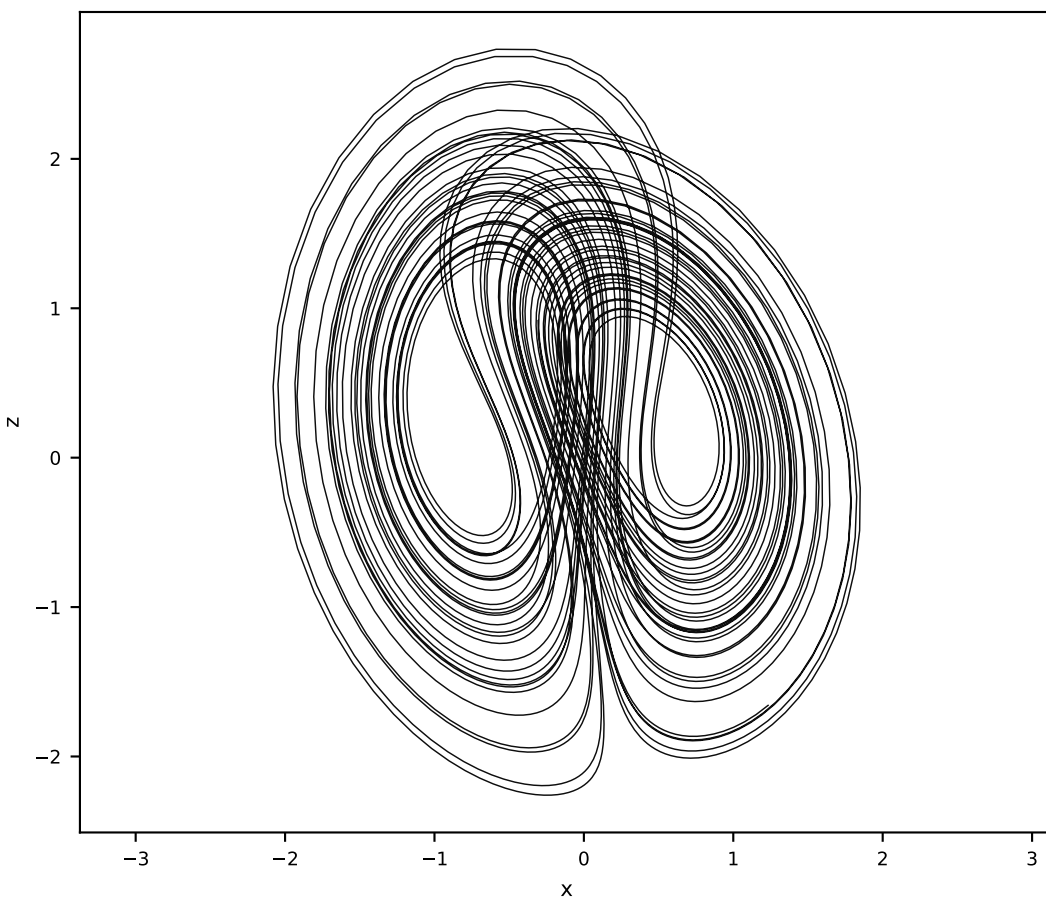
Vanilla AL-RNN (seed 0)



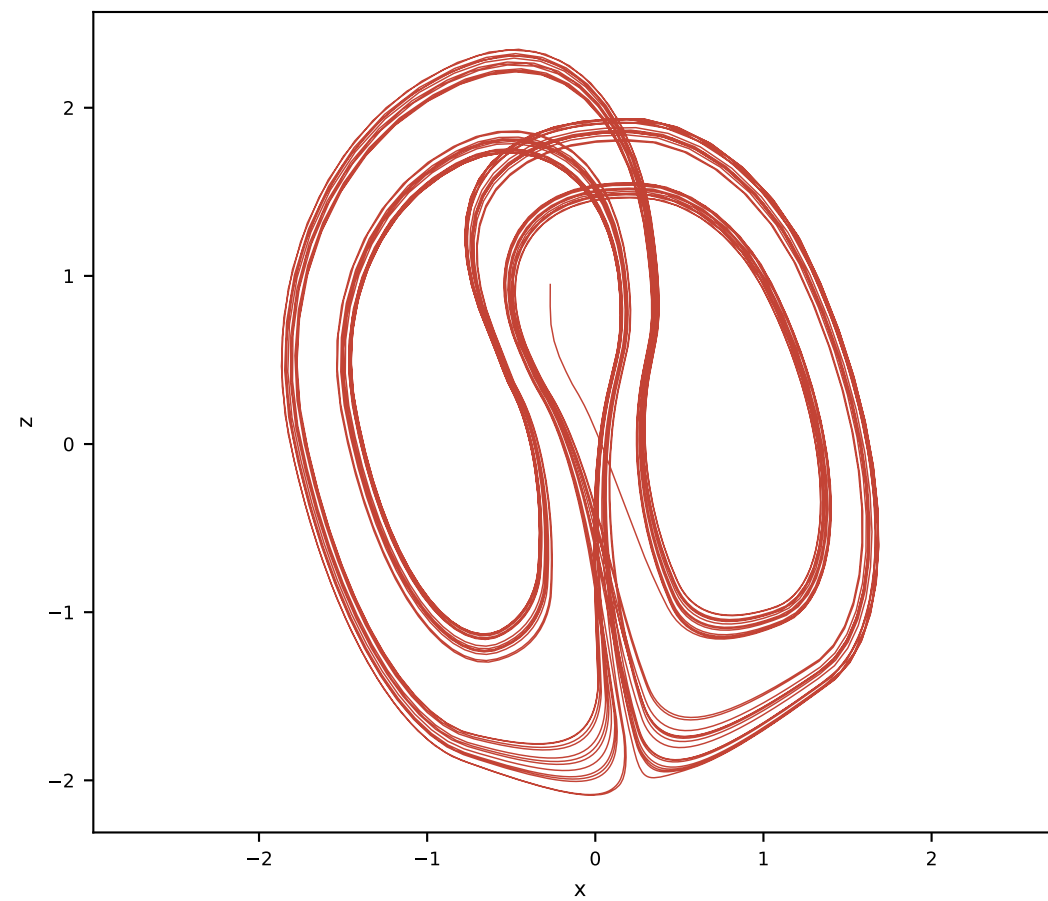
Orthogonal AL-RNN (seed 0)



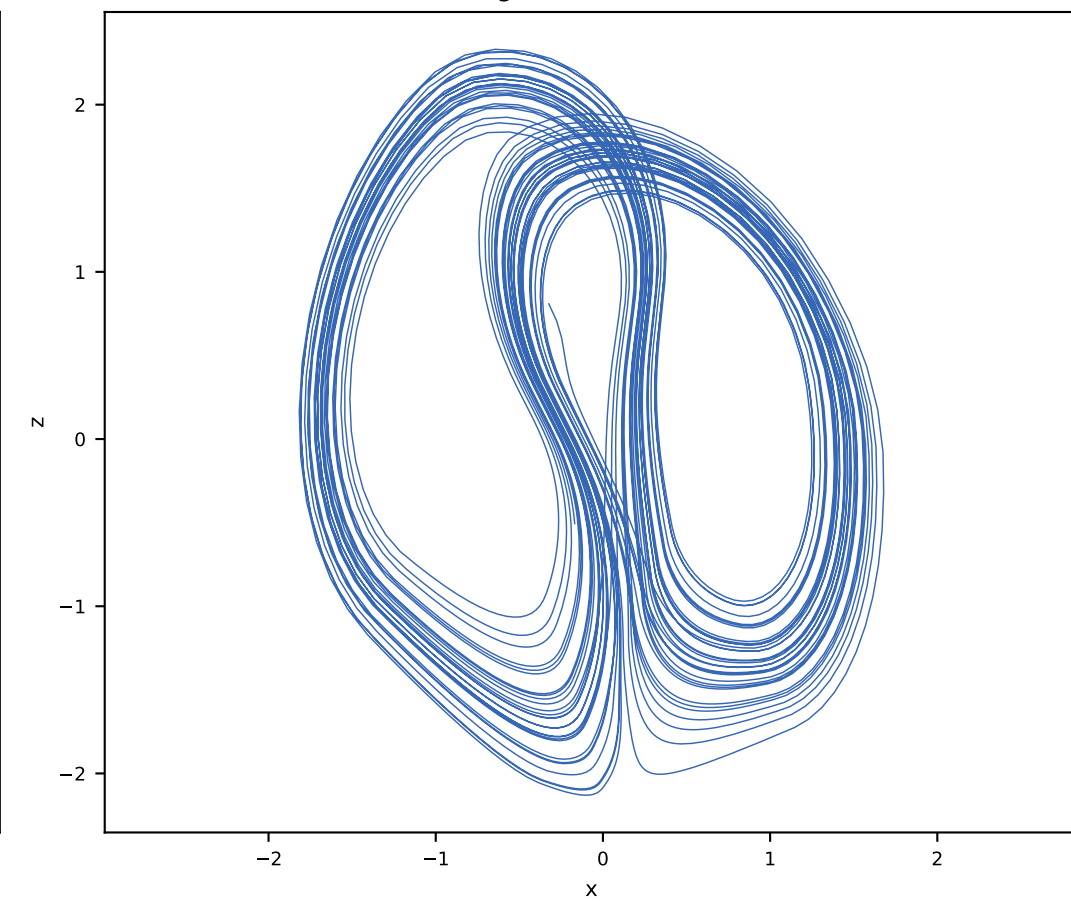
Ground truth x-z



Vanilla x-z (seed 0)

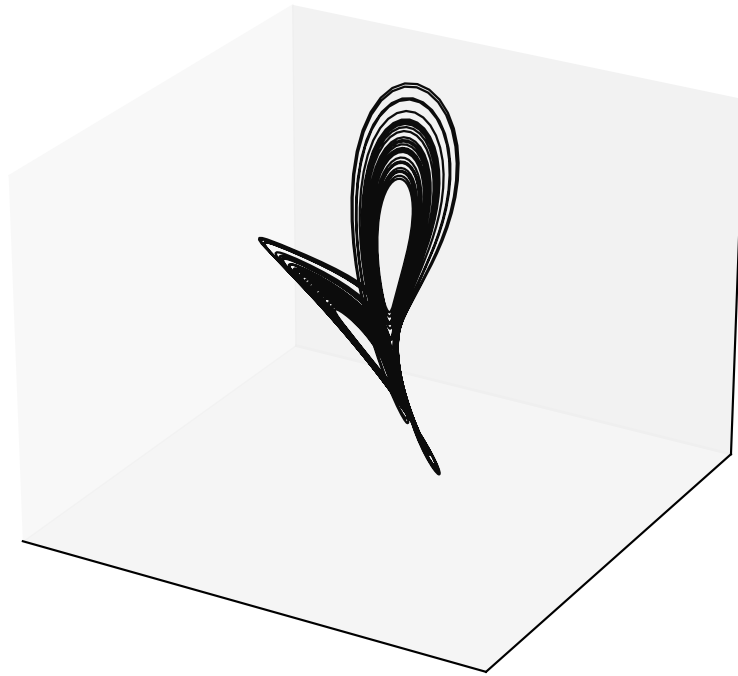


Orthogonal x-z (seed 0)

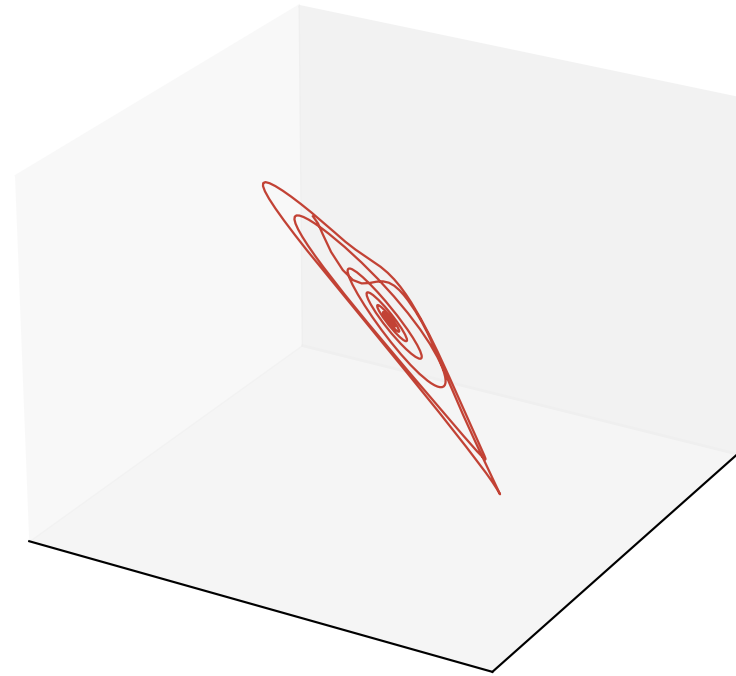


M = 14 P = 2 (median-PSE seed = 0 of [0, 1, 2])
PSE vanilla 0.3195 ± 0.0505 ortho 0.2588 ± 0.1620 mean rel gap -19.0% D_H_state vanilla 0.928 ± 0.068 ortho 0.896 ± 0.035 mean rel gap -3.4%

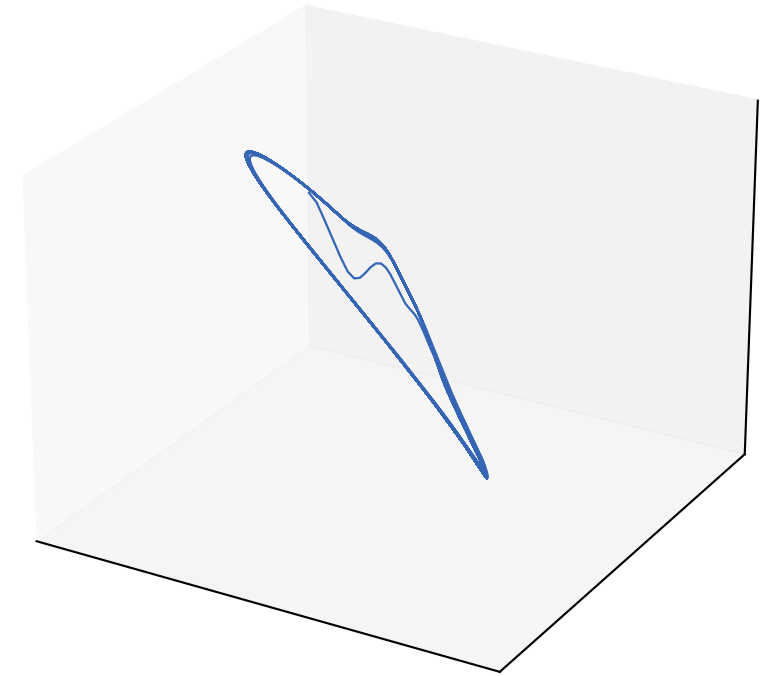
Ground truth (rotated)



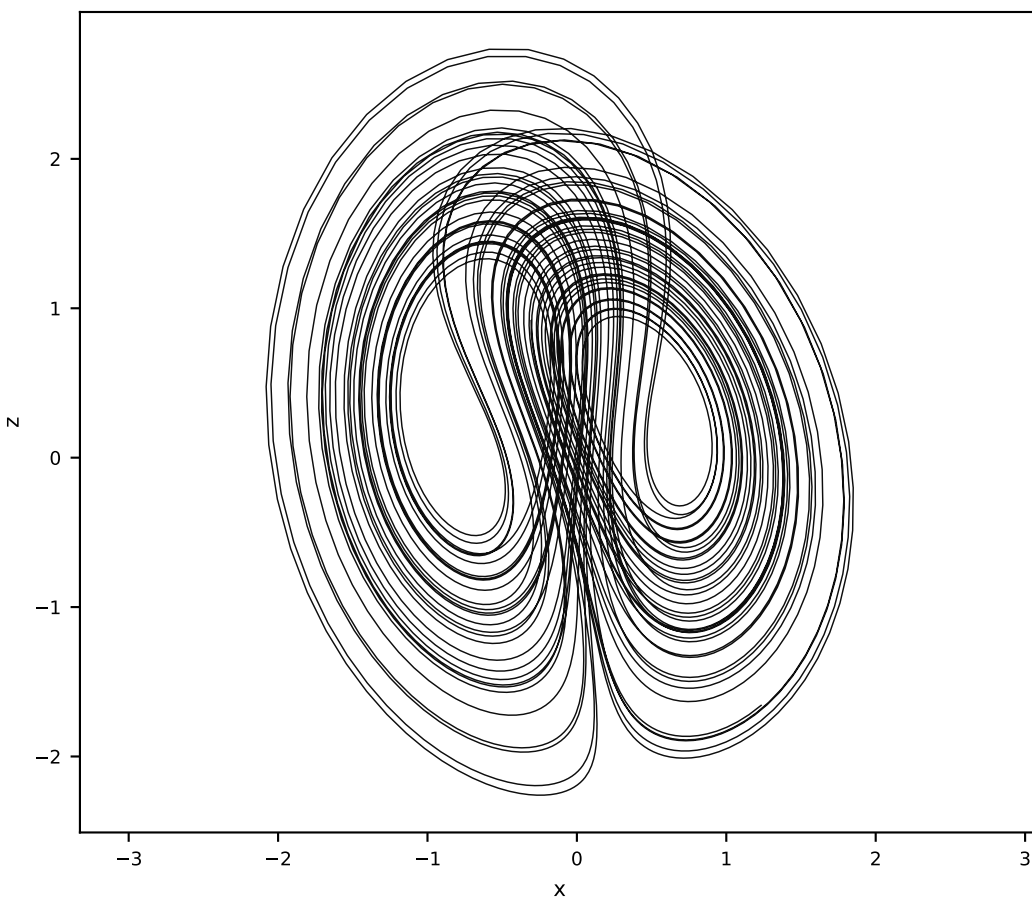
Vanilla AL-RNN (seed 0)



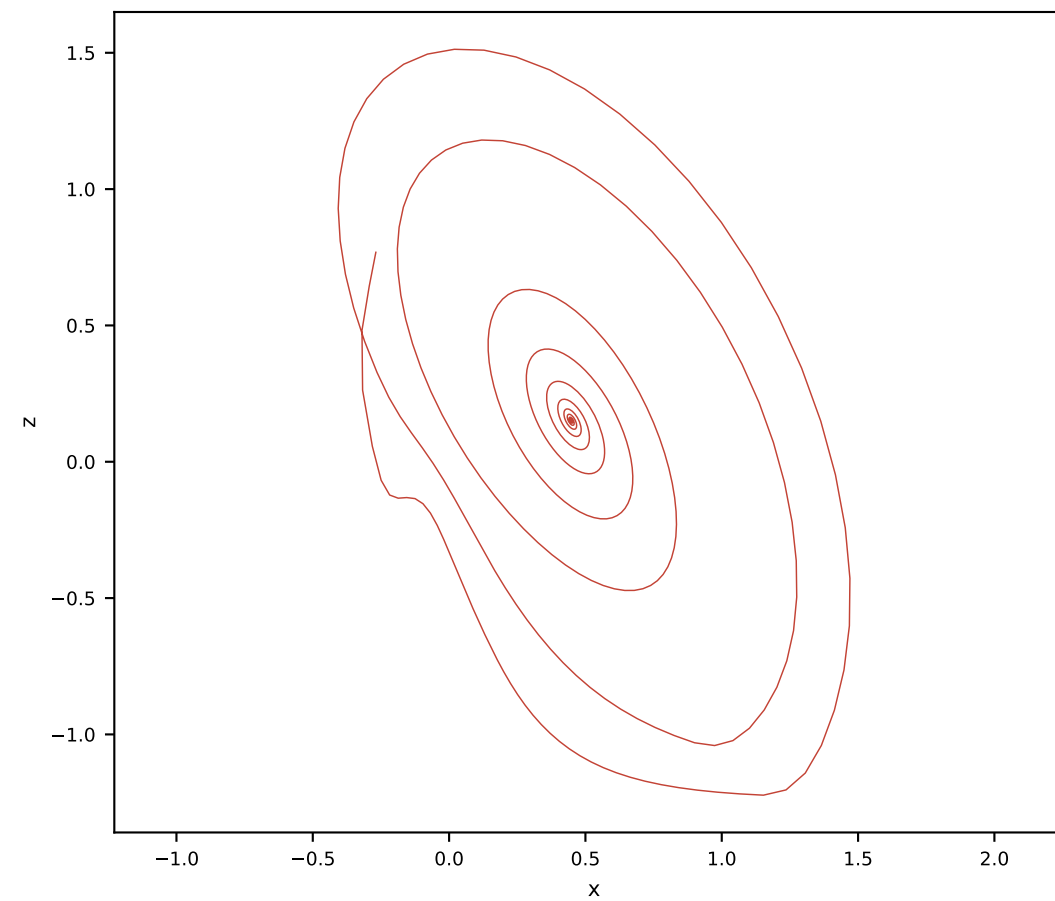
Orthogonal AL-RNN (seed 0)



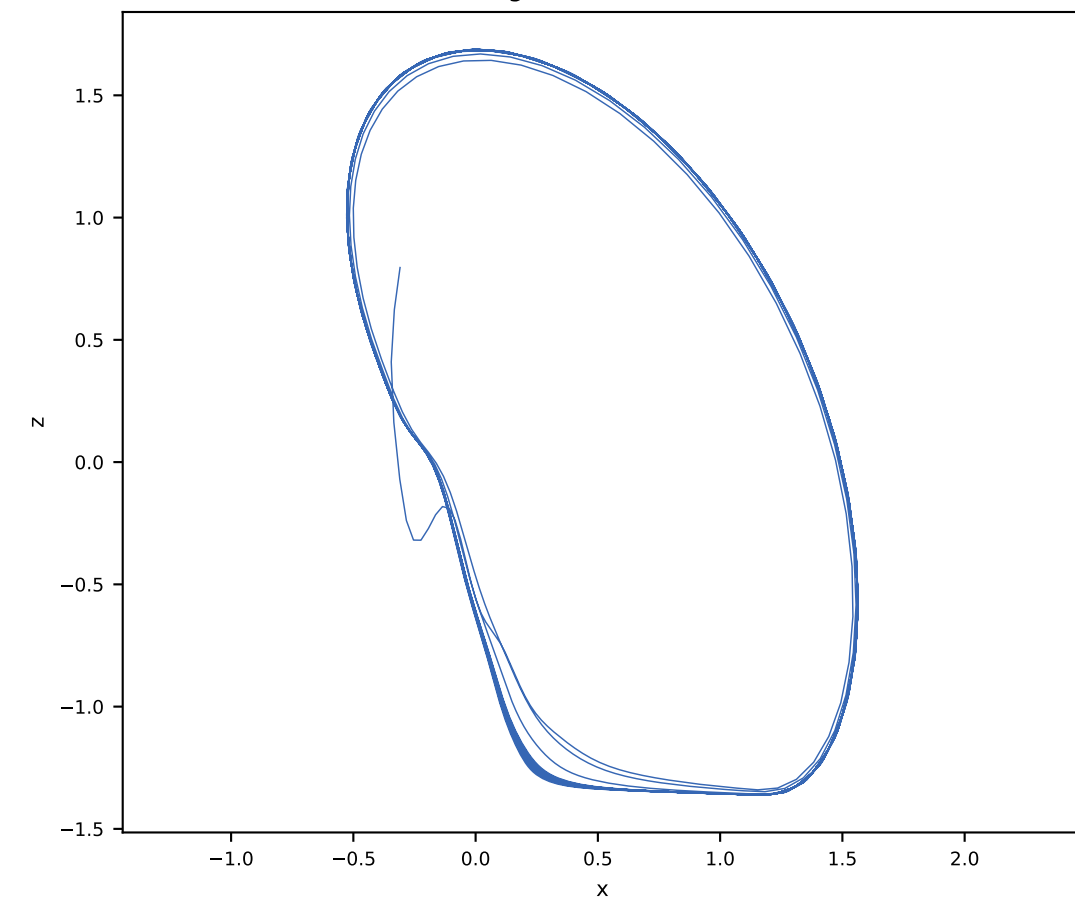
Ground truth x-z



Vanilla x-z (seed 0)

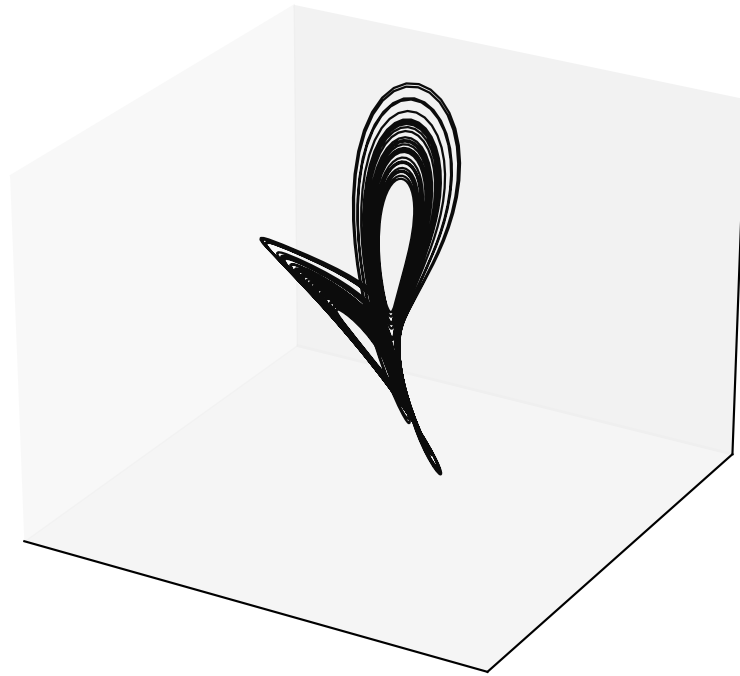


Orthogonal x-z (seed 0)

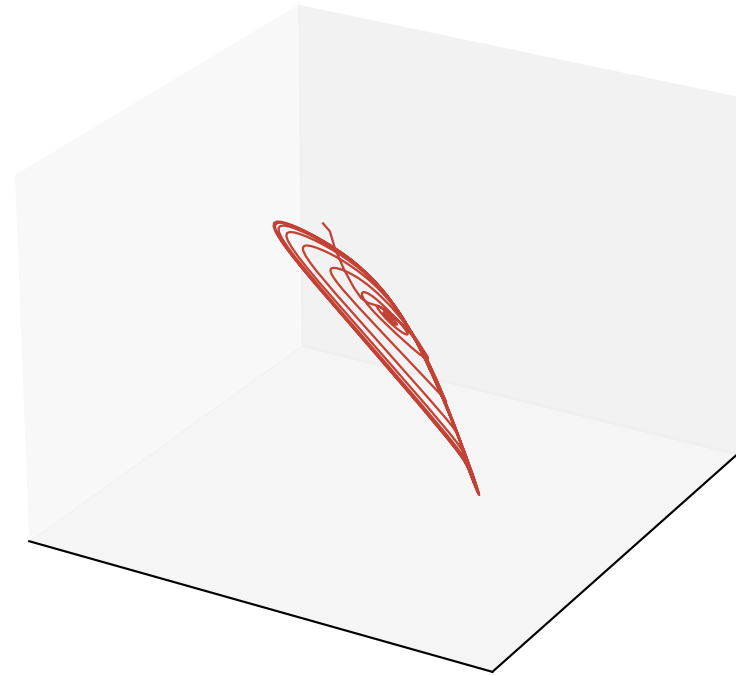


M = 14 P = 3 (median-PSE seed = 0 of [0, 1, 2])
PSE vanilla 0.1959 ± 0.1212 ortho 0.3473 ± 0.0862 mean rel gap +77.3% D_H_state vanilla 0.821 ± 0.139 ortho 0.949 ± 0.035 mean rel gap +15.6%

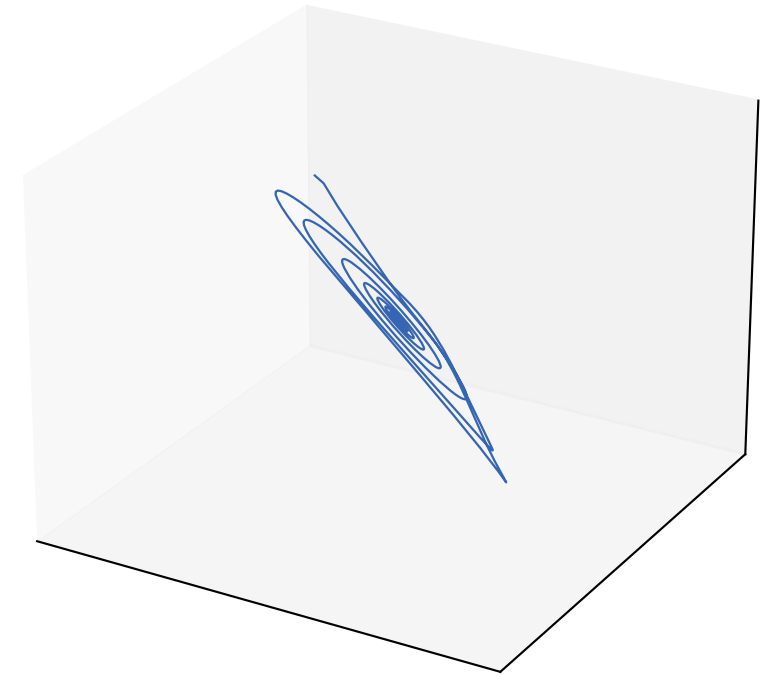
Ground truth (rotated)



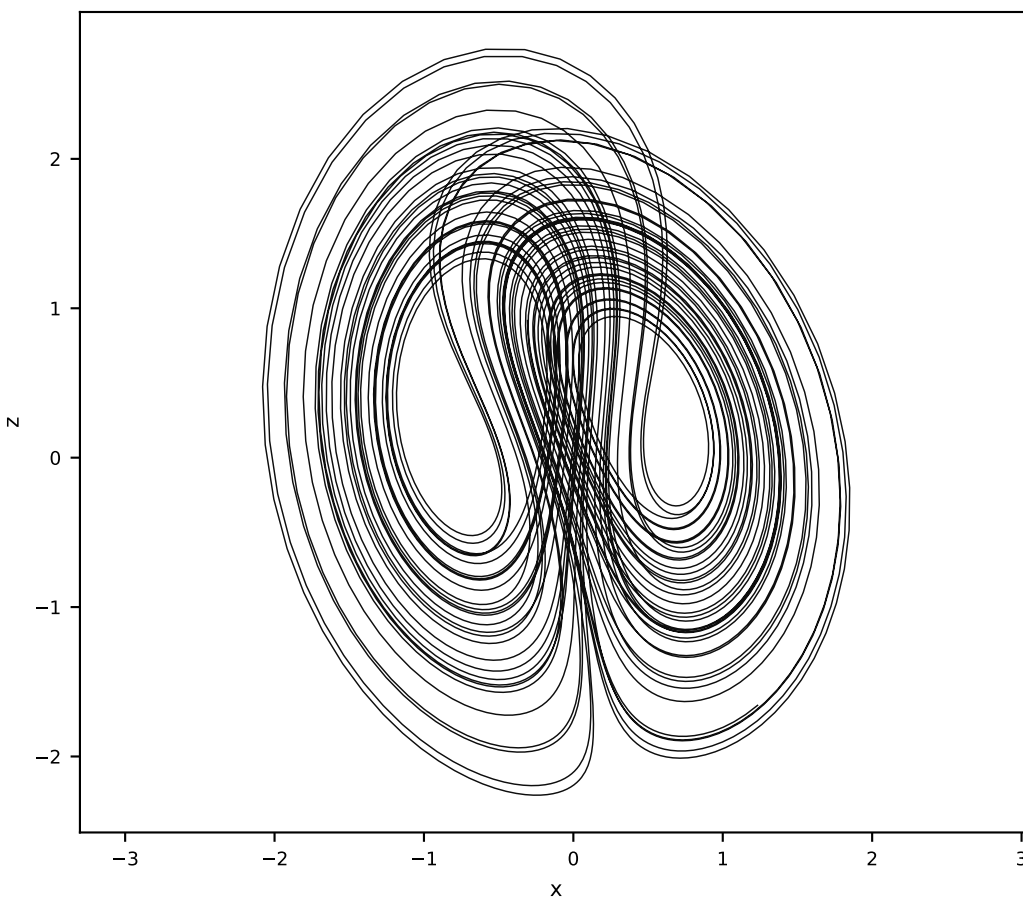
Vanilla AL-RNN (seed 0)



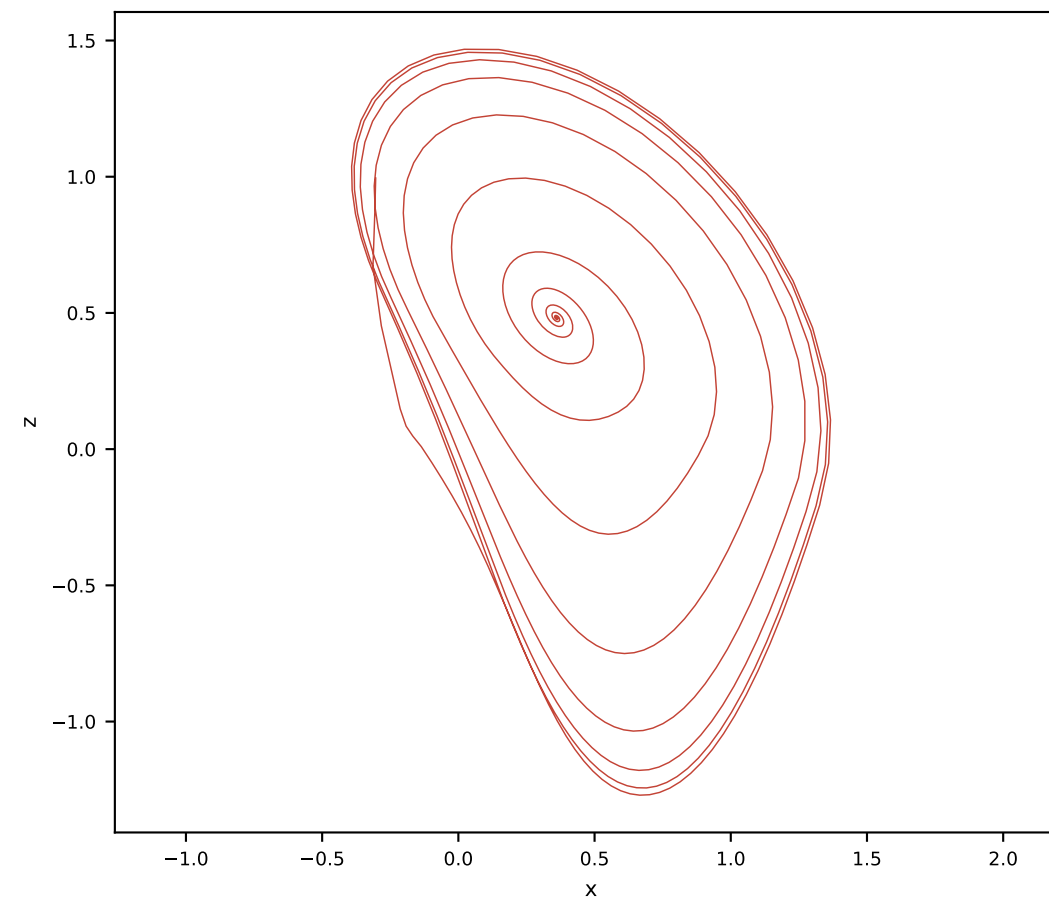
Orthogonal AL-RNN (seed 0)



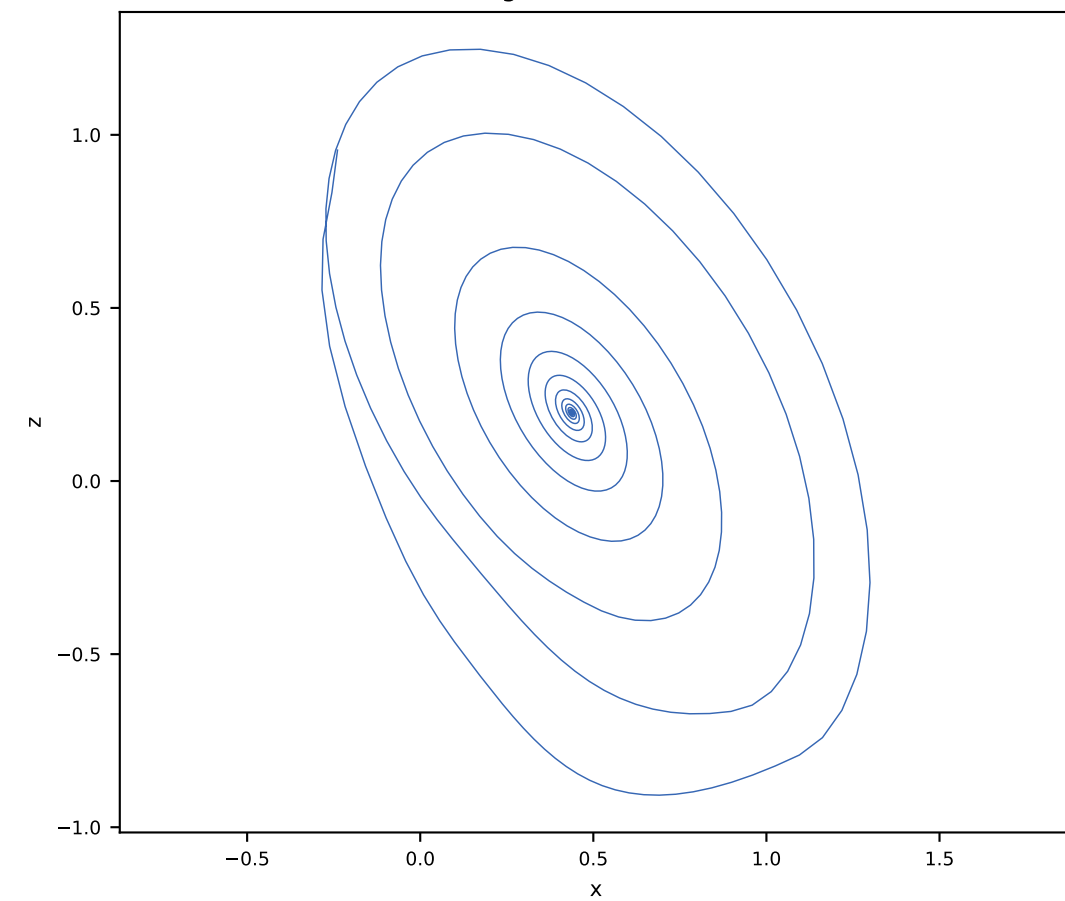
Ground truth x-z



Vanilla x-z (seed 0)

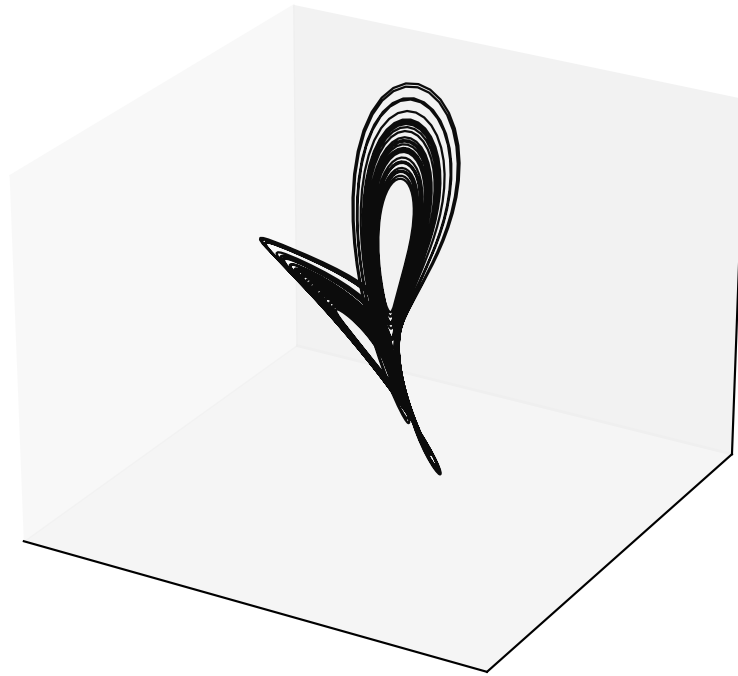


Orthogonal x-z (seed 0)

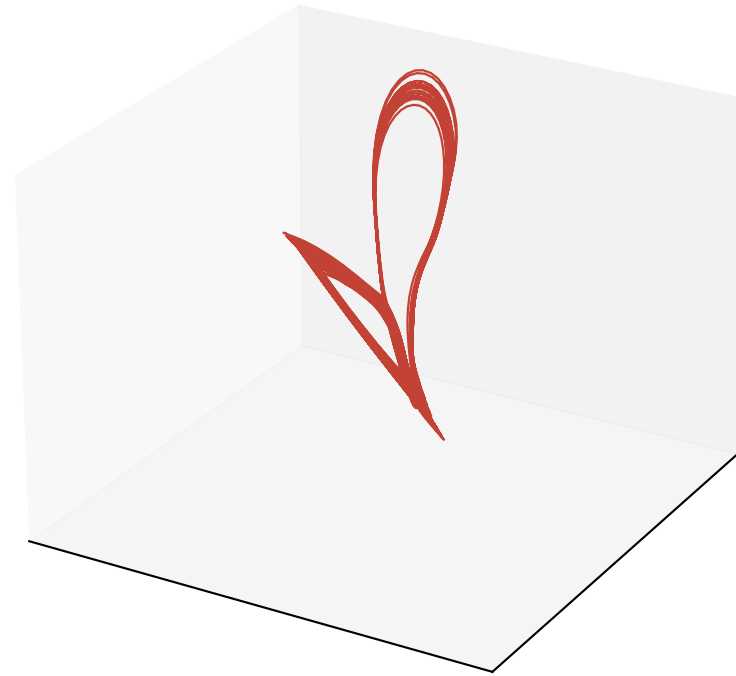


M = 14 P = 4 (median-PSE seed = 1 of [0, 1, 2])
PSE vanilla 0.0809 ± 0.0031 ortho 0.0658 ± 0.0191 mean rel gap -18.7% D_H_state vanilla 0.626 ± 0.009 ortho 0.561 ± 0.072 mean rel gap -10.4%

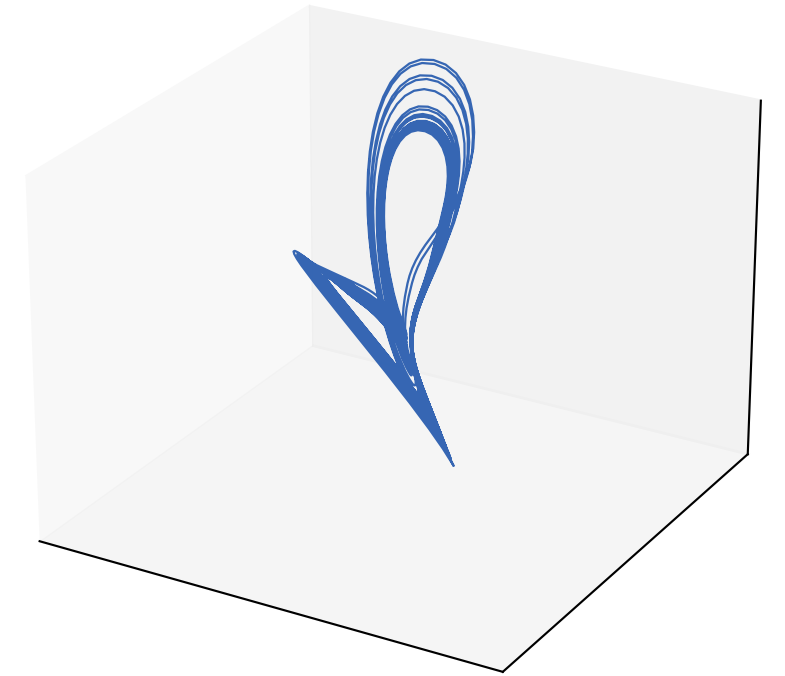
Ground truth (rotated)



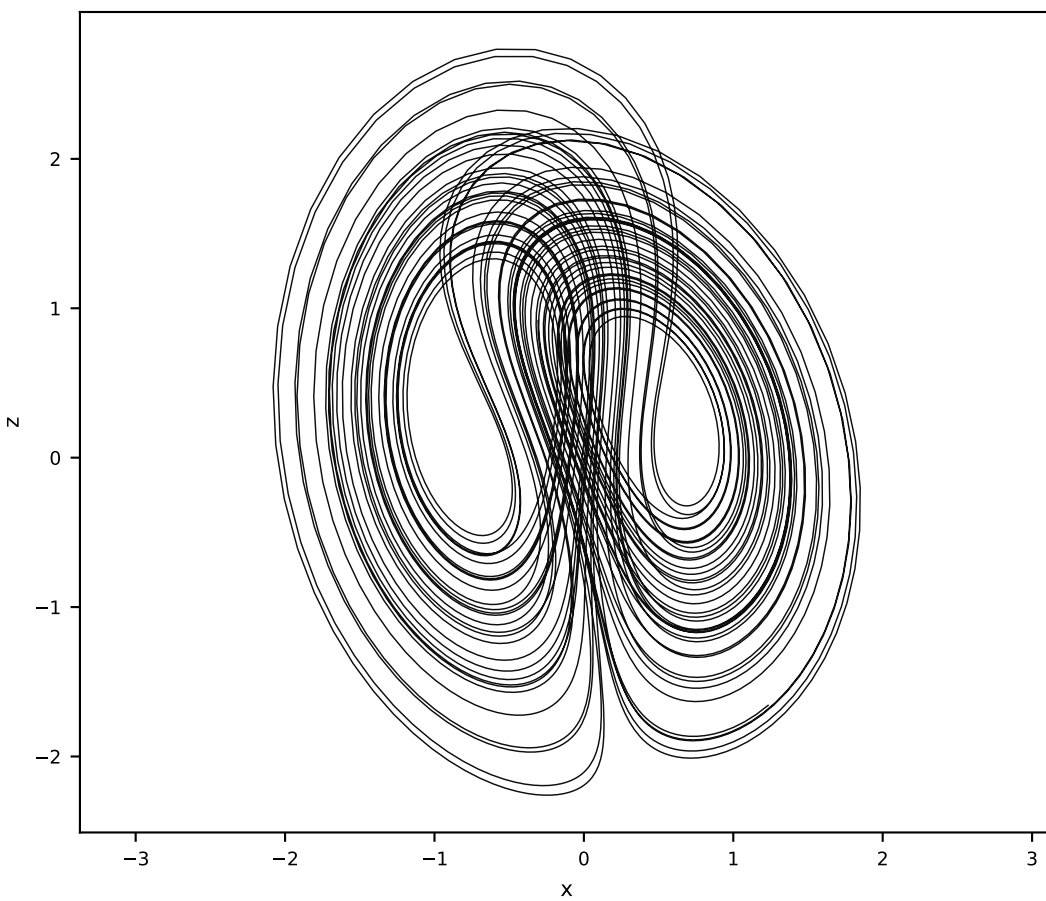
Vanilla AL-RNN (seed 1)



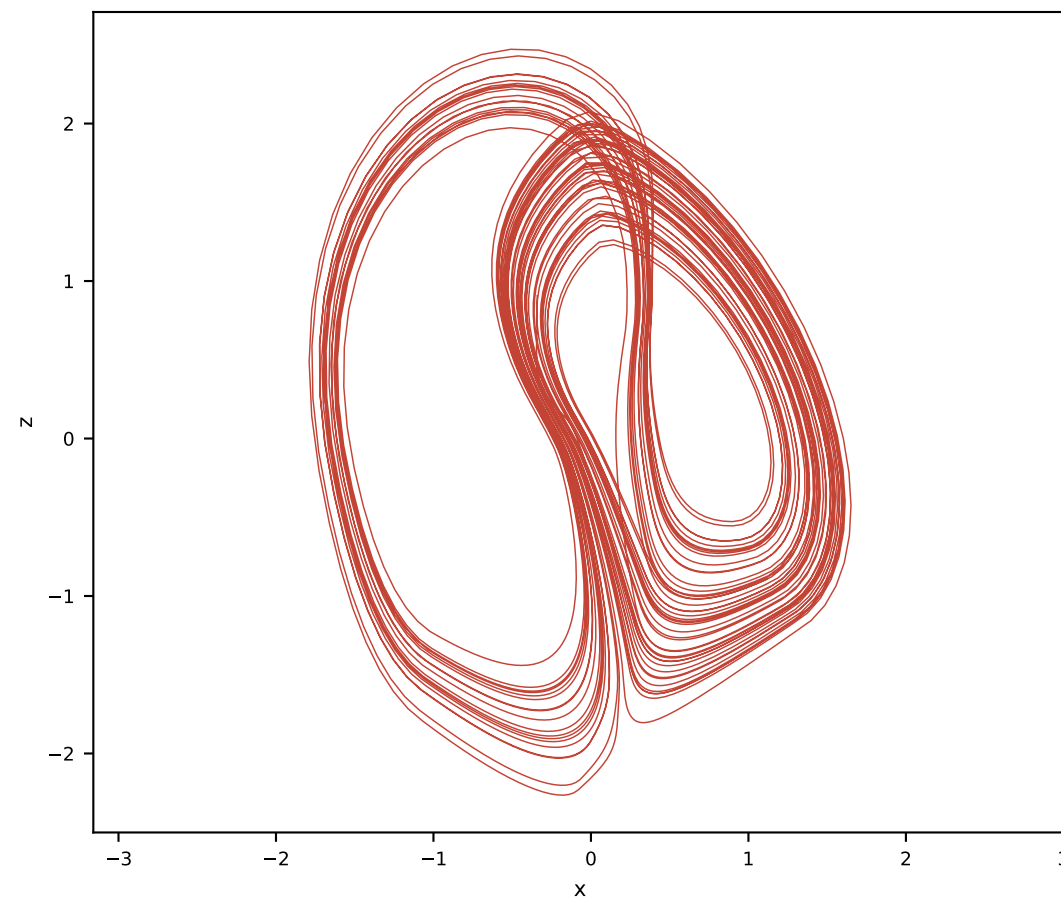
Orthogonal AL-RNN (seed 1)



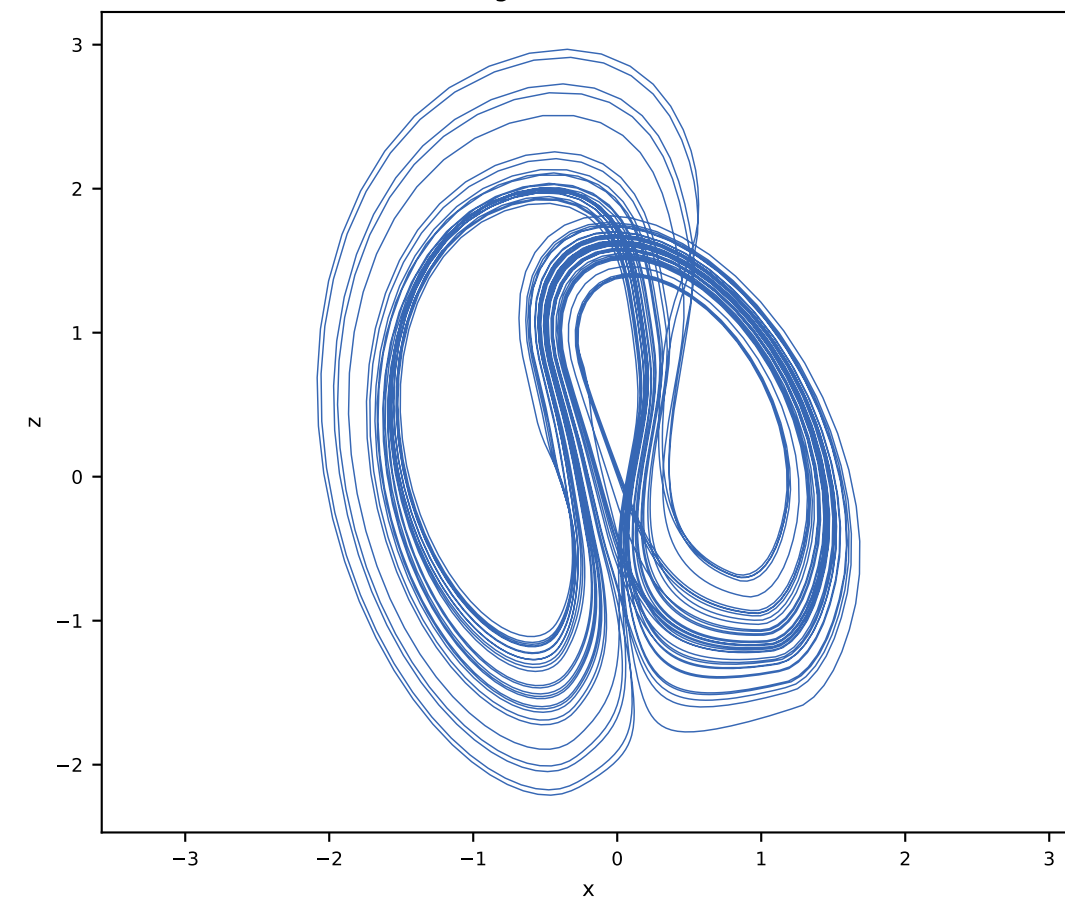
Ground truth x-z



Vanilla x-z (seed 1)

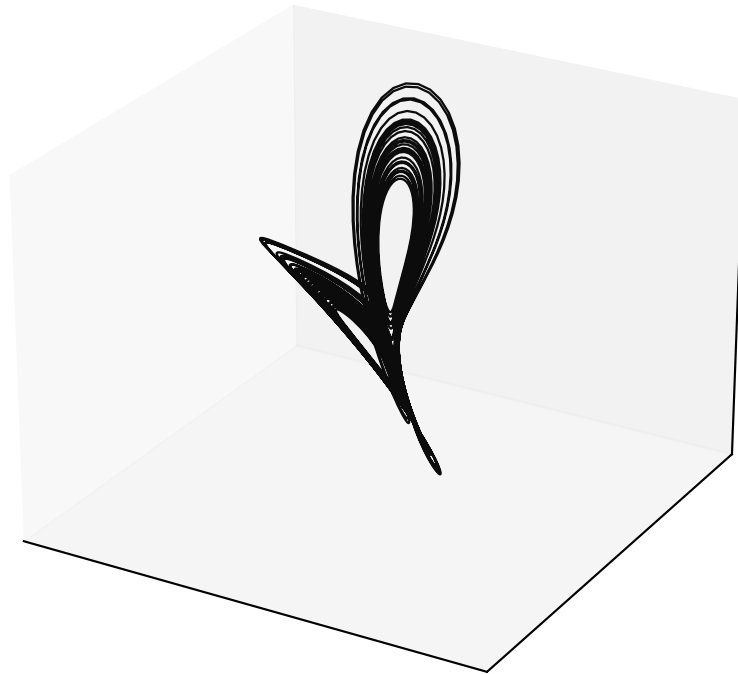


Orthogonal x-z (seed 1)

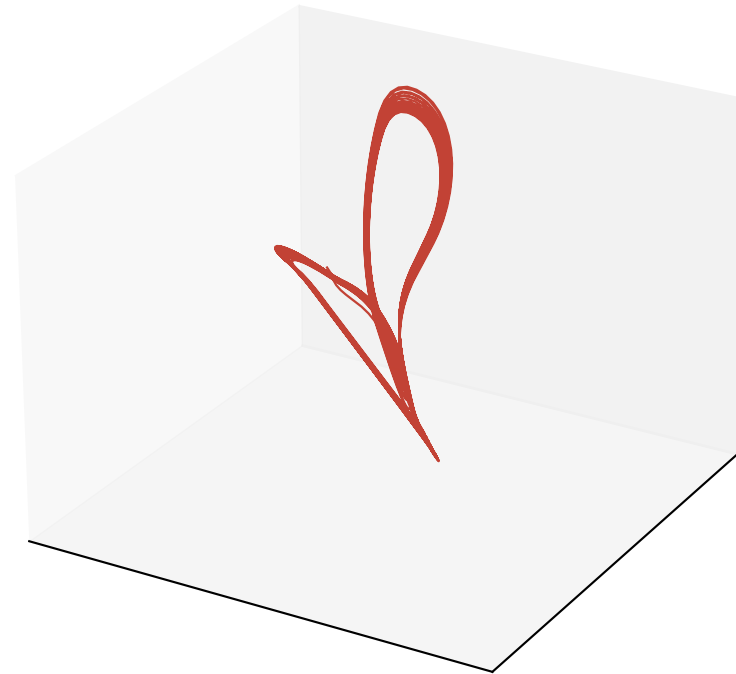


M = 14 P = 5 (median-PSE seed = 2 of [0, 1, 2])
PSE vanilla 0.0862 ± 0.0061 ortho 0.0627 ± 0.0256 mean rel gap -27.2% D_H_state vanilla 0.644 ± 0.039 ortho 0.540 ± 0.126 mean rel gap -16.1%

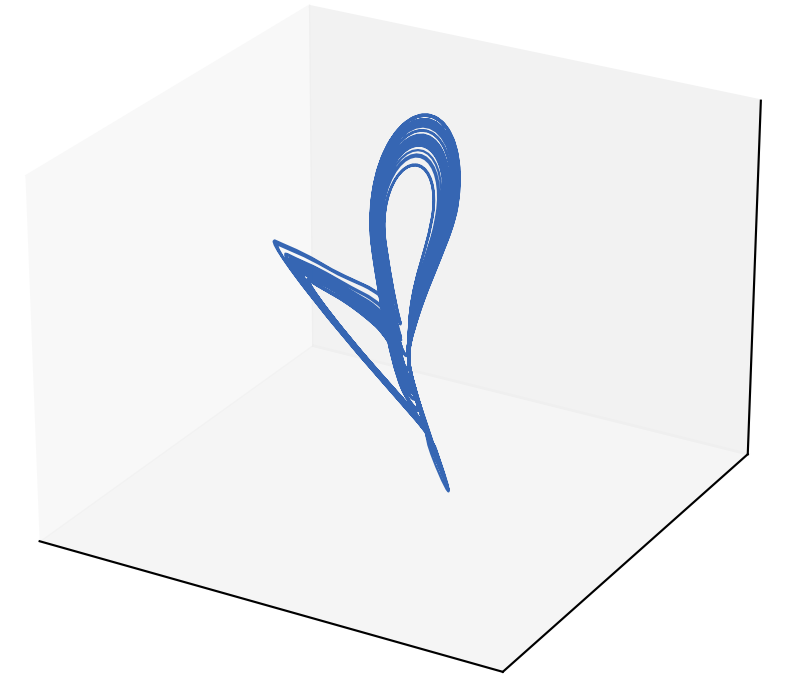
Ground truth (rotated)



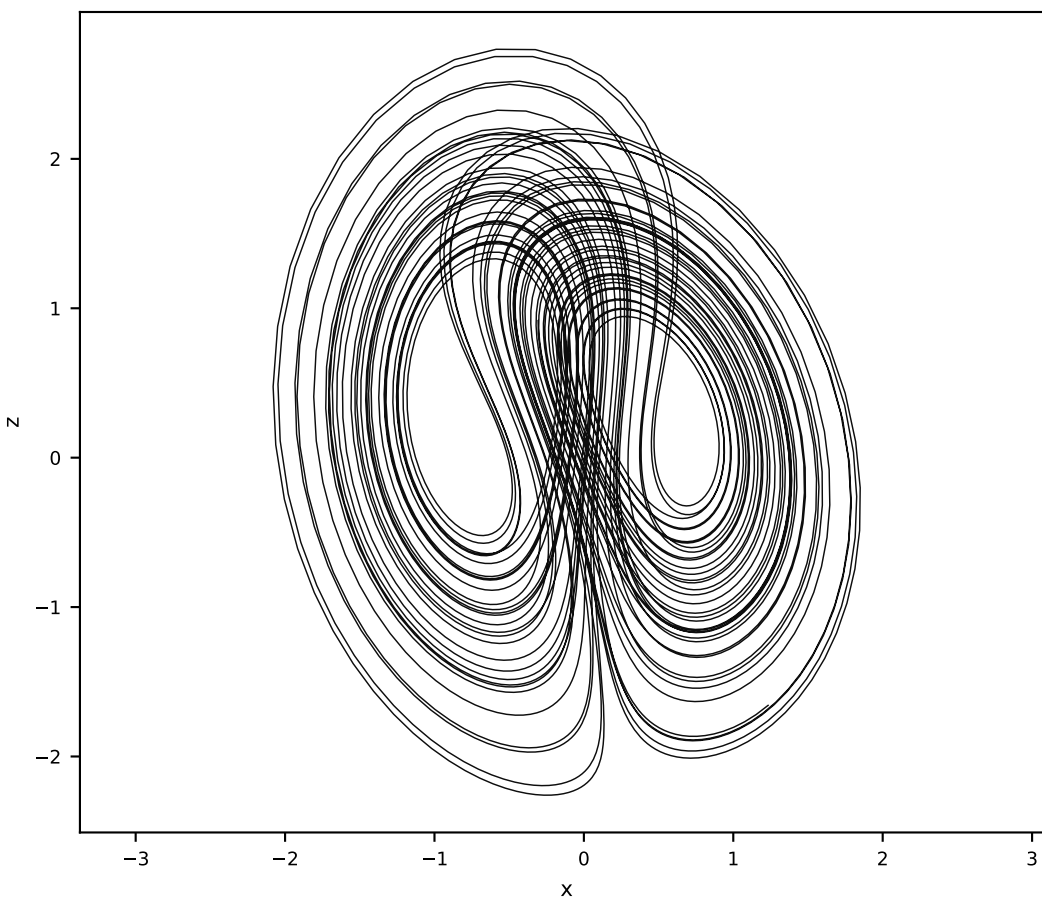
Vanilla AL-RNN (seed 2)



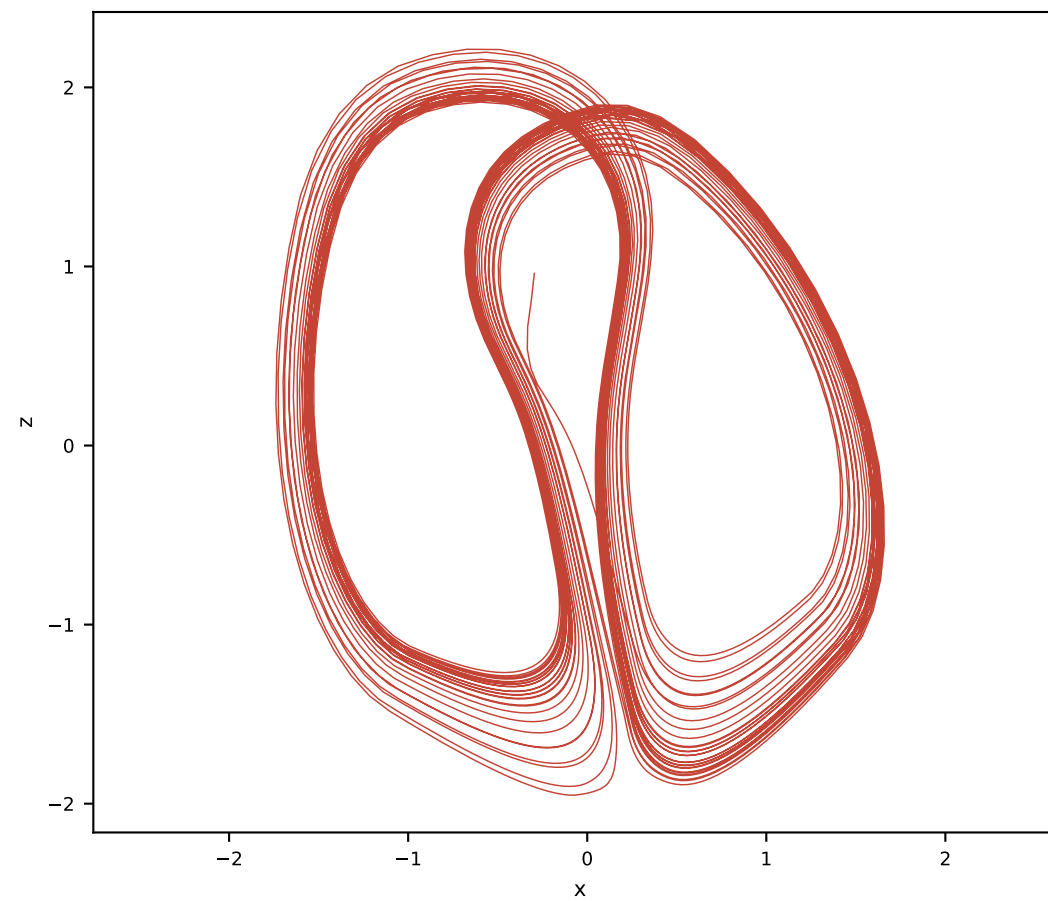
Orthogonal AL-RNN (seed 2)



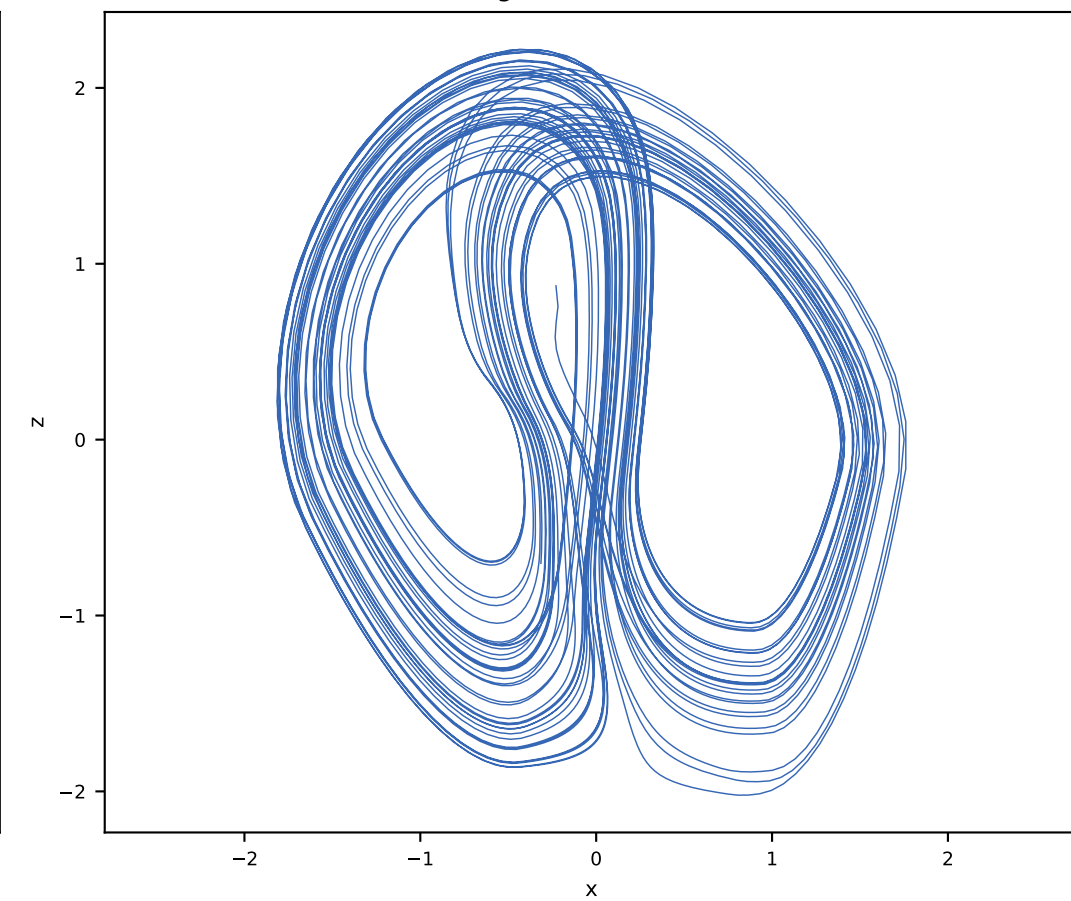
Ground truth x-z



Vanilla x-z (seed 2)

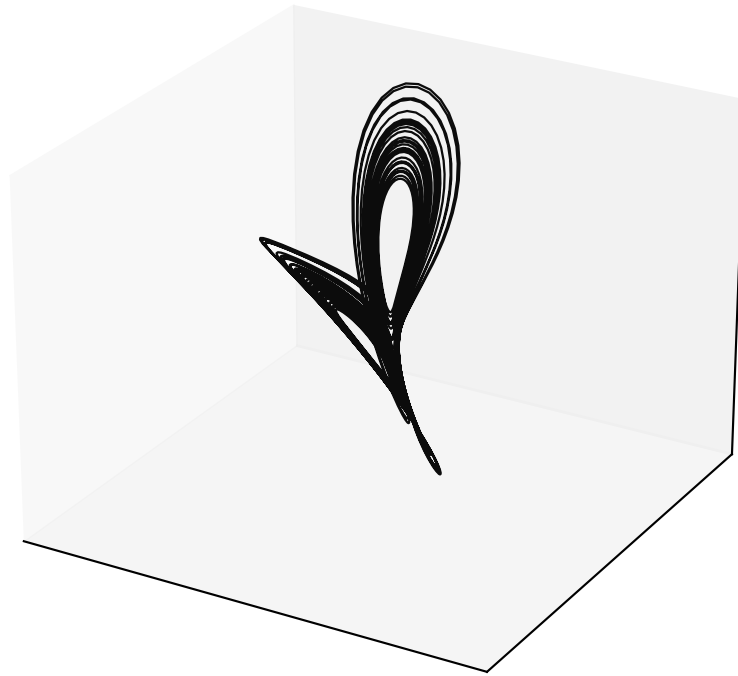


Orthogonal x-z (seed 2)

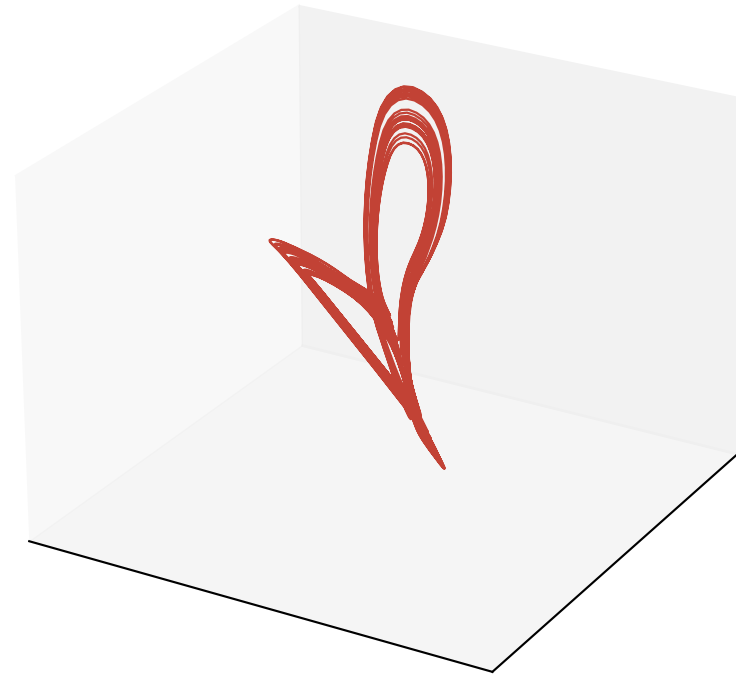


M = 14 P = 6 (median-PSE seed = 0 of [0, 1, 2])
PSE vanilla 0.0613 ± 0.0122 ortho 0.0702 ± 0.0253 mean rel gap +14.5% D_H_state vanilla 0.533 ± 0.026 ortho 0.576 ± 0.121 mean rel gap +8.0%

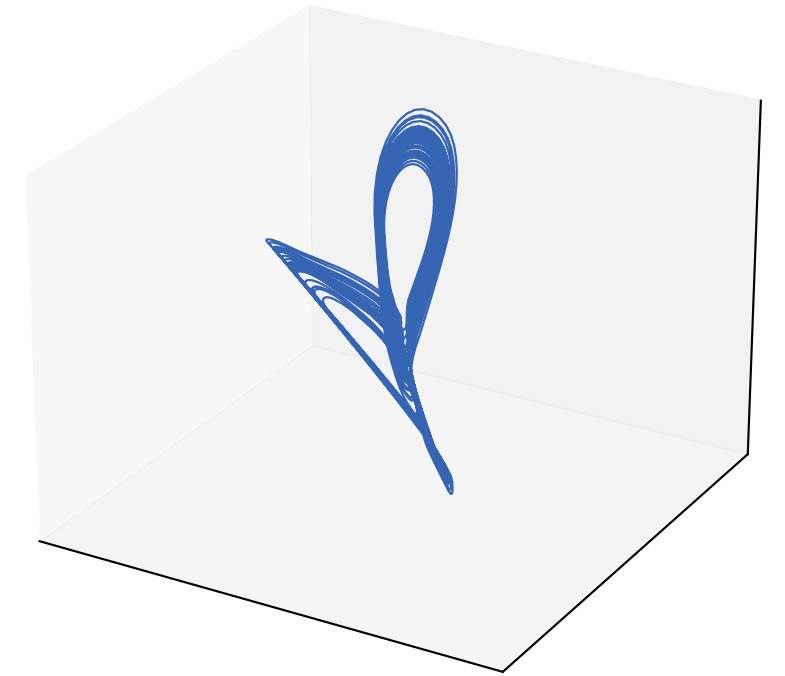
Ground truth (rotated)



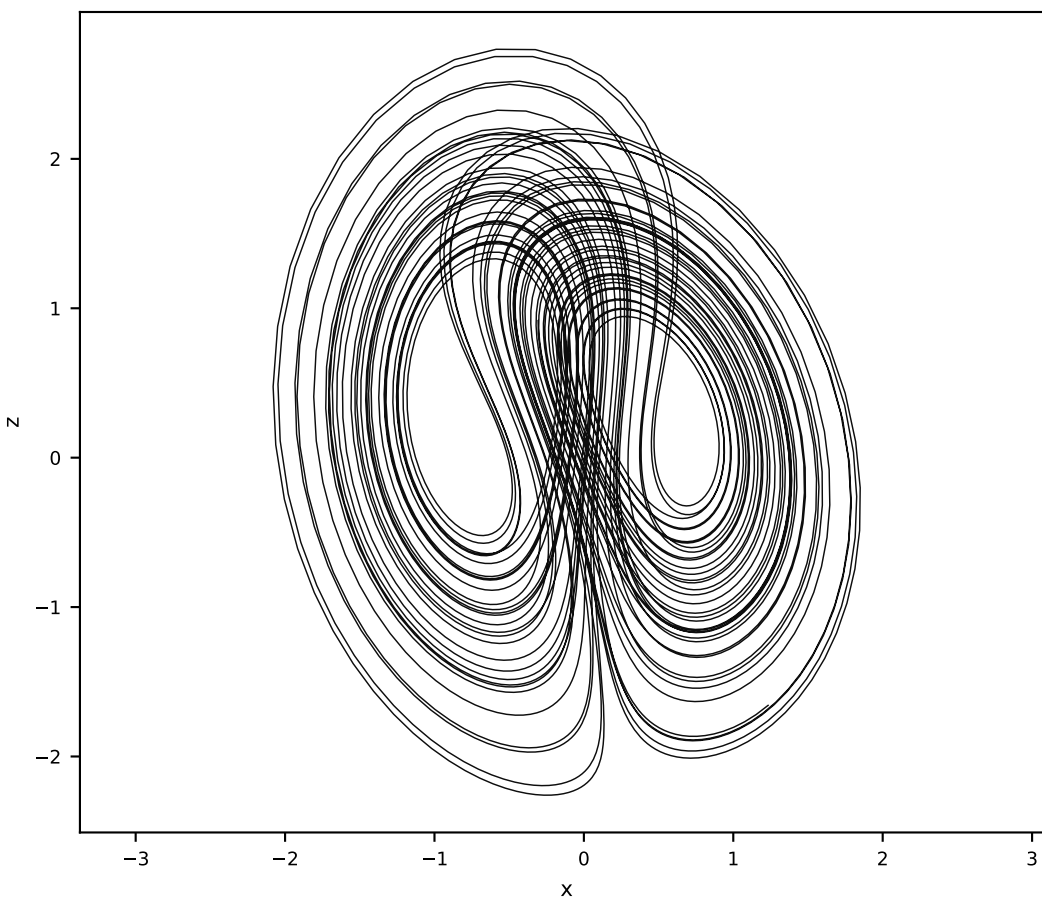
Vanilla AL-RNN (seed 0)



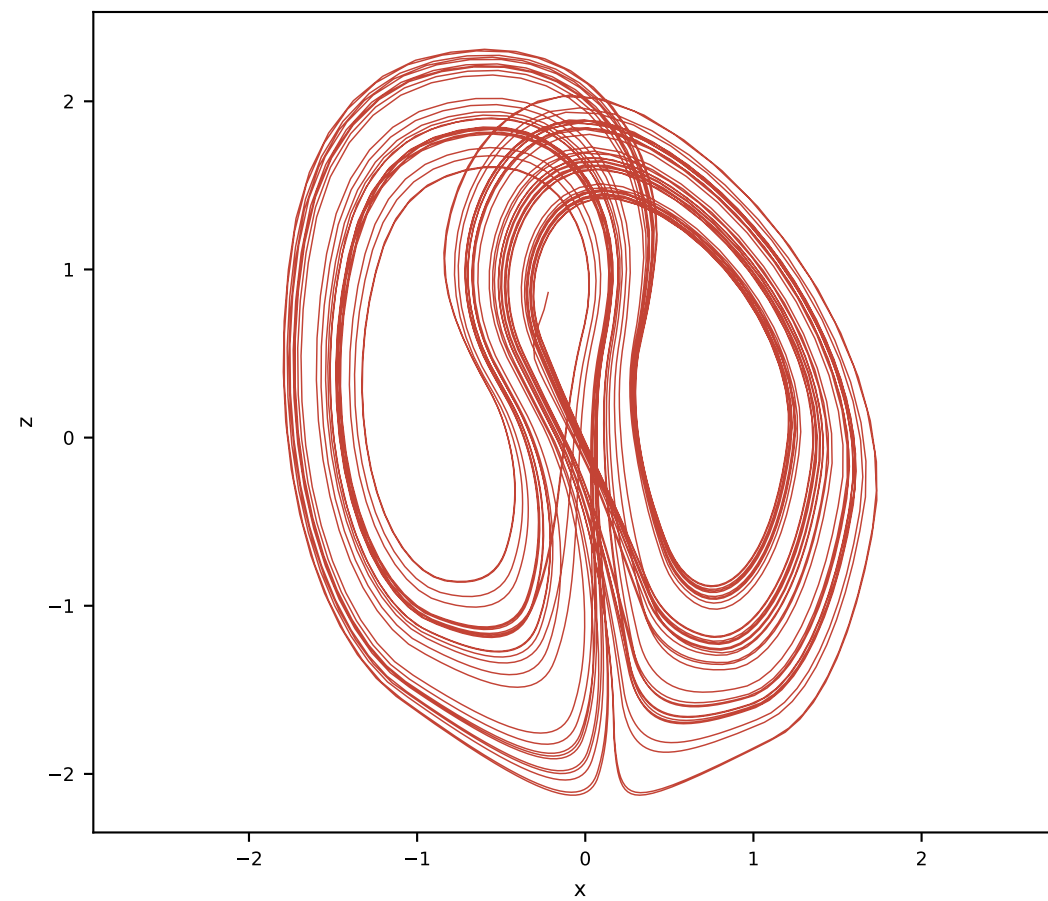
Orthogonal AL-RNN (seed 0)



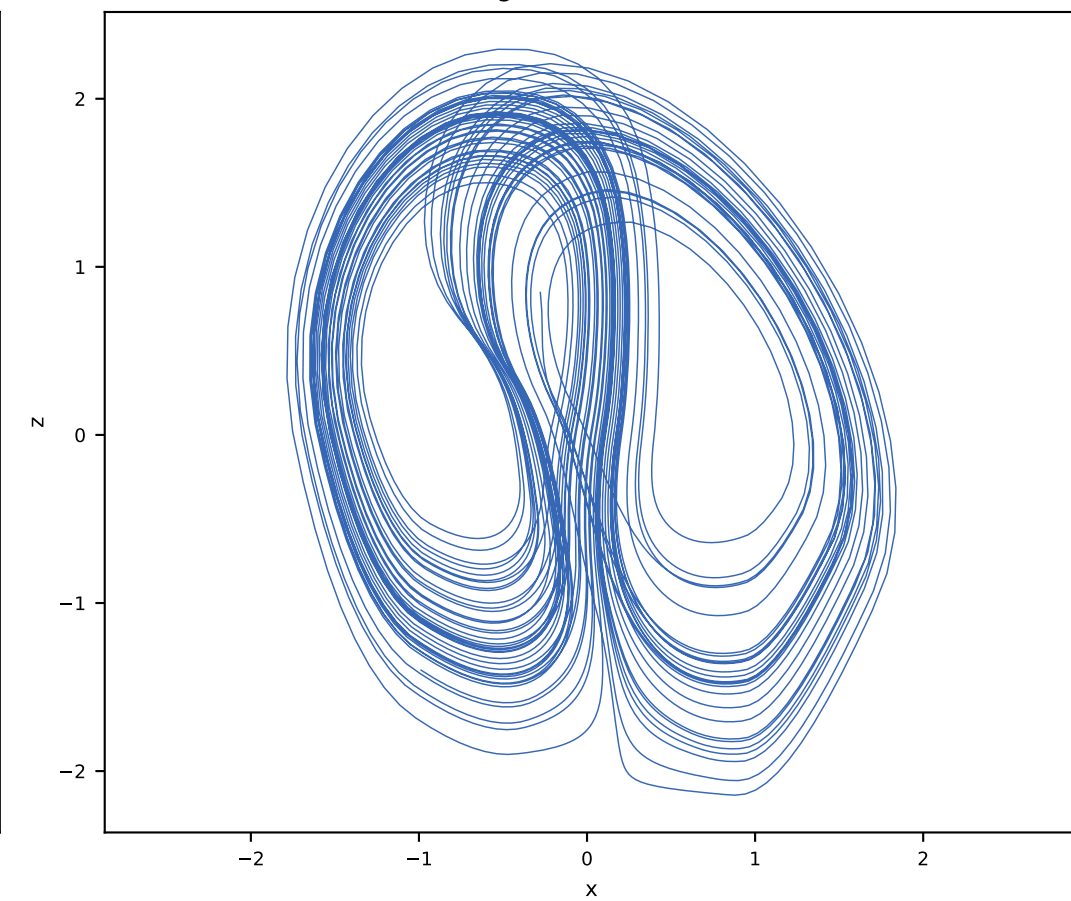
Ground truth x-z



Vanilla x-z (seed 0)

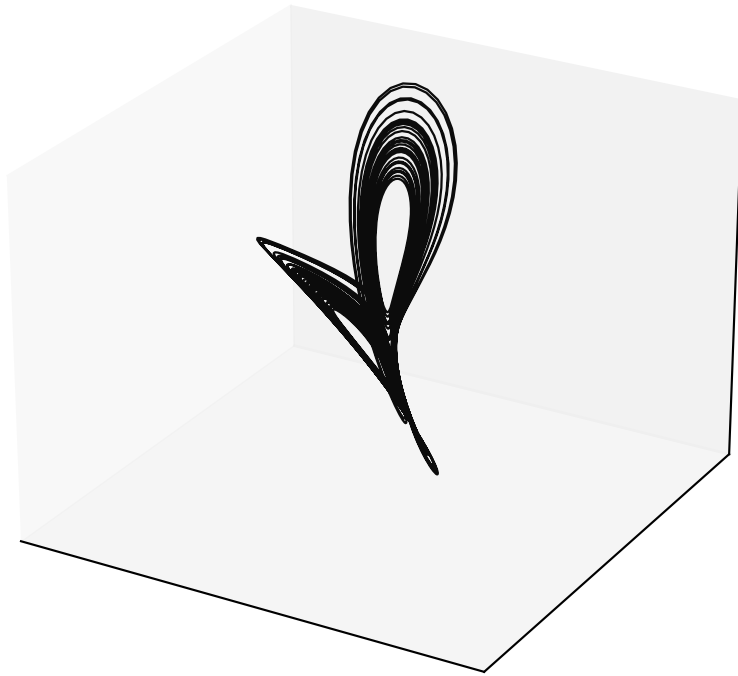


Orthogonal x-z (seed 0)

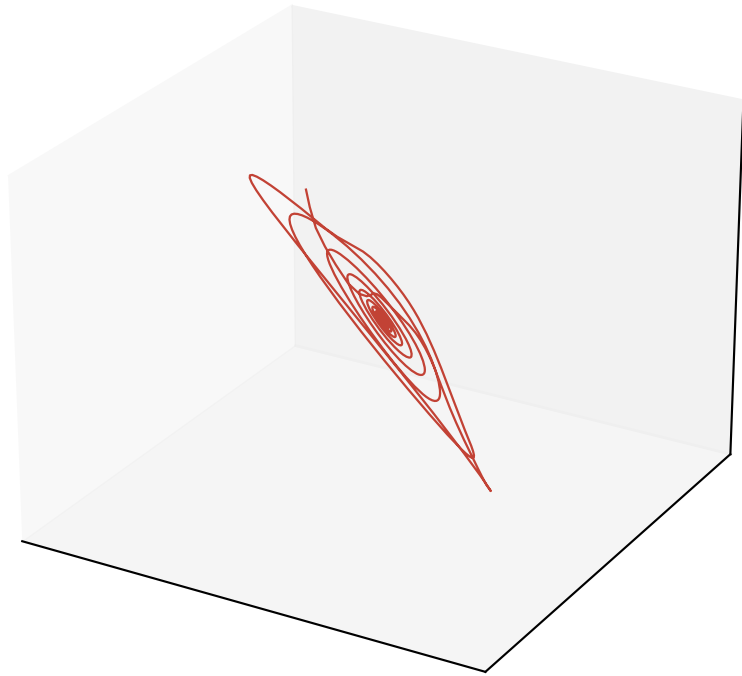


M = 15 P = 2 (median-PSE seed = 1 of [0, 1, 2])
PSE vanilla 0.2275 ± 0.1418 ortho 0.2355 ± 0.1547 mean rel gap +3.5% D_H_state vanilla 0.872 ± 0.173 ortho 0.858 ± 0.199 mean rel gap -1.6%

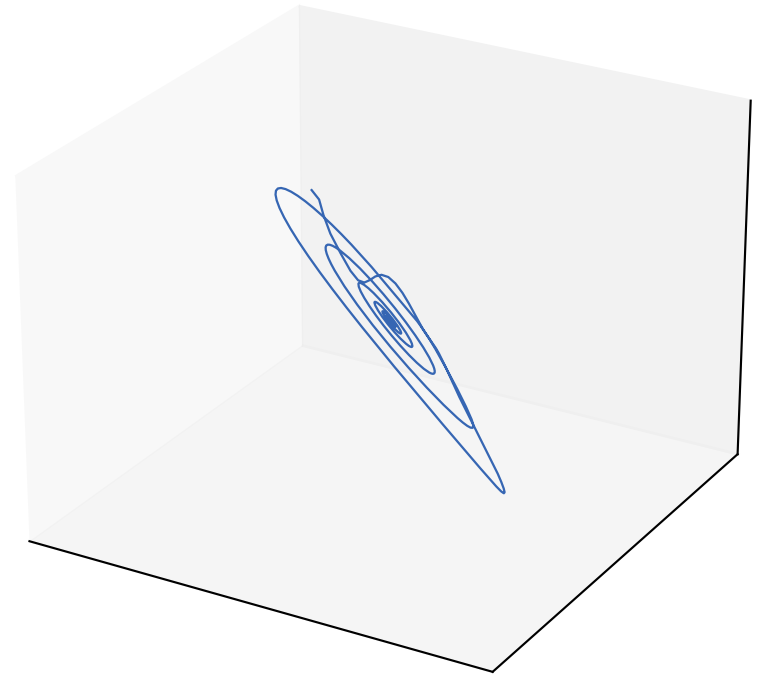
Ground truth (rotated)



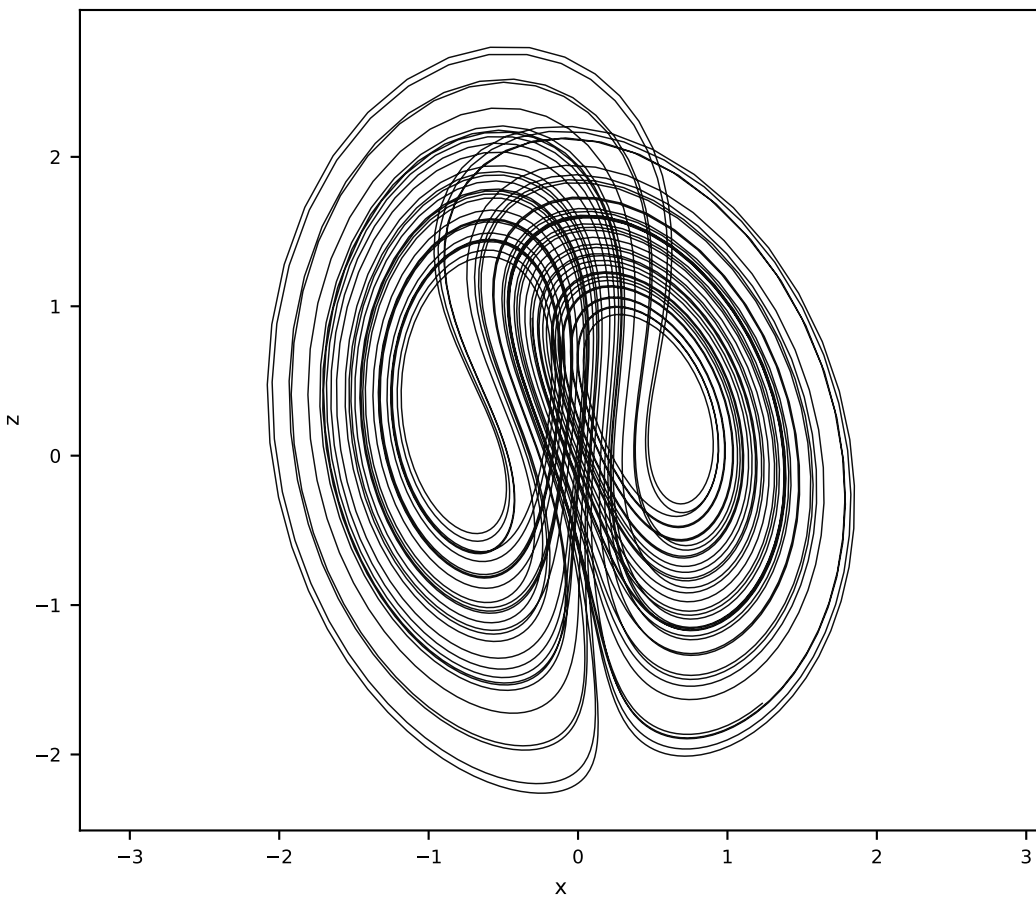
Vanilla AL-RNN (seed 1)



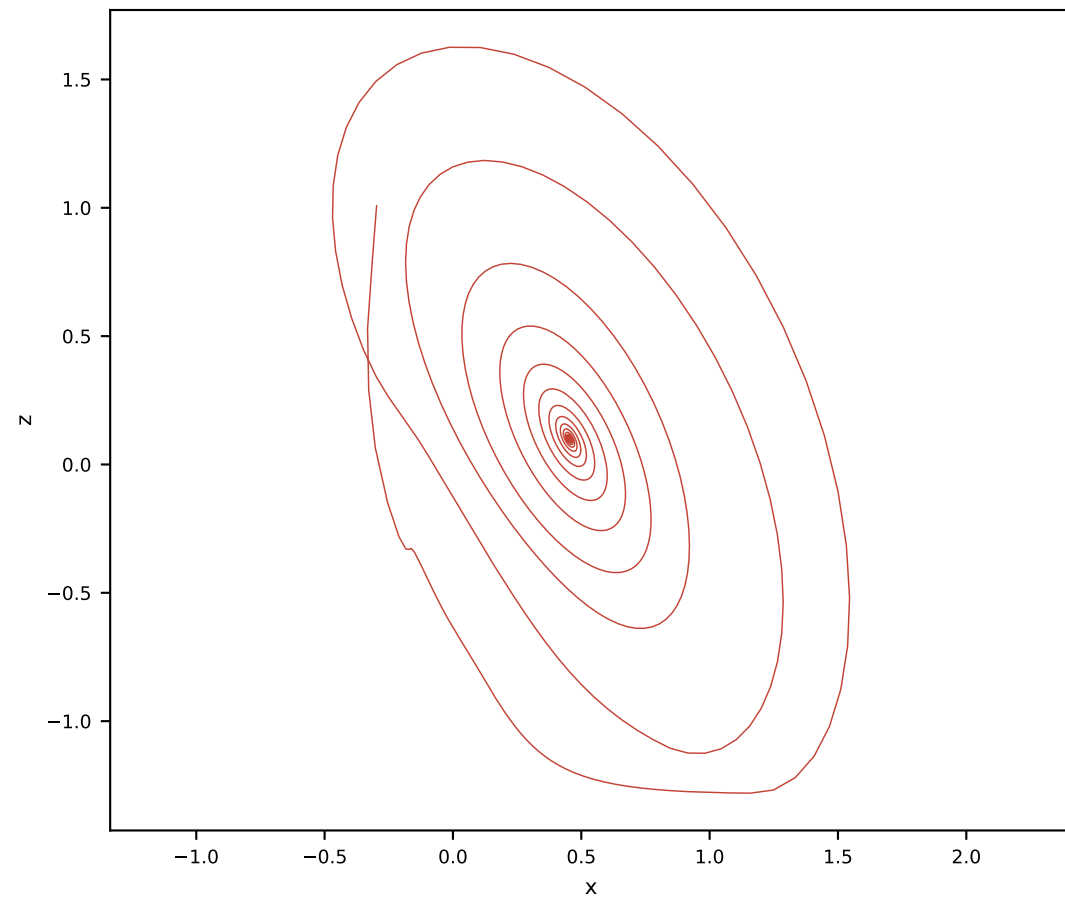
Orthogonal AL-RNN (seed 1)



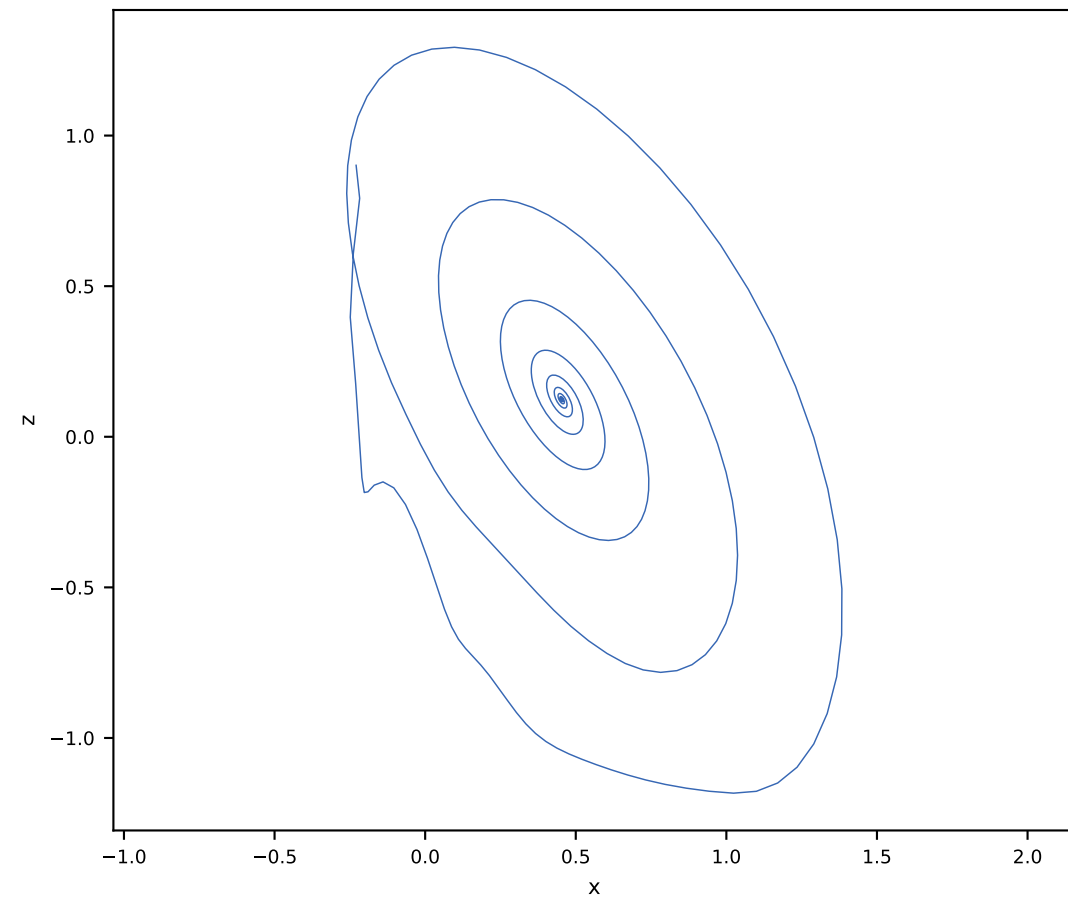
Ground truth x-z



Vanilla x-z (seed 1)

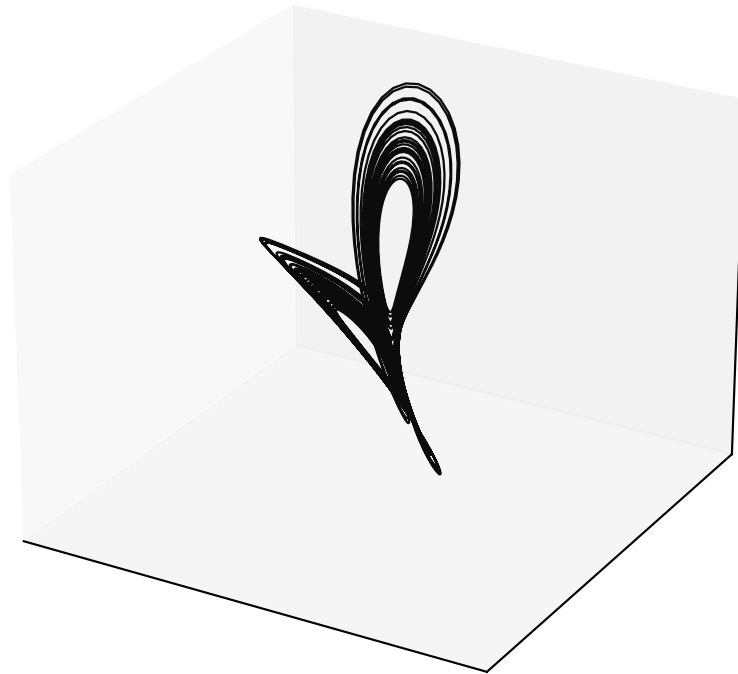


Orthogonal x-z (seed 1)

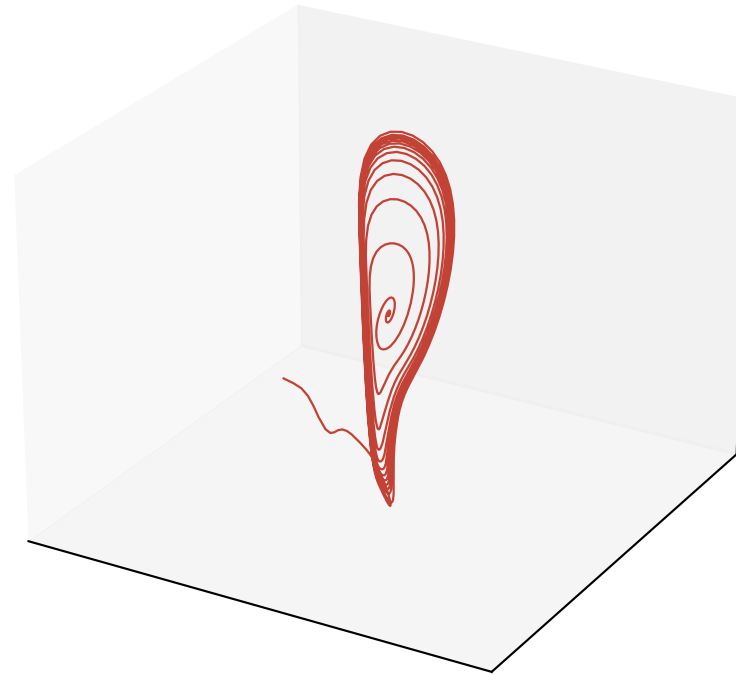


M = 15 P = 3 (median-PSE seed = 0 of [0, 1, 2])
PSE vanilla 0.2415±0.1102 ortho 0.1728±0.1676 mean rel gap -28.4% D_H_state vanilla 0.945±0.030 ortho 0.785±0.170 mean rel gap -16.9%

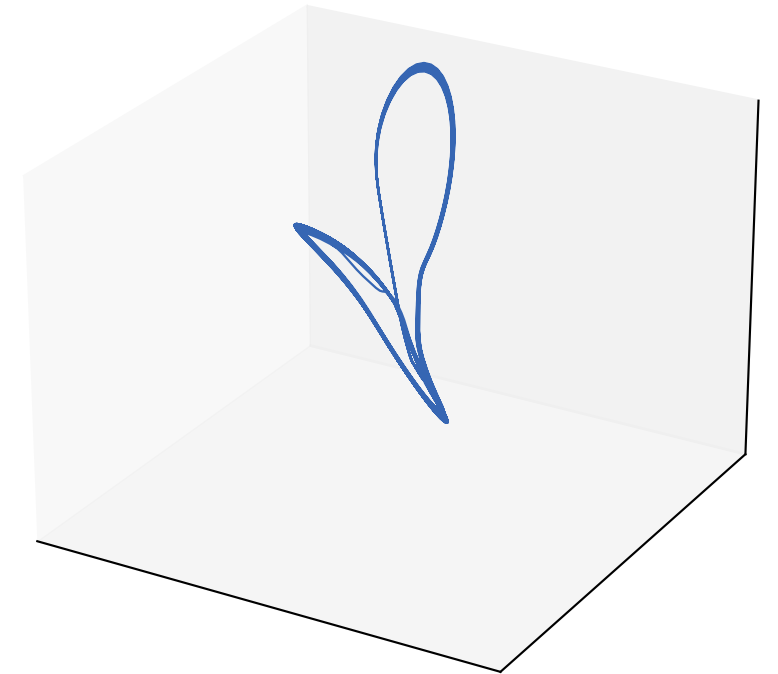
Ground truth (rotated)



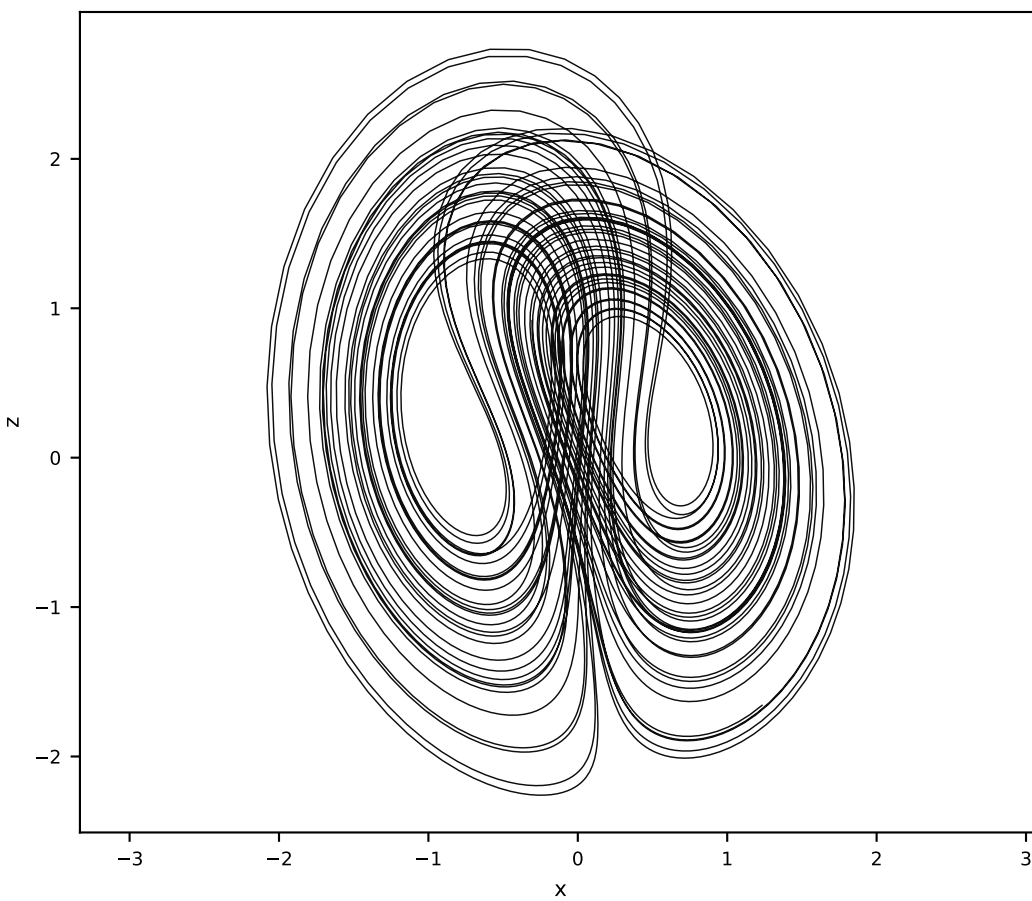
Vanilla AL-RNN (seed 0)



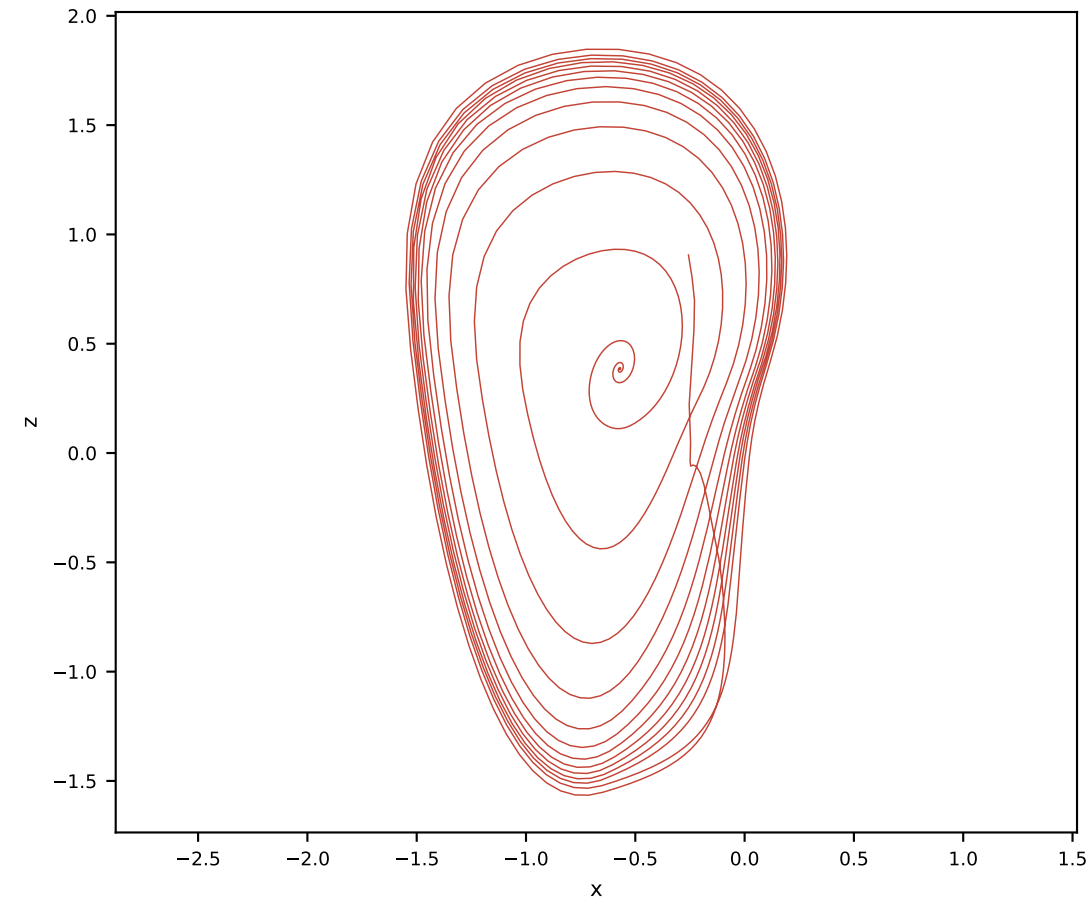
Orthogonal AL-RNN (seed 0)



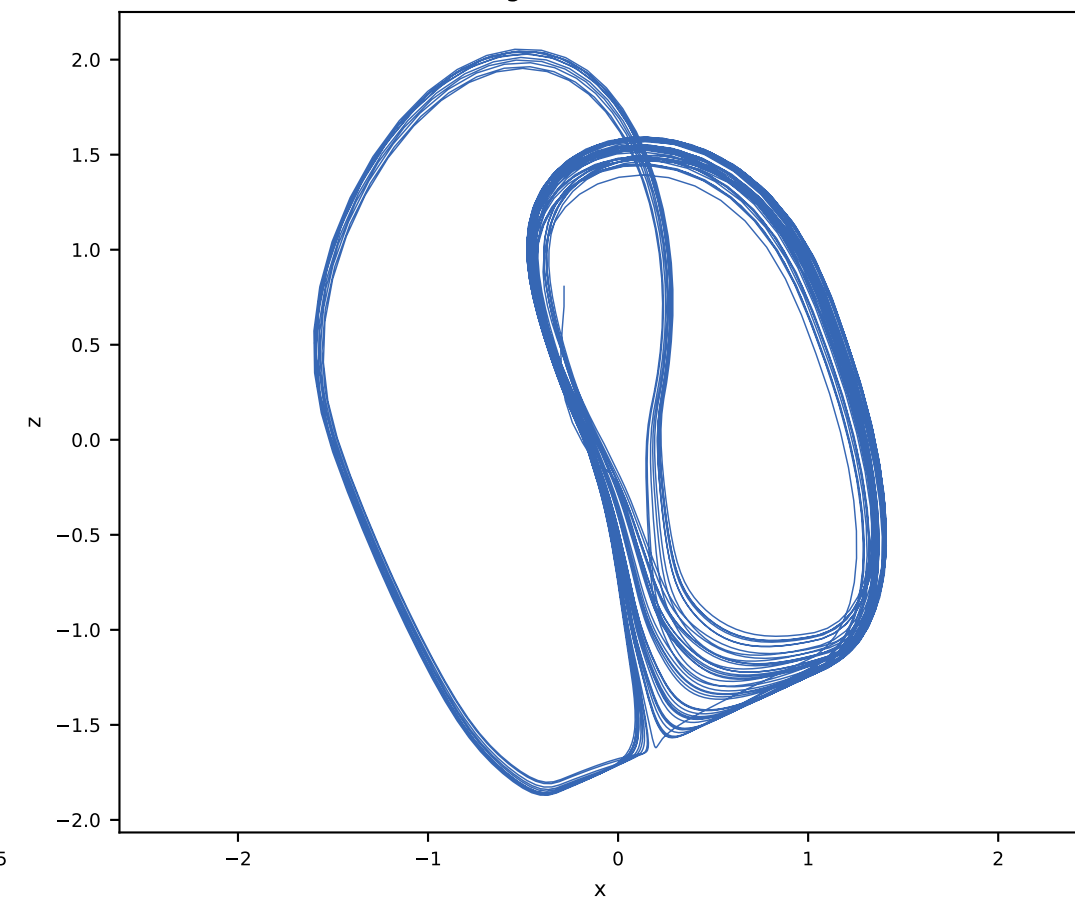
Ground truth x-z



Vanilla x-z (seed 0)

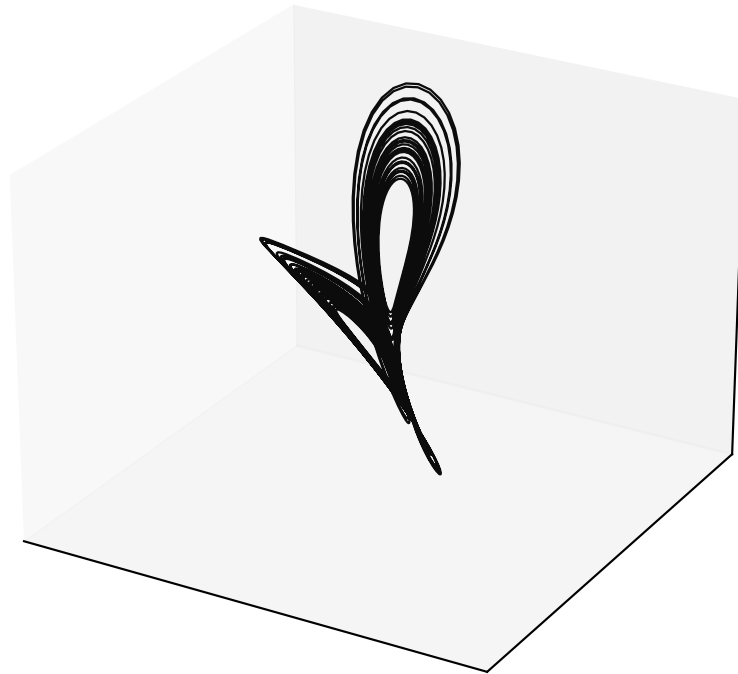


Orthogonal x-z (seed 0)

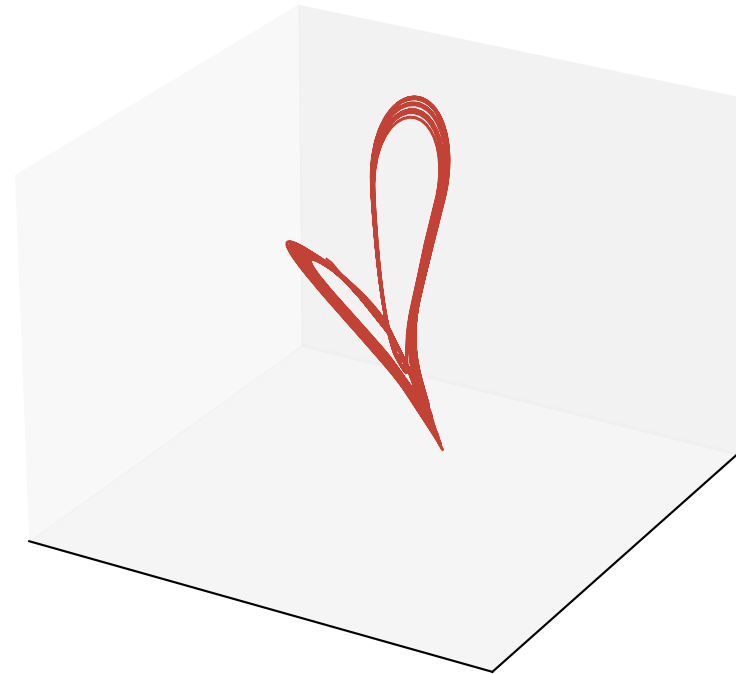


M = 15 P = 4 (median-PSE seed = 0 of [0, 1, 2])
PSE vanilla 0.0864 ± 0.0128 ortho 0.0600 ± 0.0345 mean rel gap -30.5% D_H_state vanilla 0.702 ± 0.014 ortho 0.556 ± 0.058 mean rel gap -20.9%

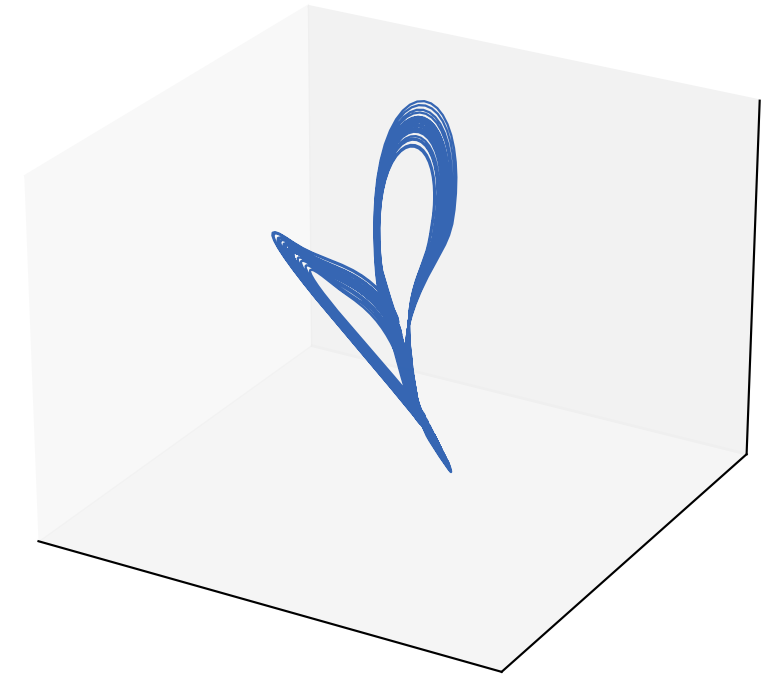
Ground truth (rotated)



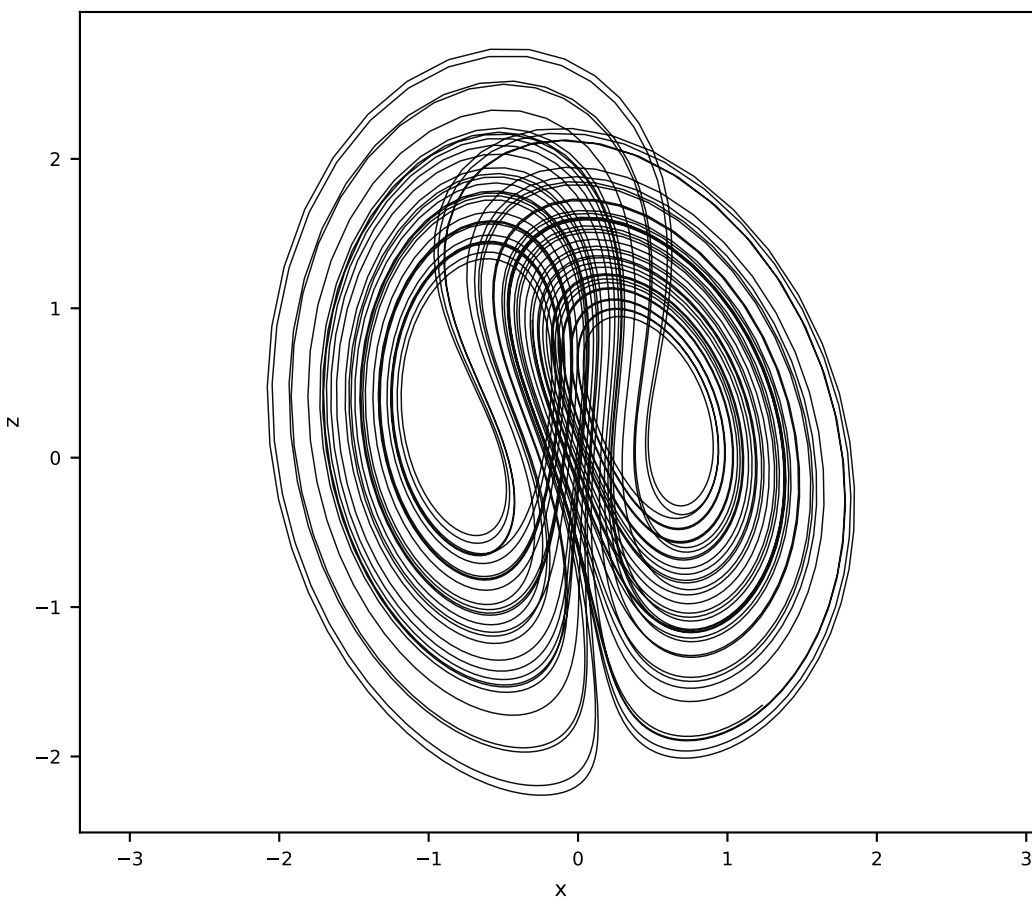
Vanilla AL-RNN (seed 0)



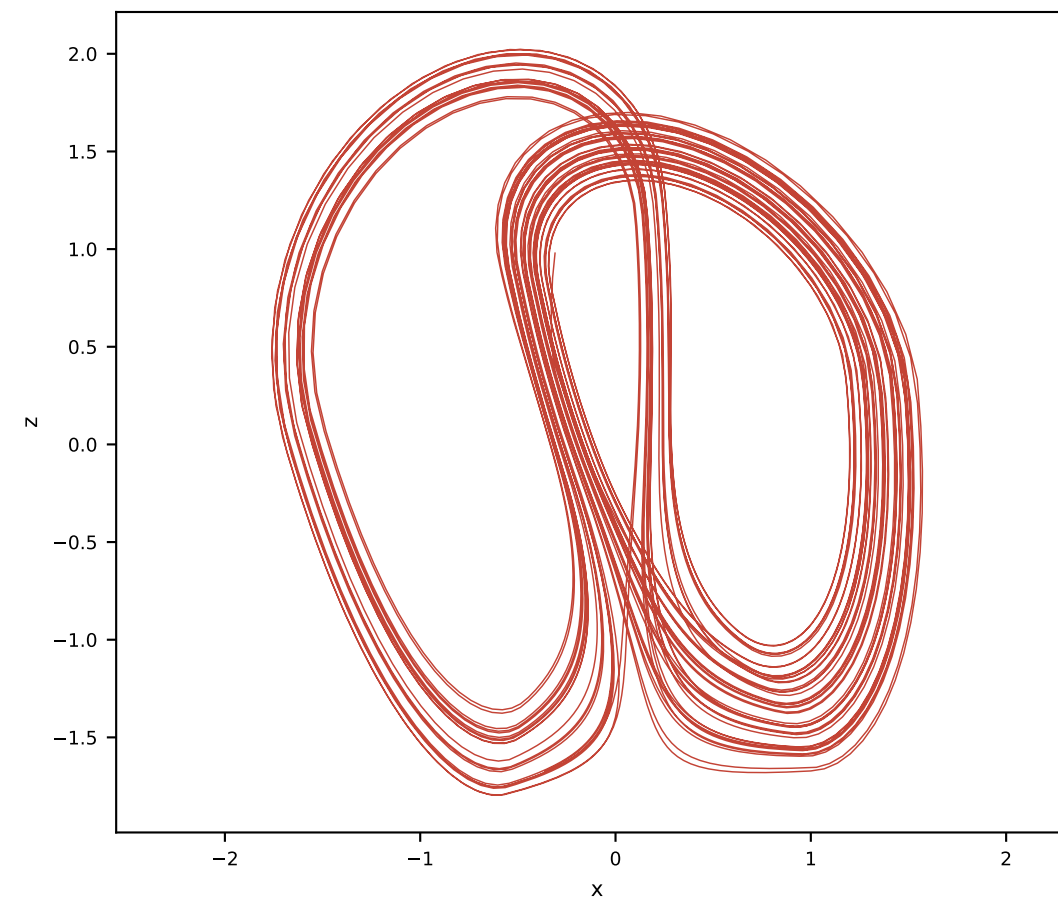
Orthogonal AL-RNN (seed 0)



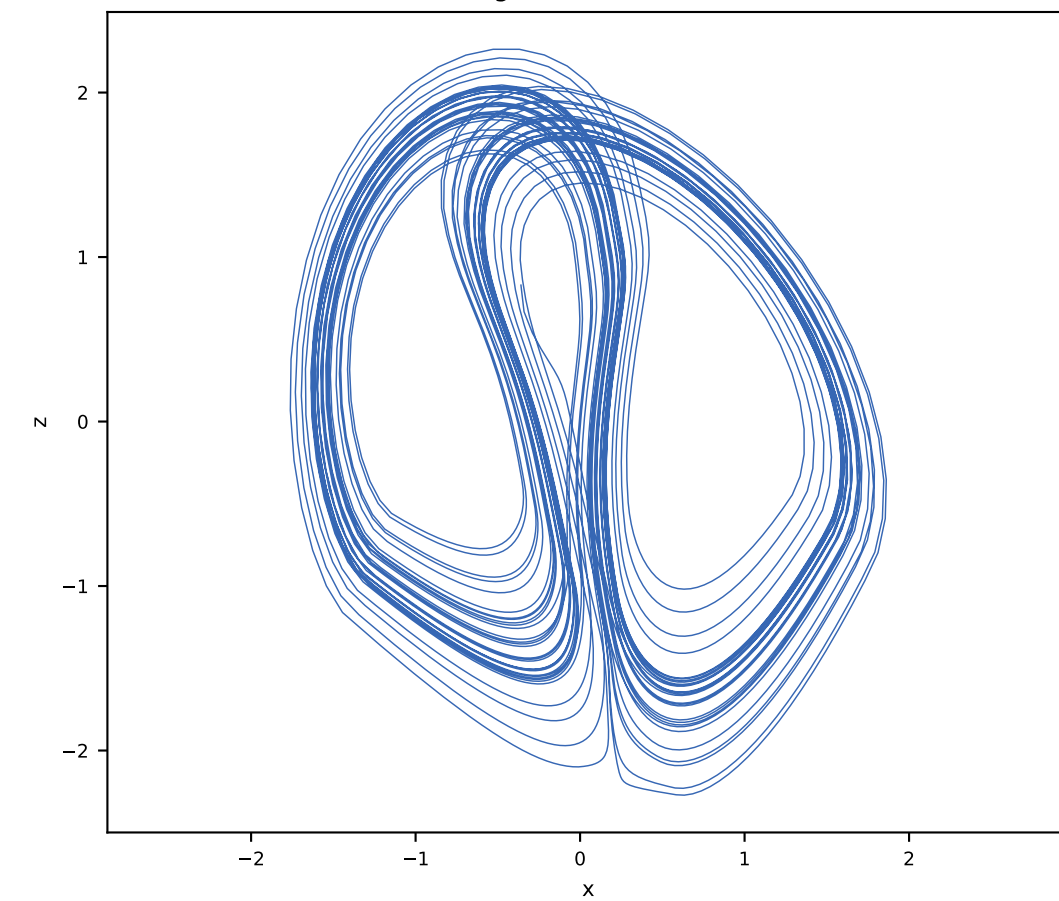
Ground truth x-z



Vanilla x-z (seed 0)

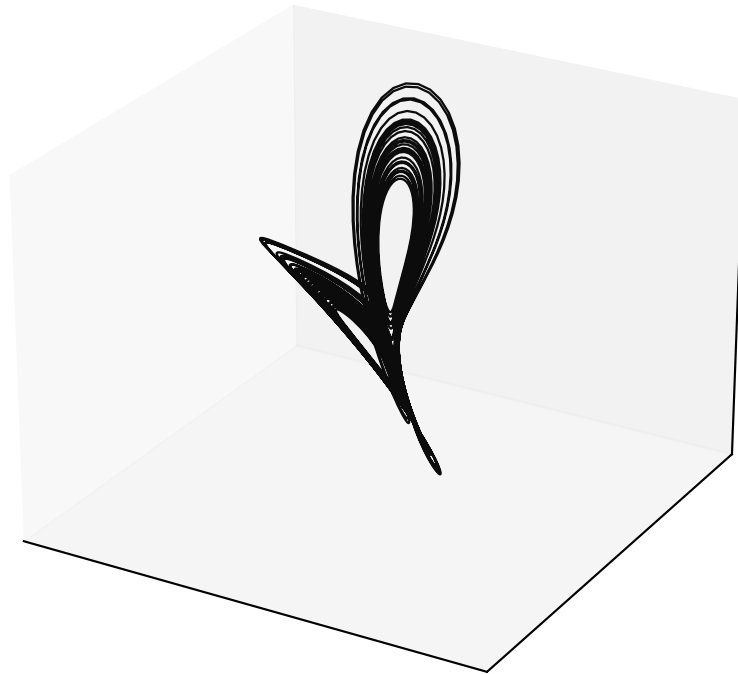


Orthogonal x-z (seed 0)

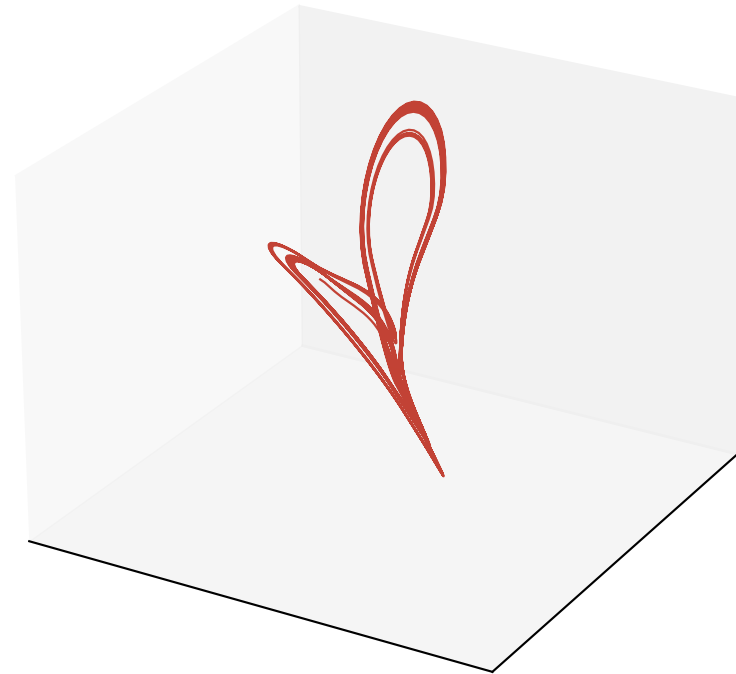


M = 15 P = 5 (median-PSE seed = 2 of [0, 1, 2])
PSE vanilla 0.0772 ± 0.0065 ortho 0.0757 ± 0.0024 mean rel gap -1.9% D_H_state vanilla 0.680 ± 0.064 ortho 0.646 ± 0.130 mean rel gap -5.0%

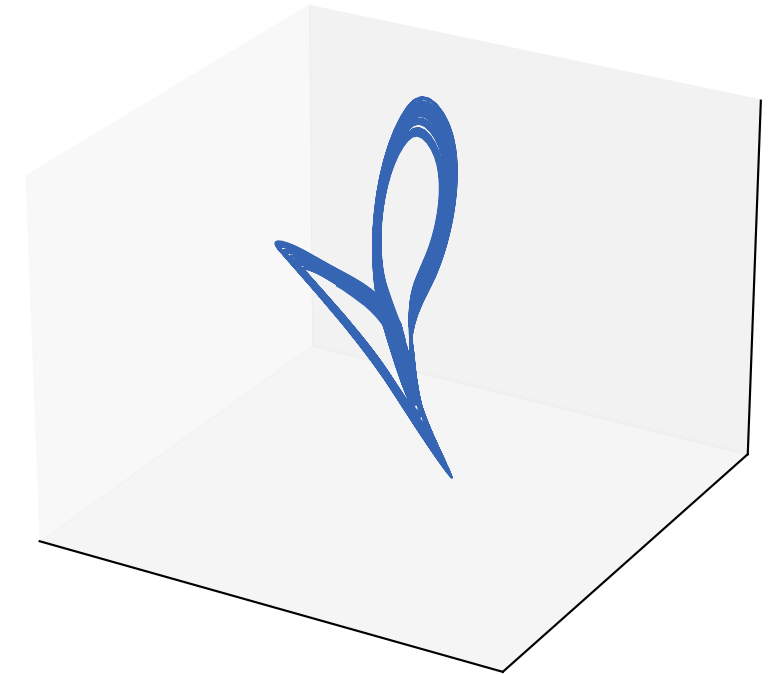
Ground truth (rotated)



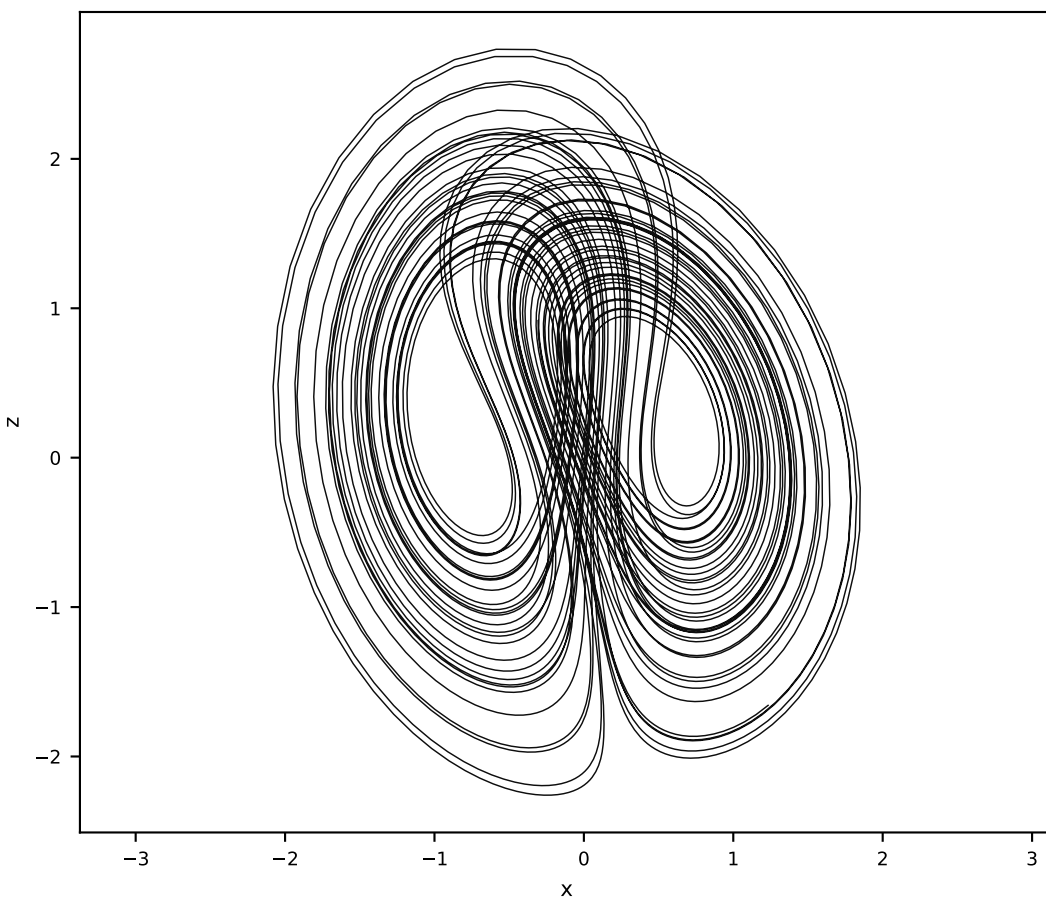
Vanilla AL-RNN (seed 2)



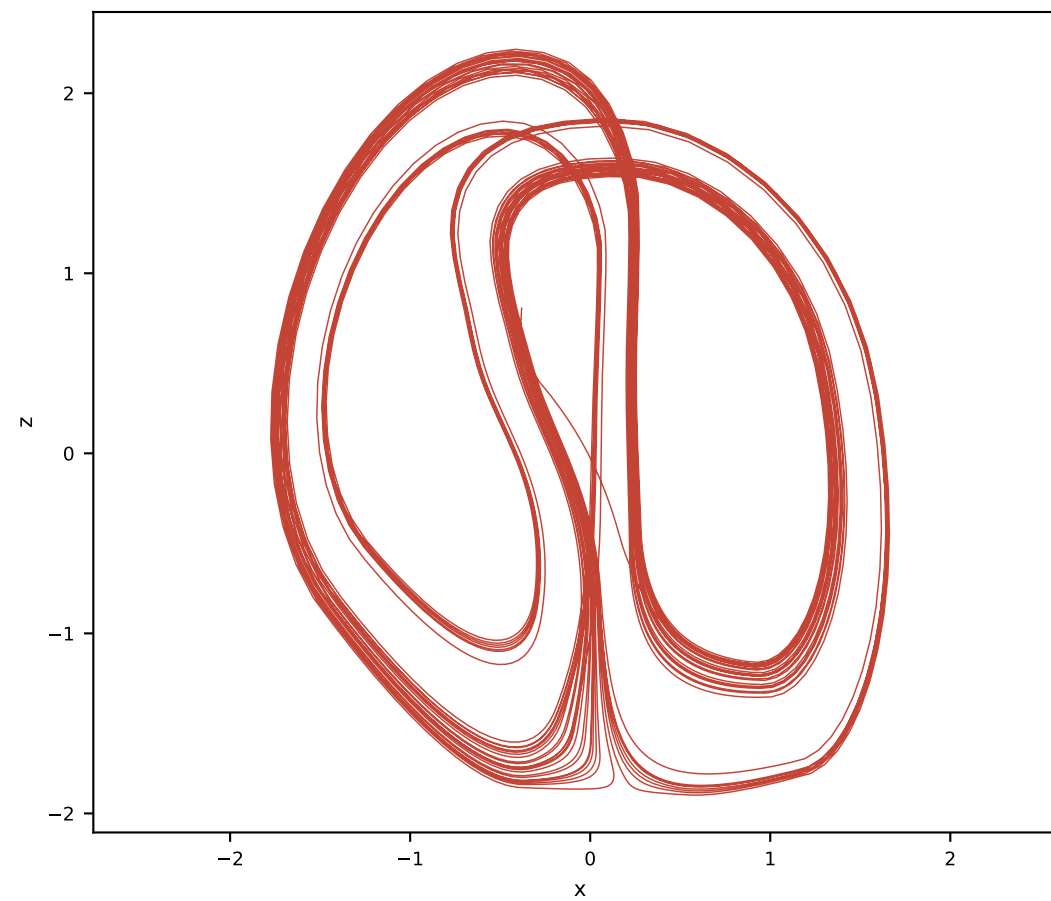
Orthogonal AL-RNN (seed 2)



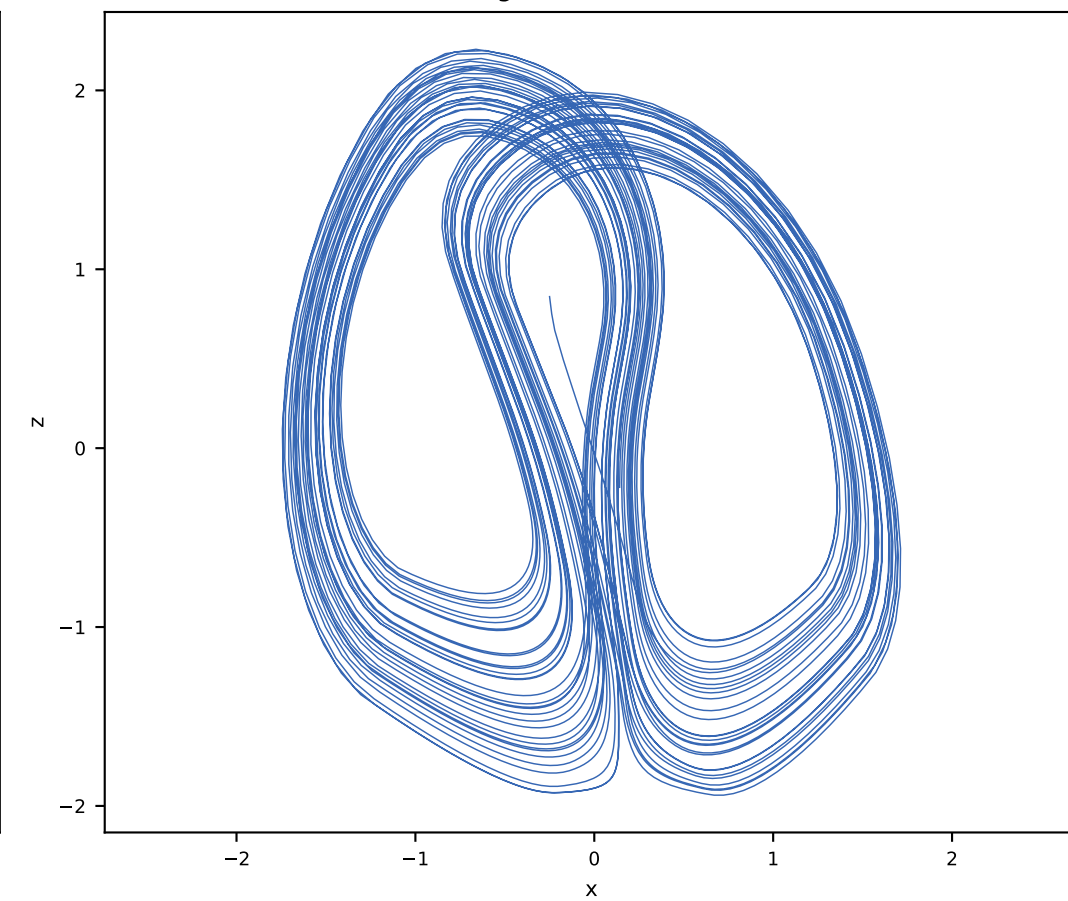
Ground truth x-z



Vanilla x-z (seed 2)

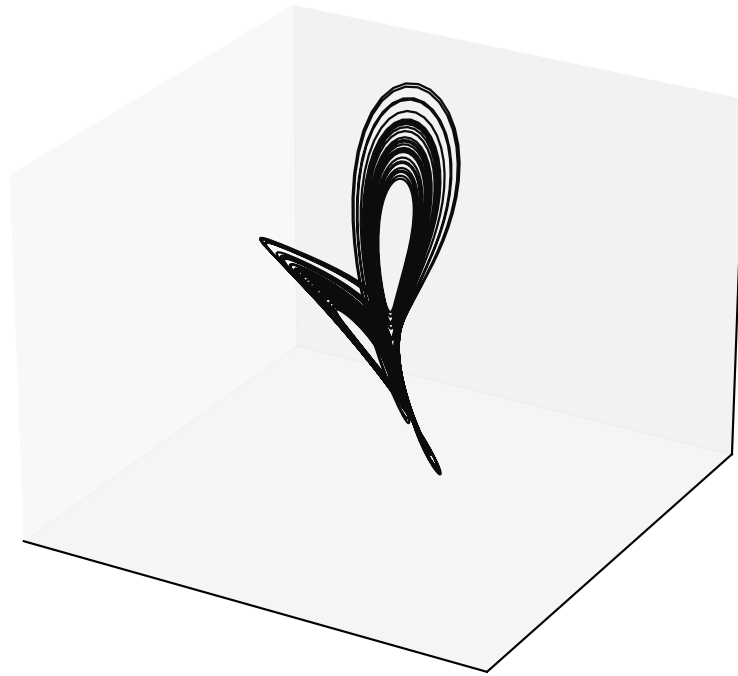


Orthogonal x-z (seed 2)

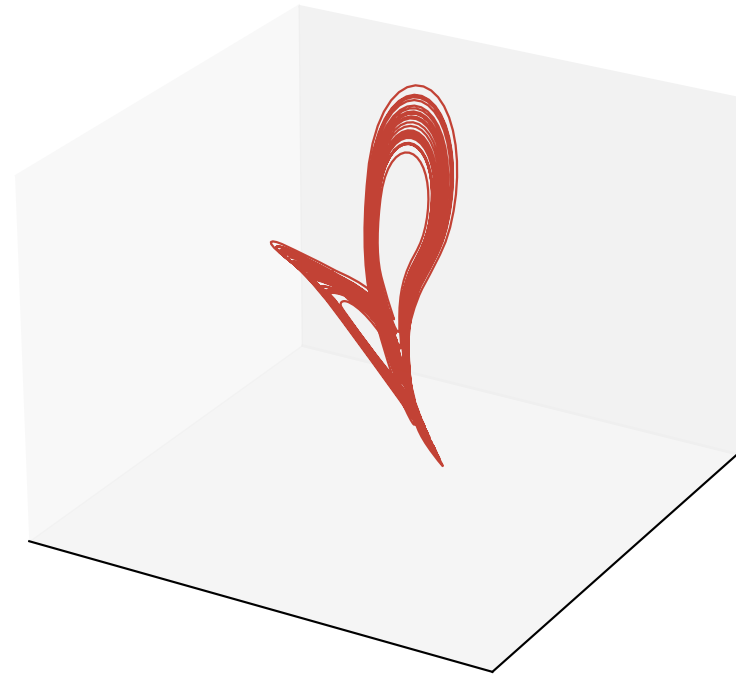


M = 15 P = 6 (median-PSE seed = 1 of [0, 1, 2])
PSE vanilla 0.0637 ± 0.0247 ortho 0.0504 ± 0.0230 mean rel gap -20.8% D_H_state vanilla 0.533 ± 0.085 ortho 0.473 ± 0.072 mean rel gap -11.3%

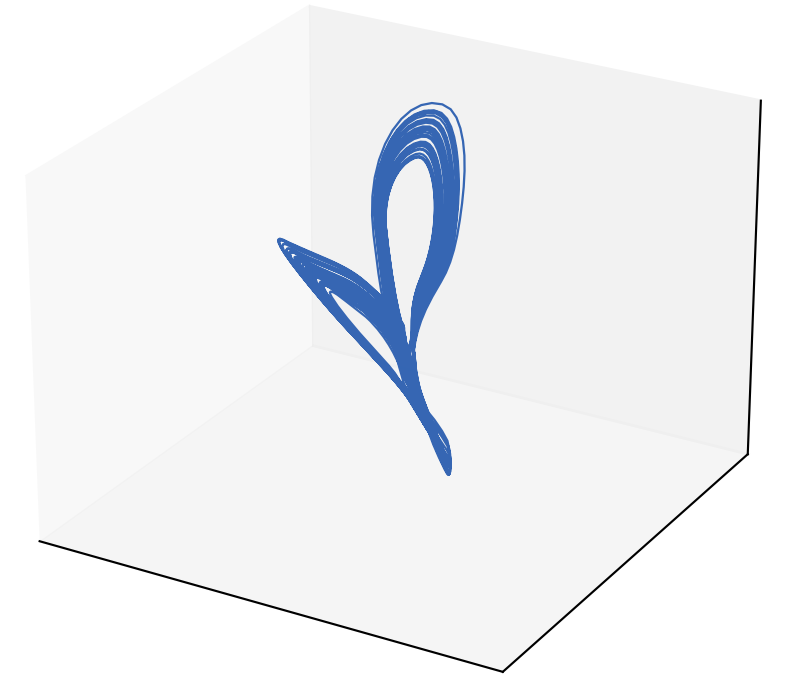
Ground truth (rotated)



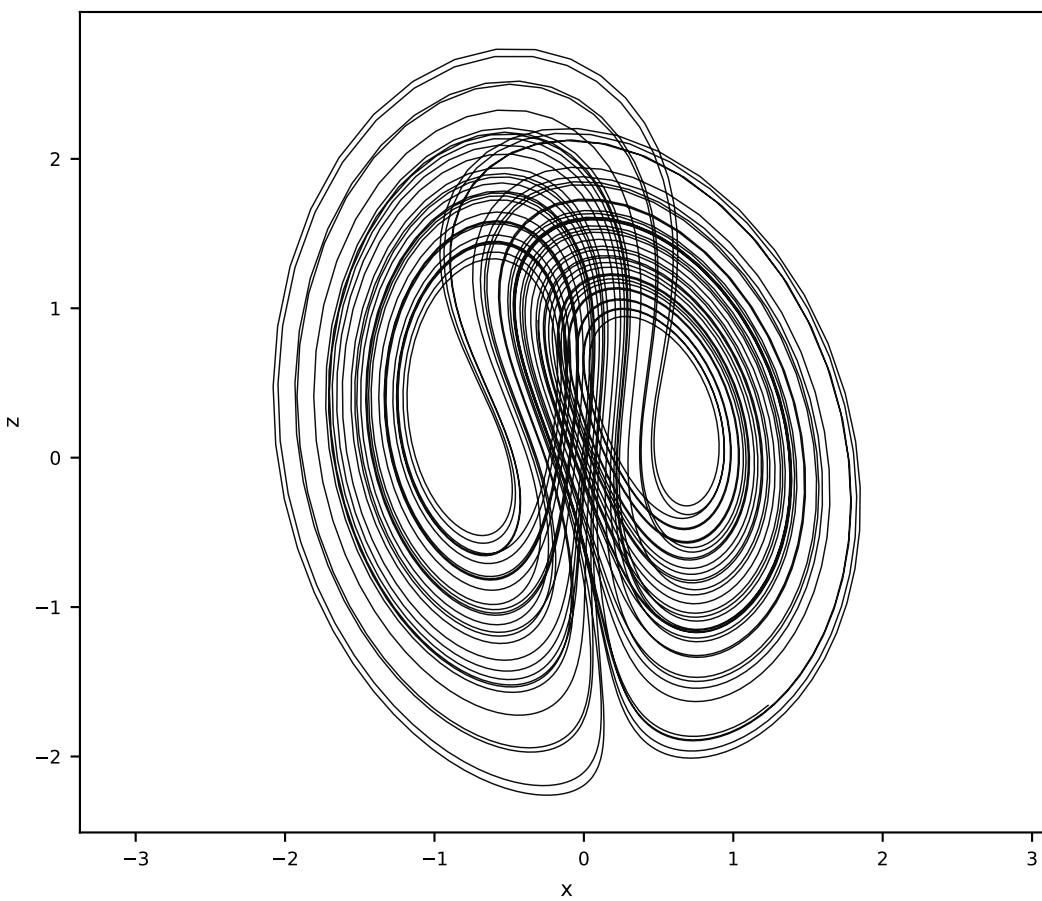
Vanilla AL-RNN (seed 1)



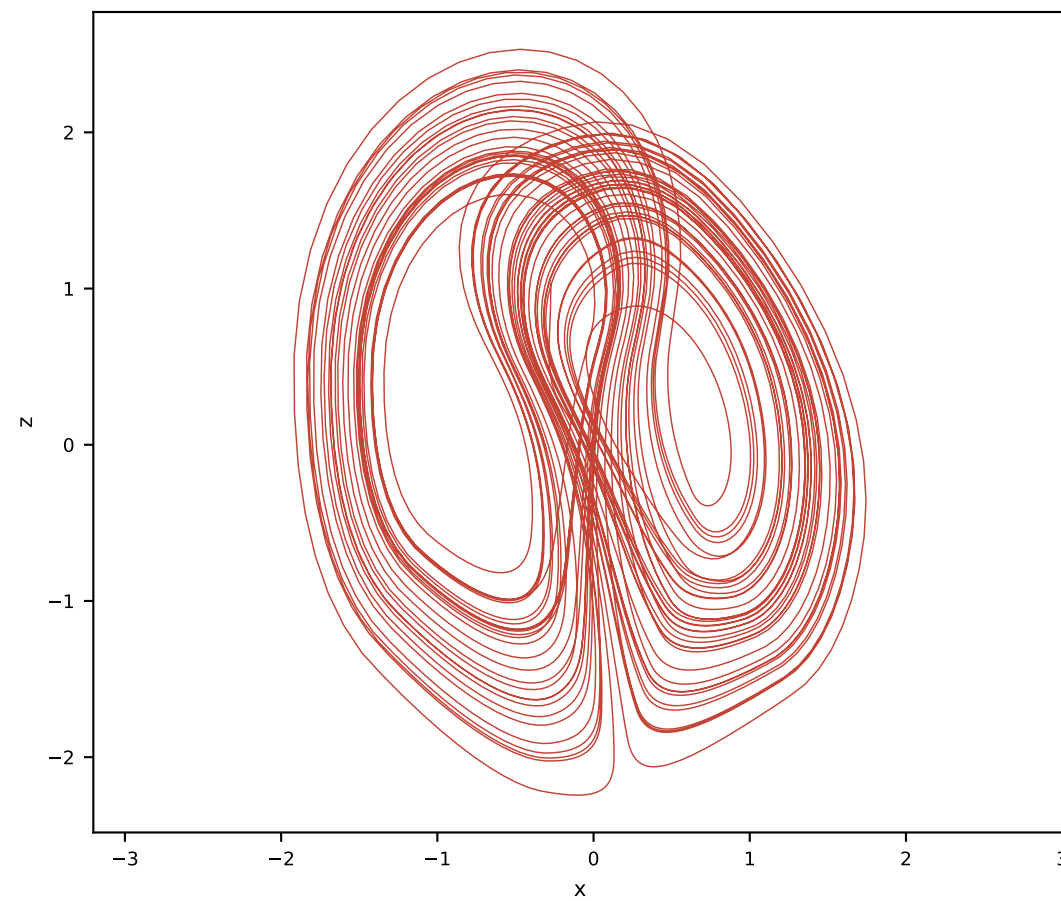
Orthogonal AL-RNN (seed 1)



Ground truth x-z



Vanilla x-z (seed 1)



Orthogonal x-z (seed 1)

